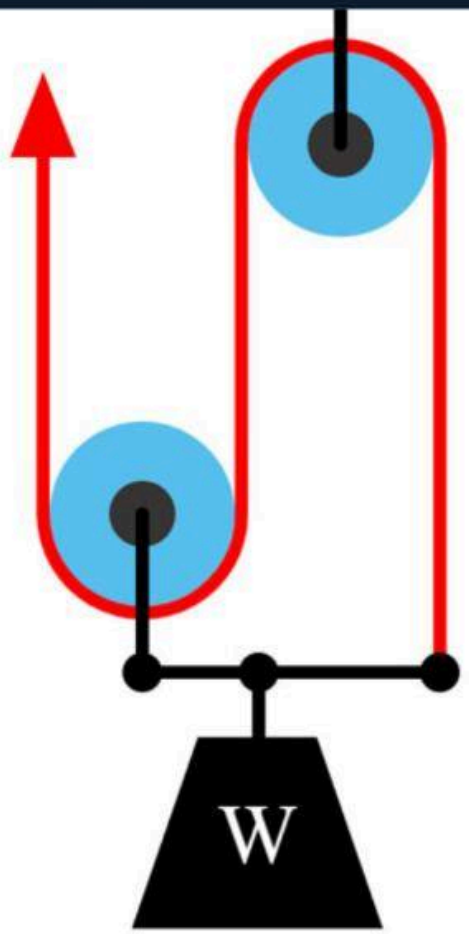


TowerCrane



Ropes are used to lift the load. They go through many pulleys. The pulley allow to evenly distribute and reduce the load on the cable.



The fixed pulley allows you to change the direction of the force. It is fixed in the wall or on a beam

The movable pulley can reduce the force required to lift the load in half. Such a pulley moves along with the cable.

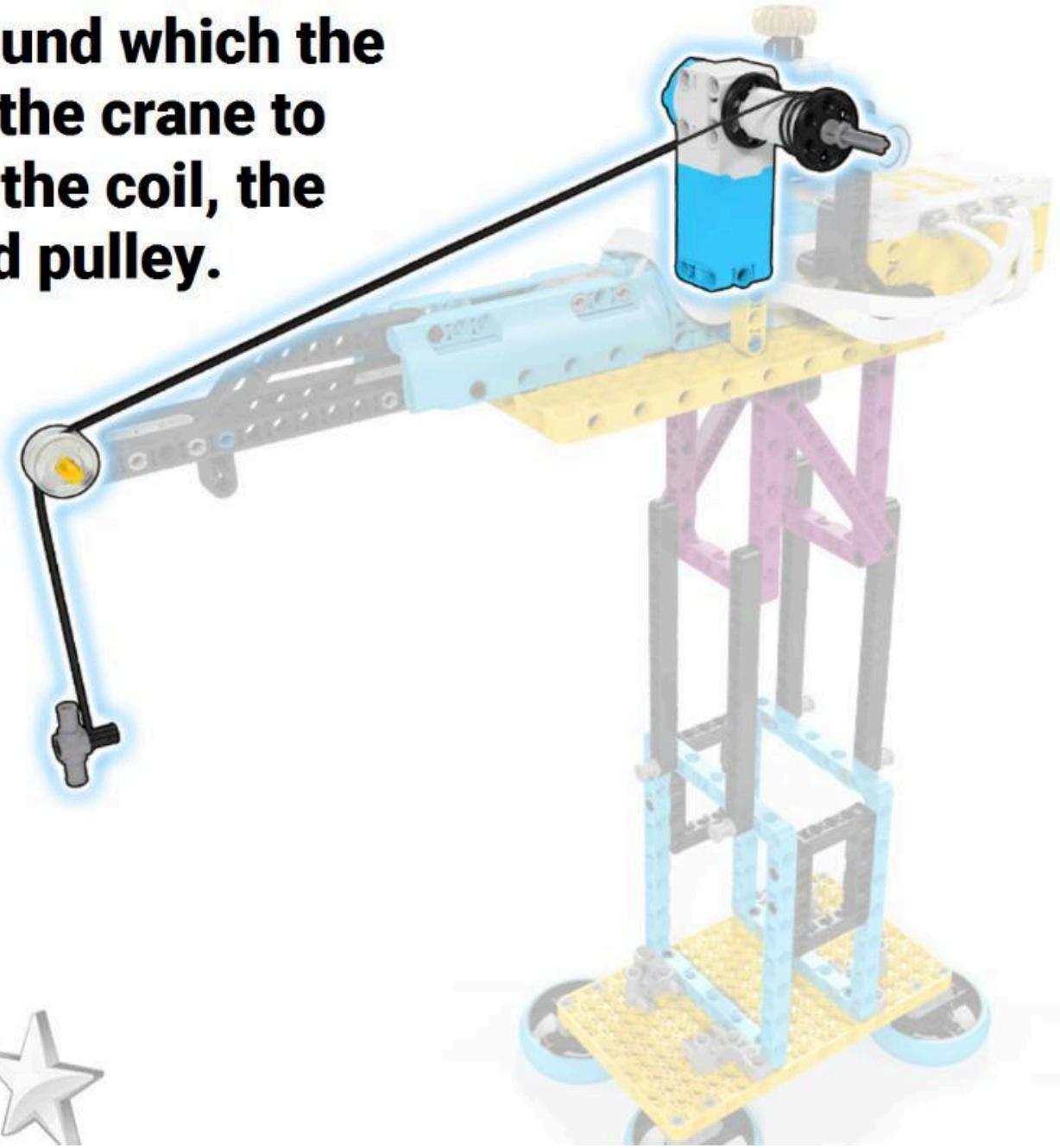
The motor spins the coil around which the cable is wound. This allows the crane to raise and lower loads. After the coil, the cable is tipped over the fixed pulley.

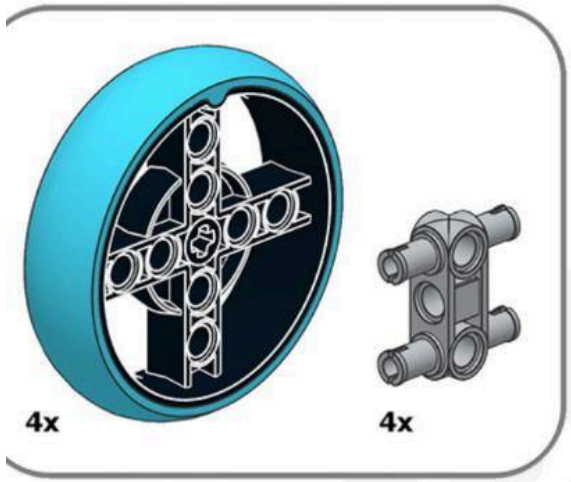


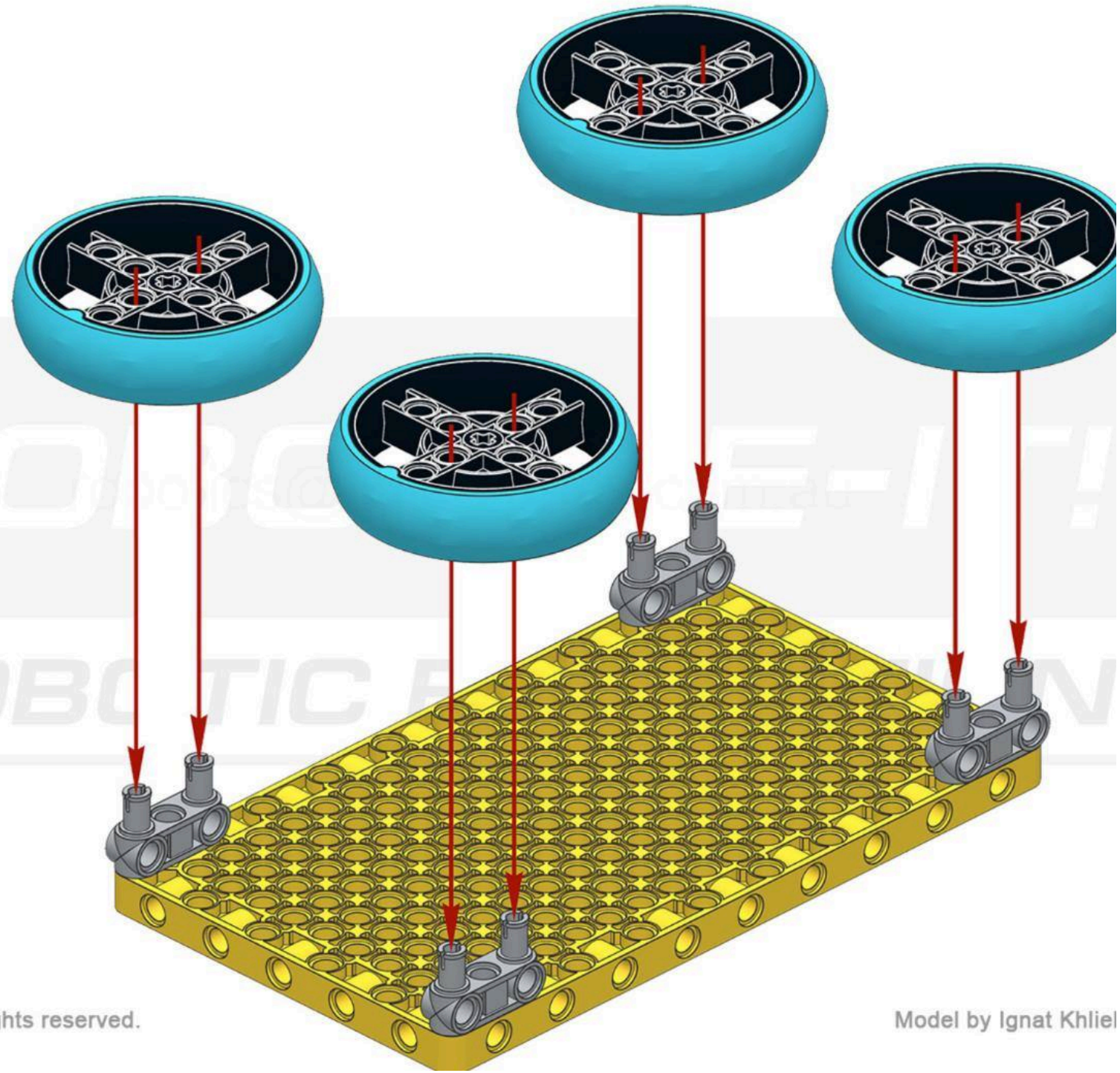
Find:

- **Coil**
- **Motor**
- **Fixed pulley**









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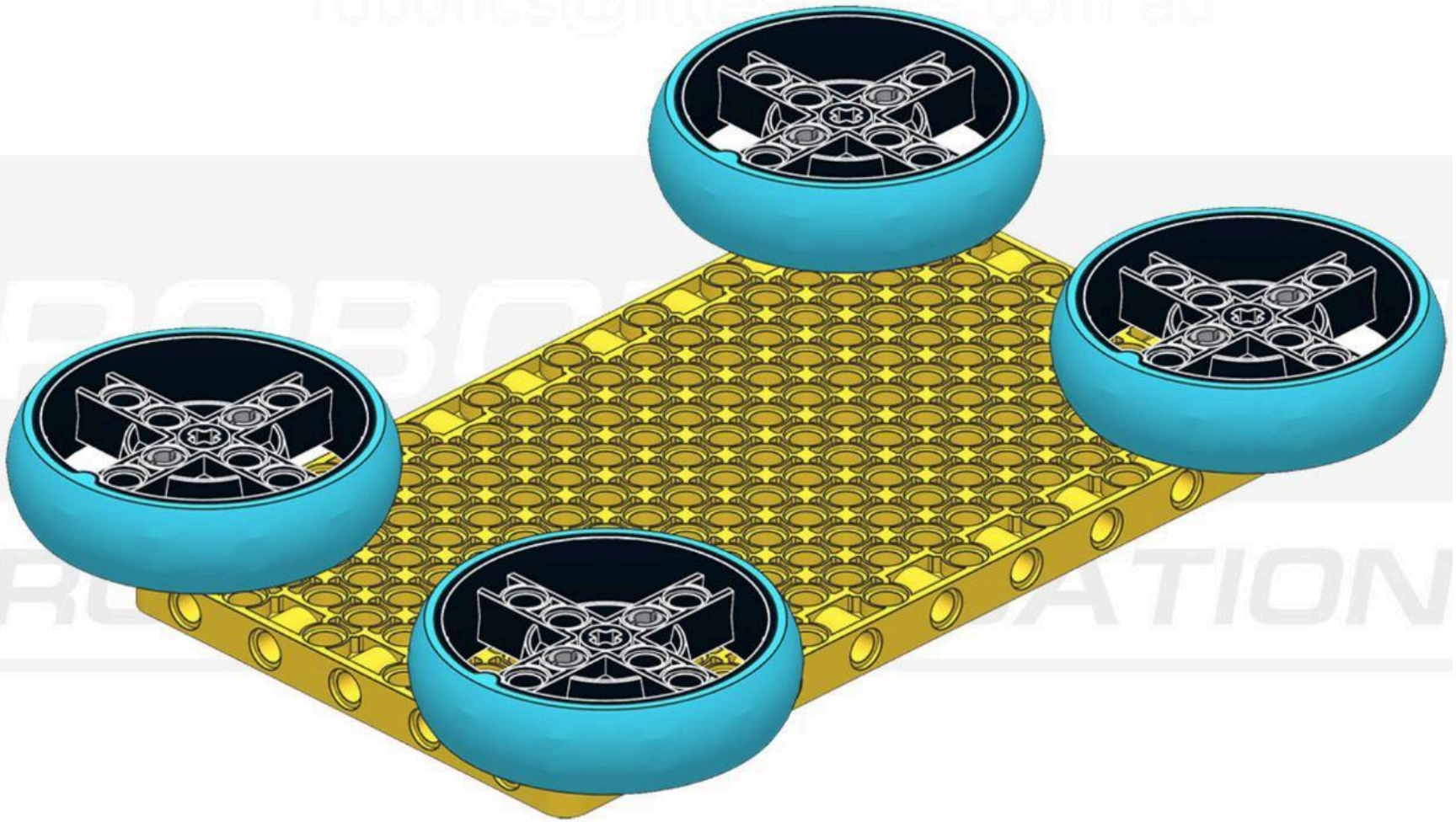
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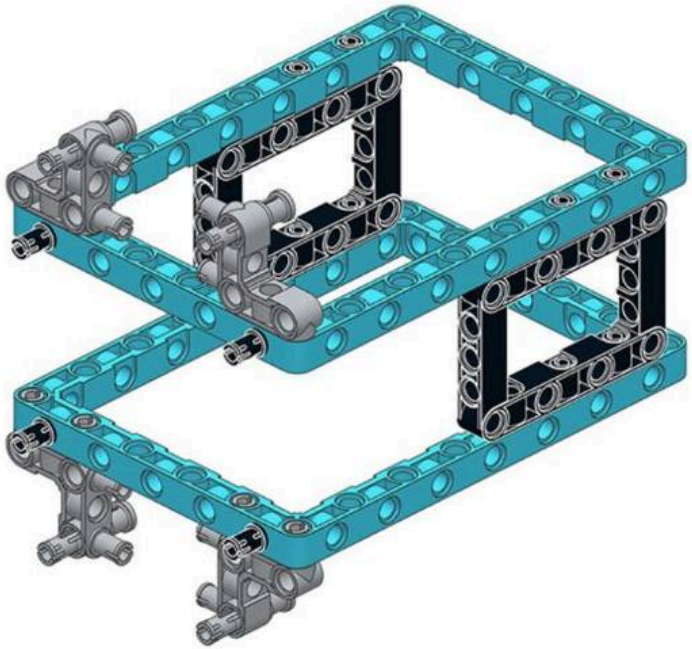
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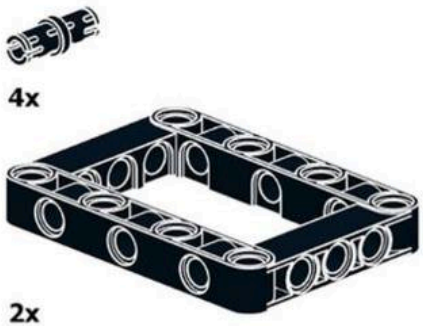
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Model by Ignat Khliel

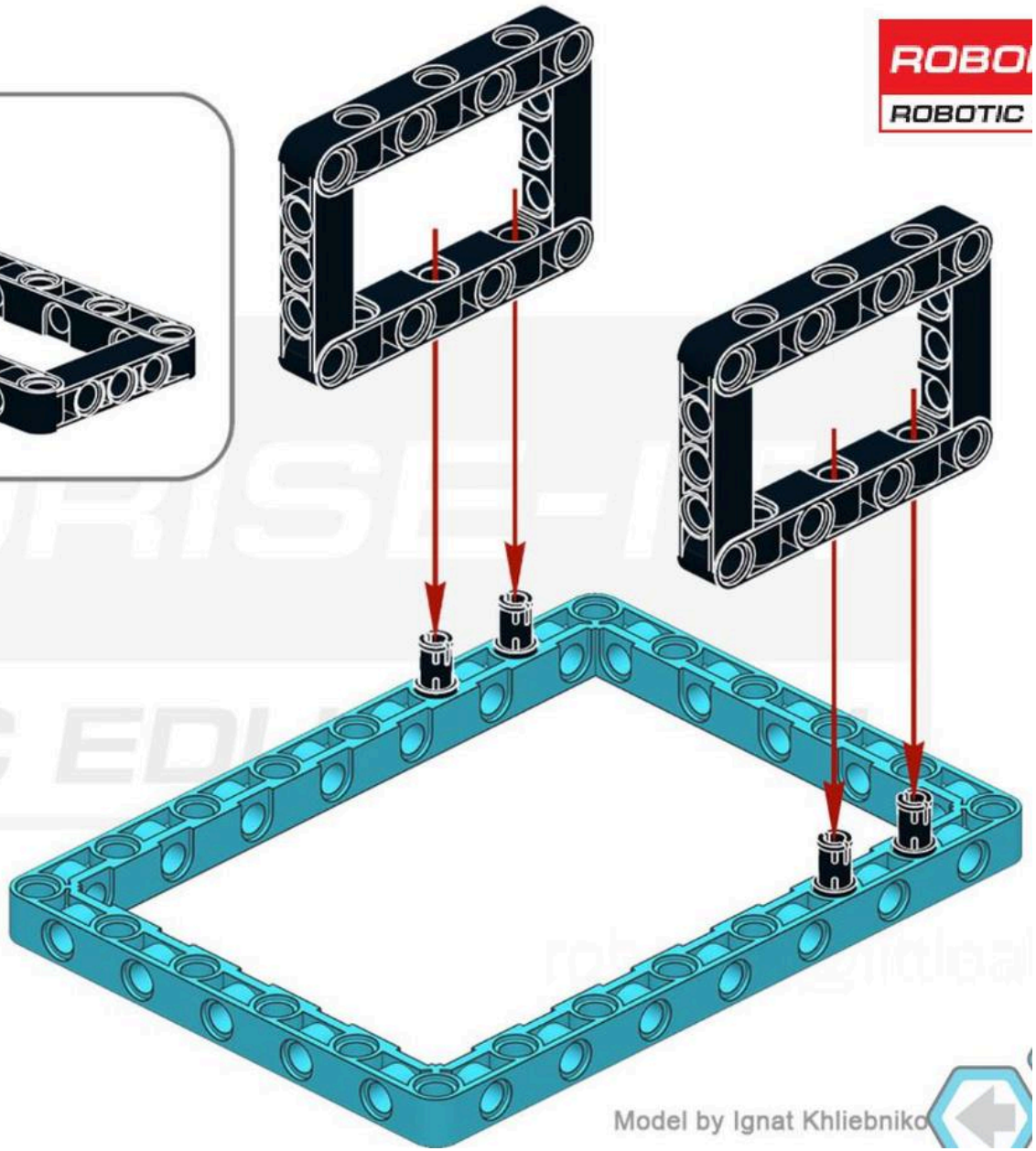
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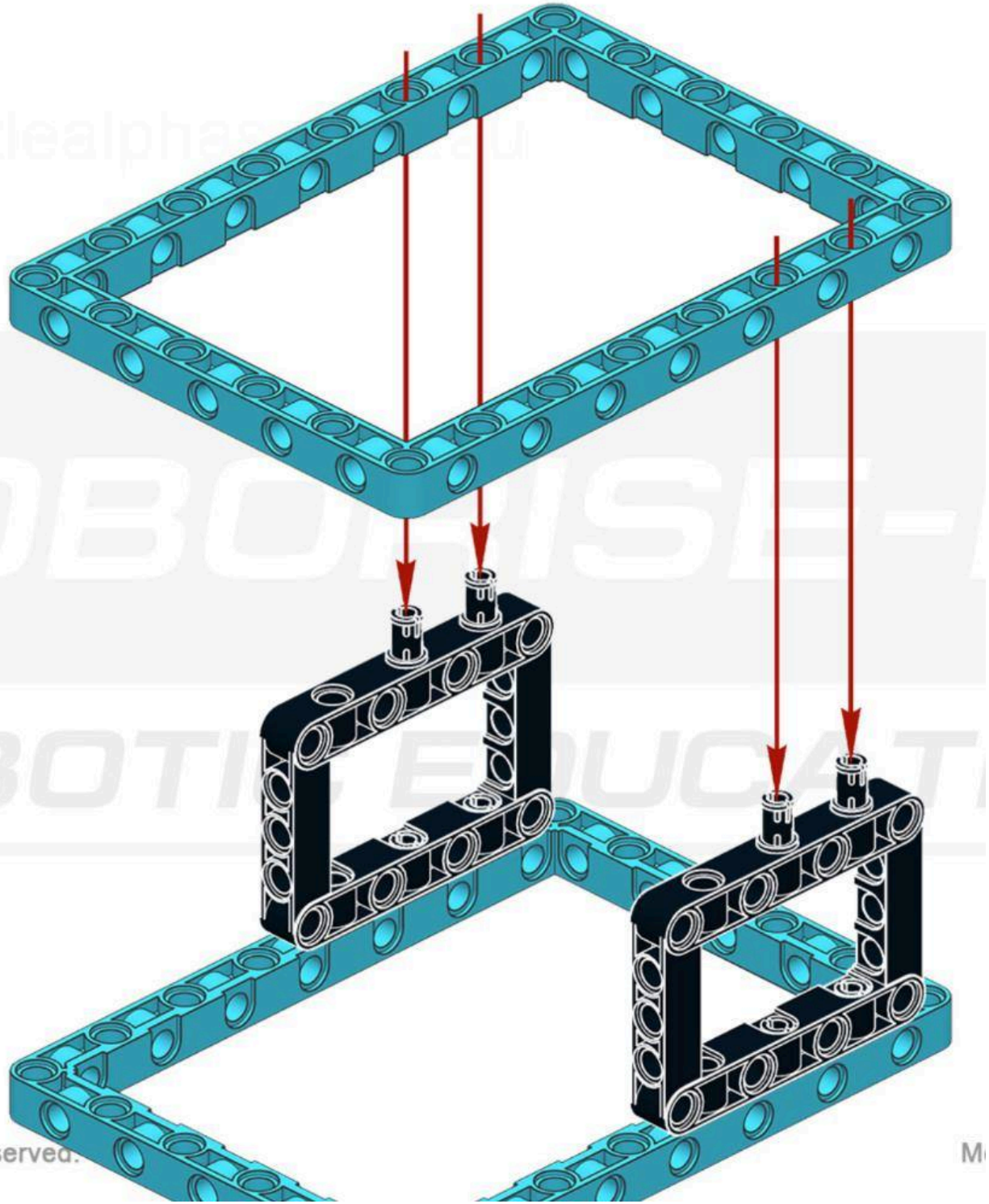
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Model by Ignat Khliebnikov



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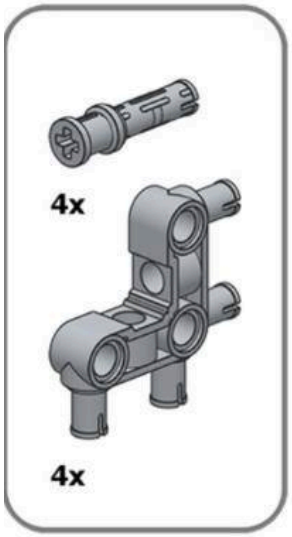




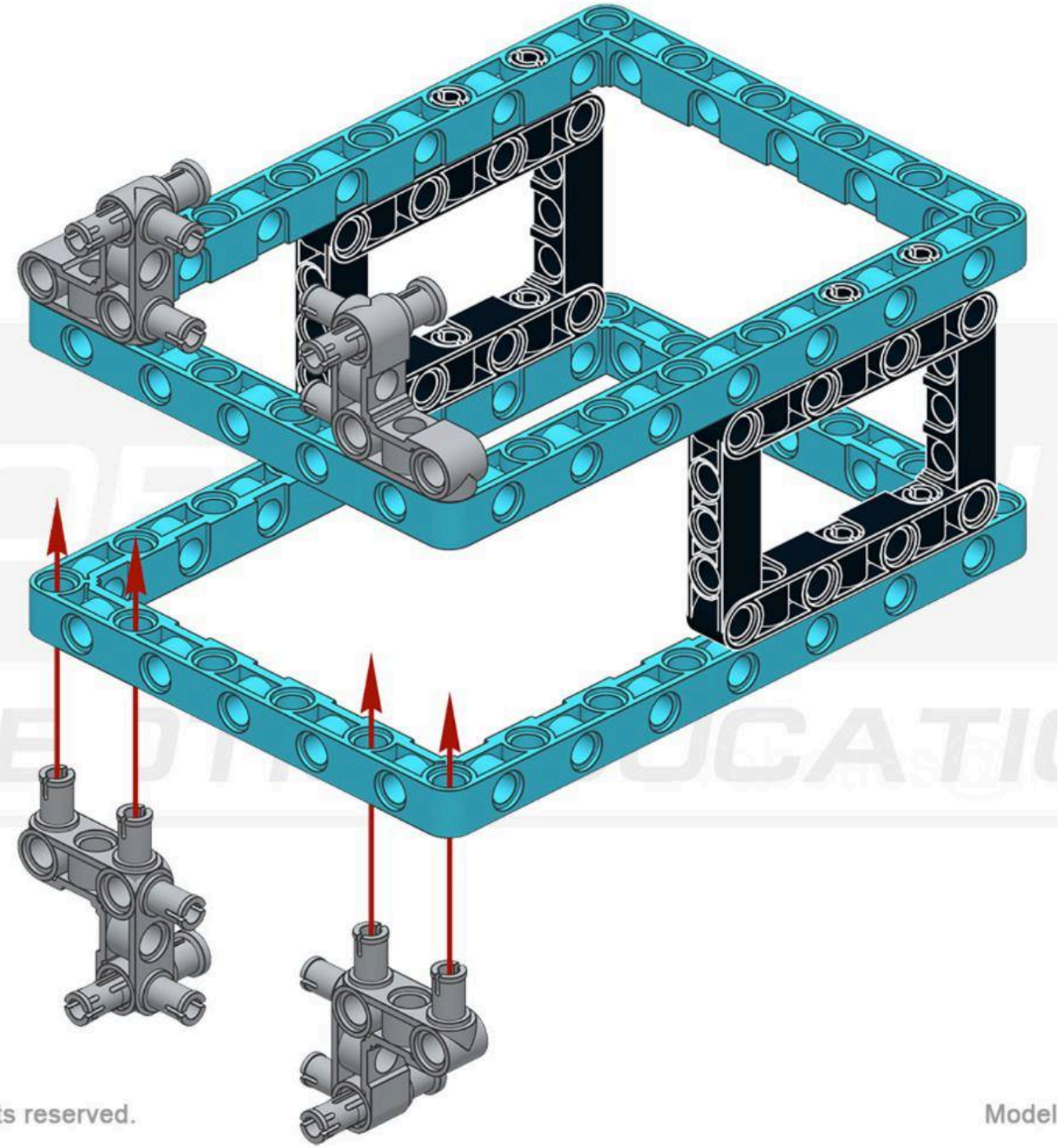




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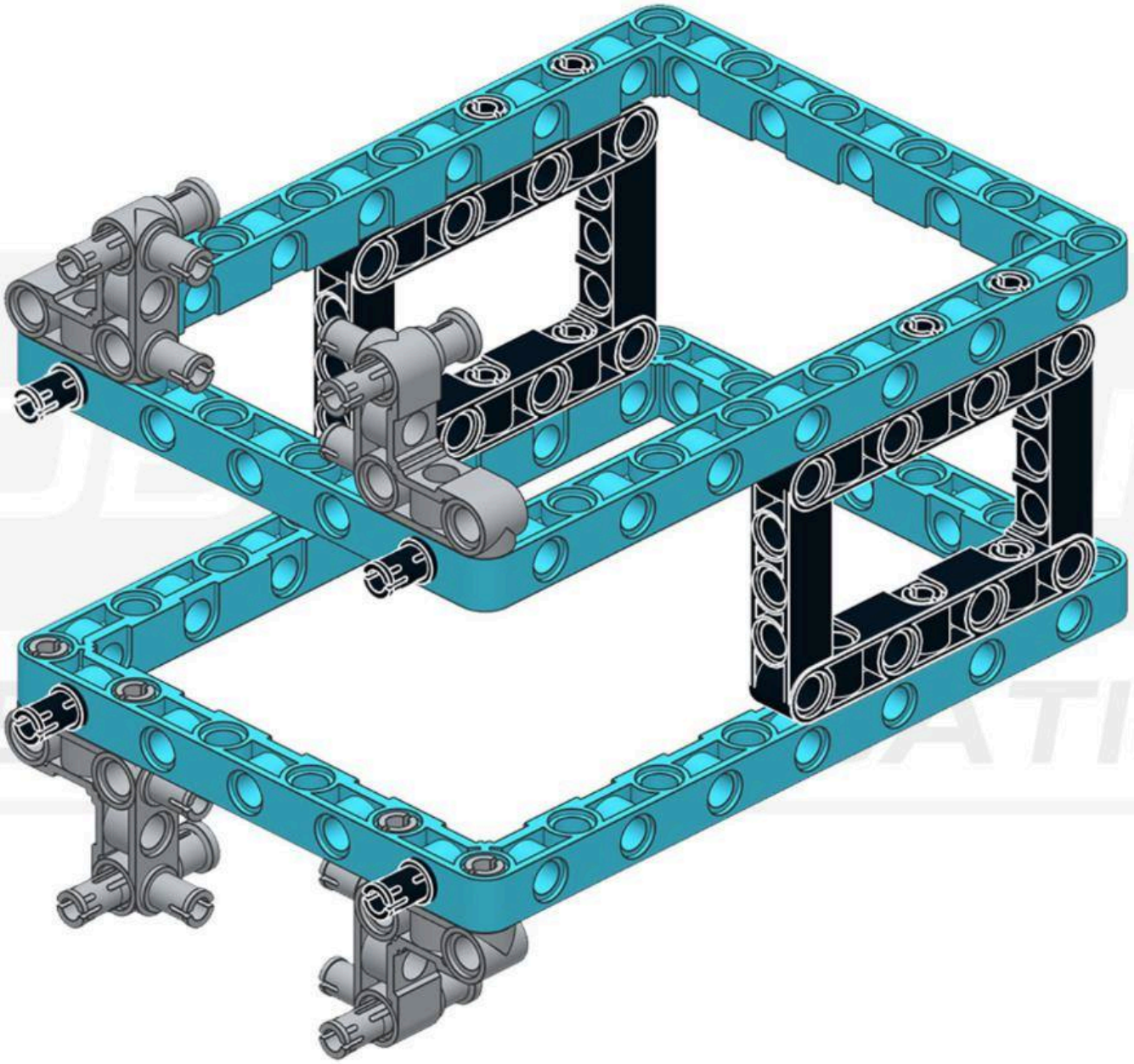
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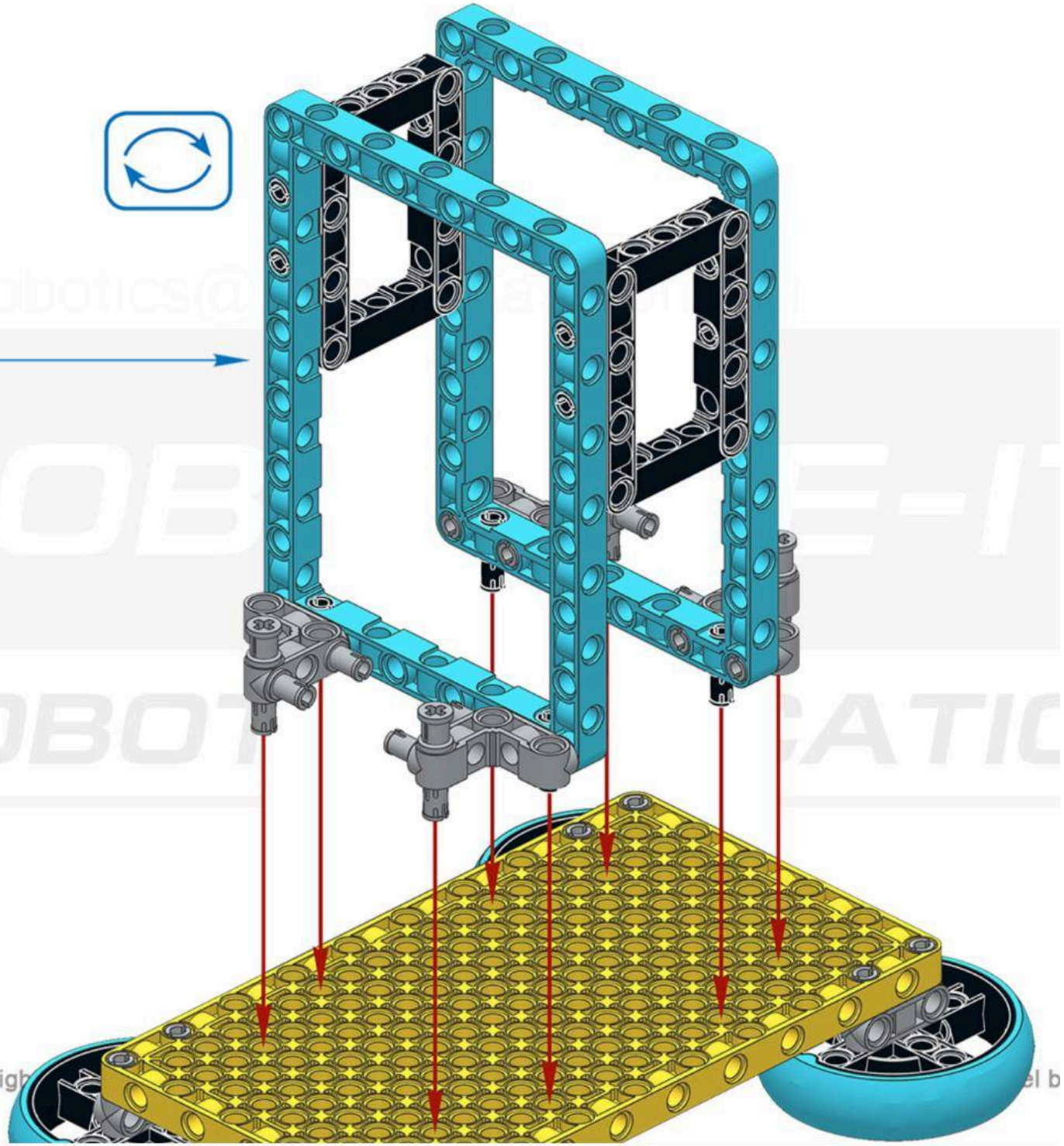


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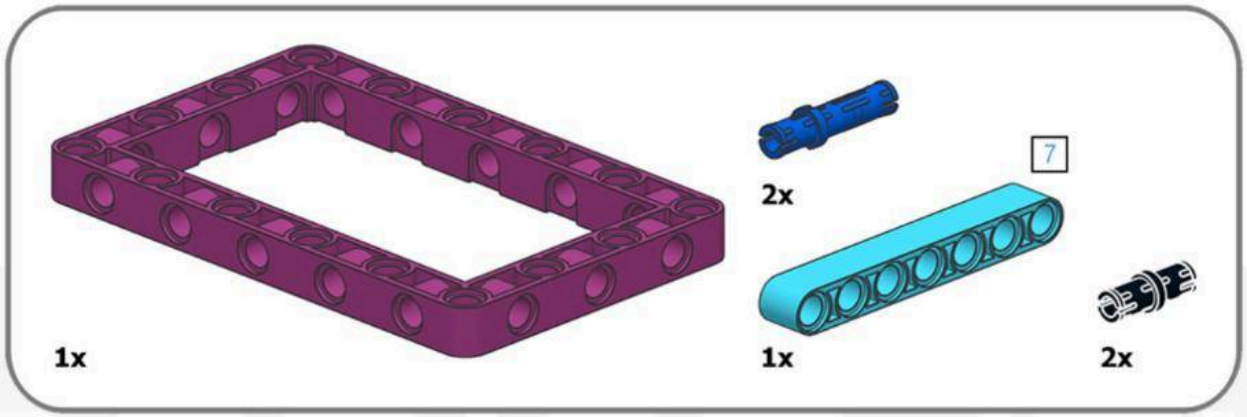
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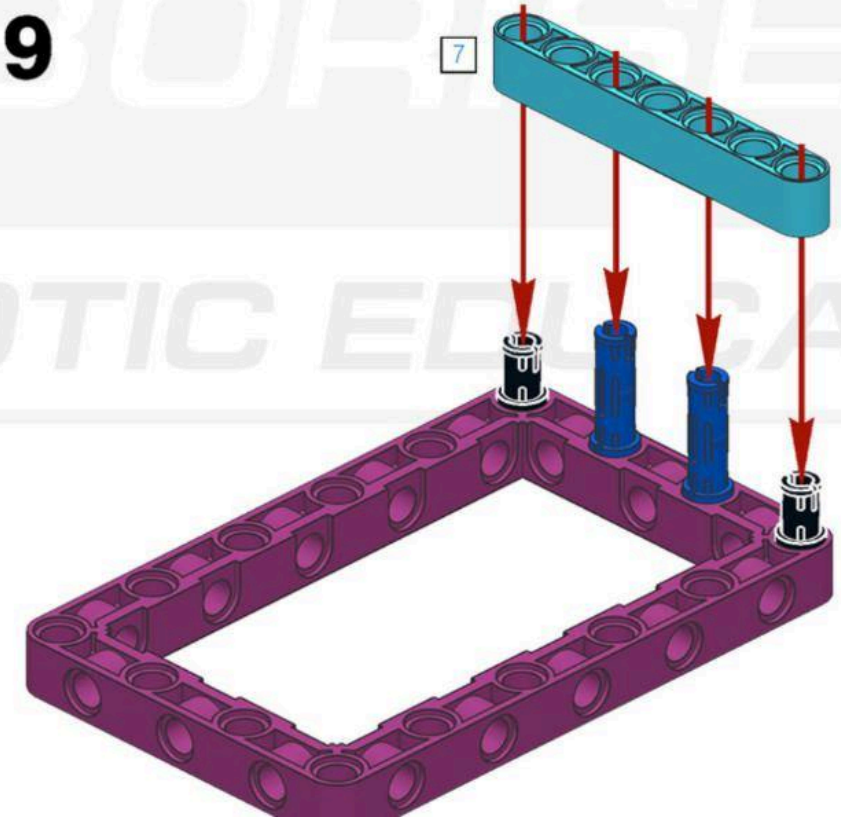
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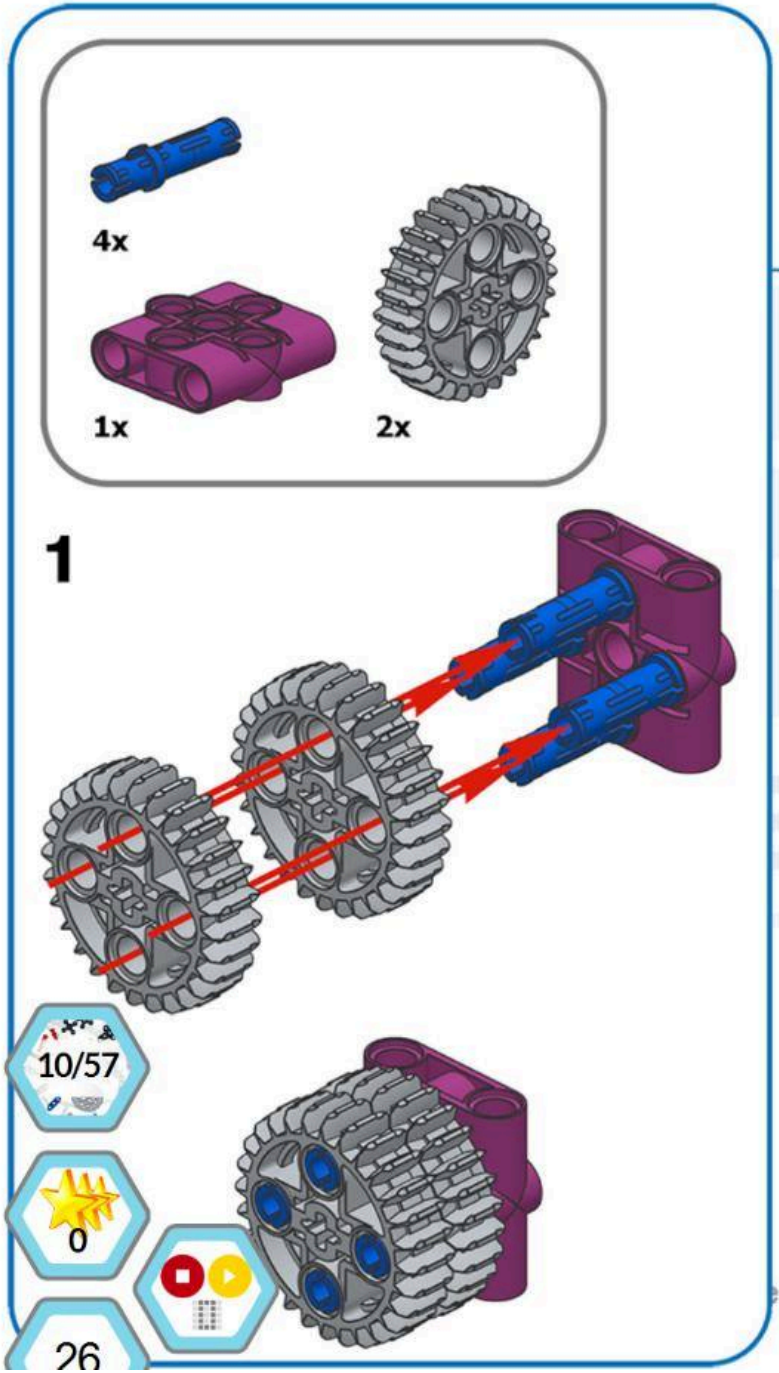


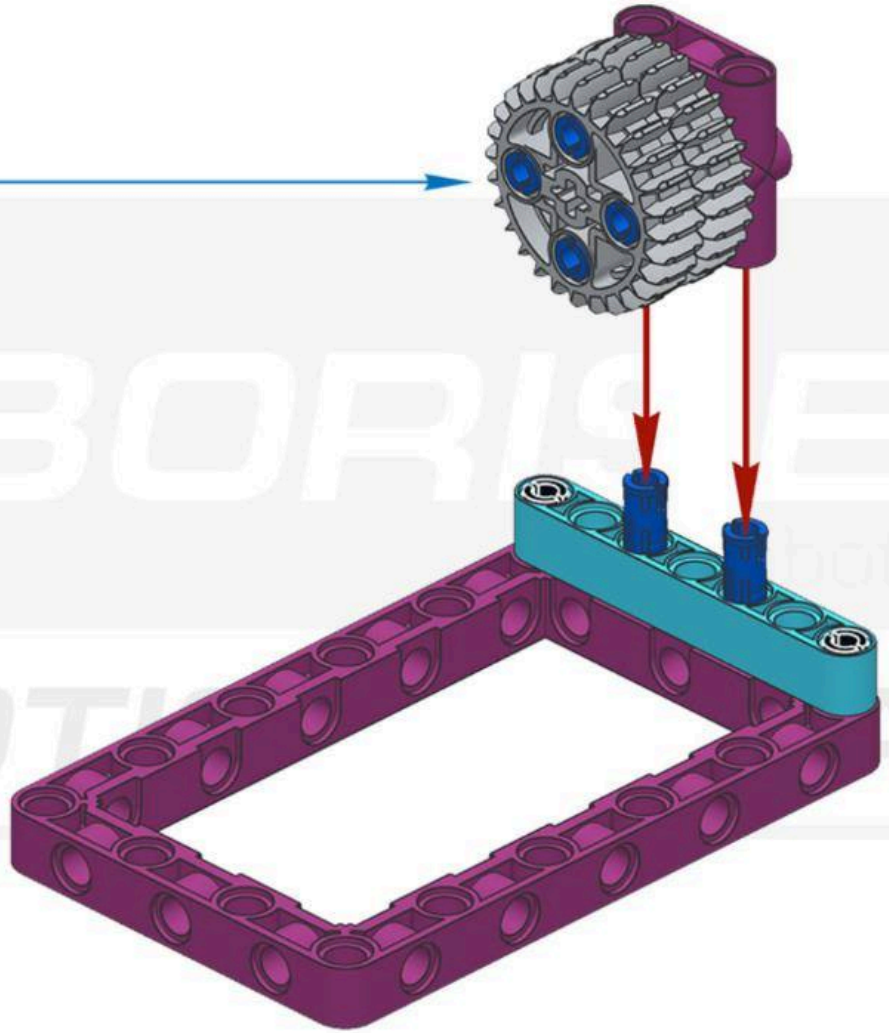
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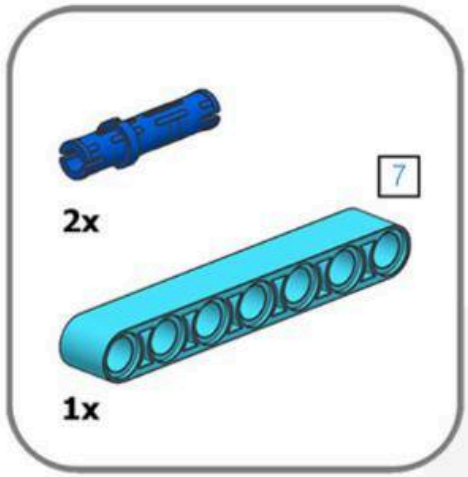
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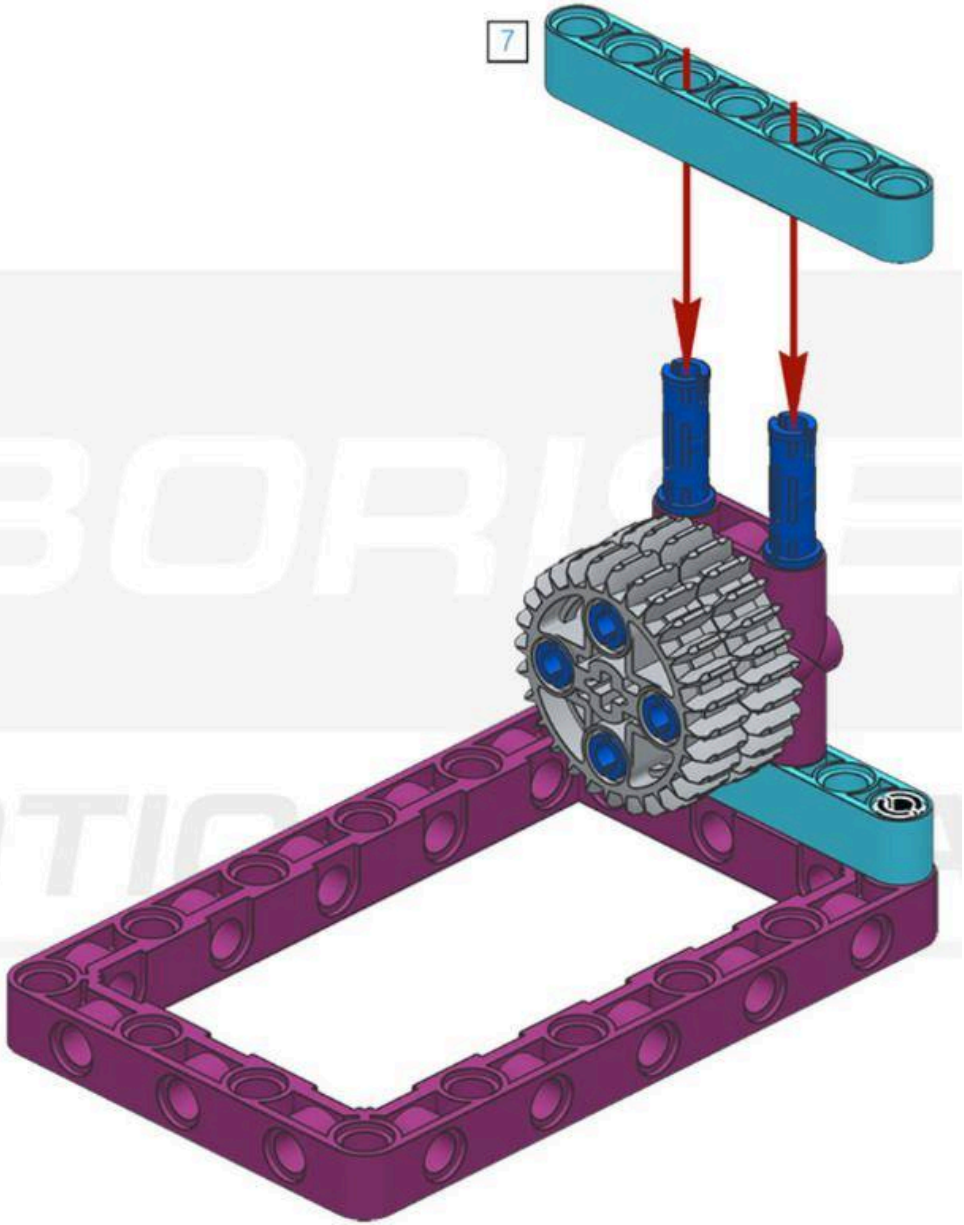




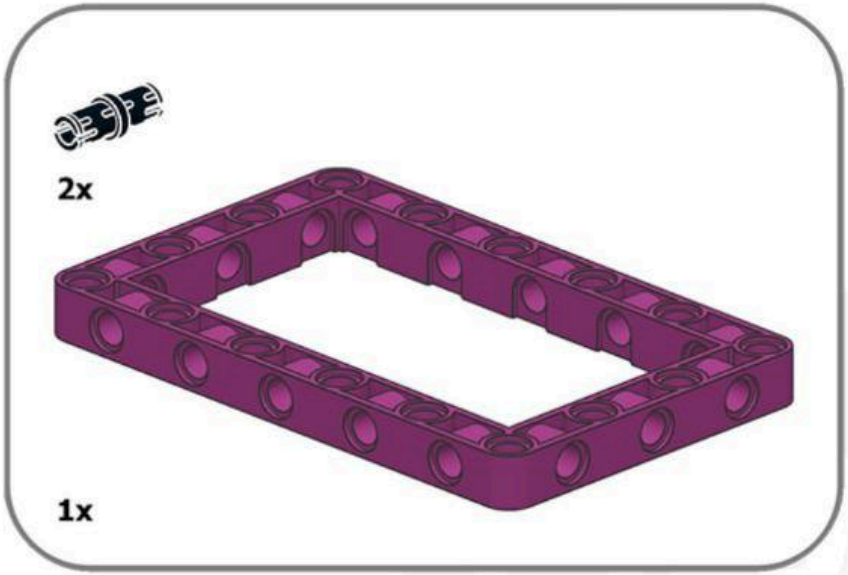
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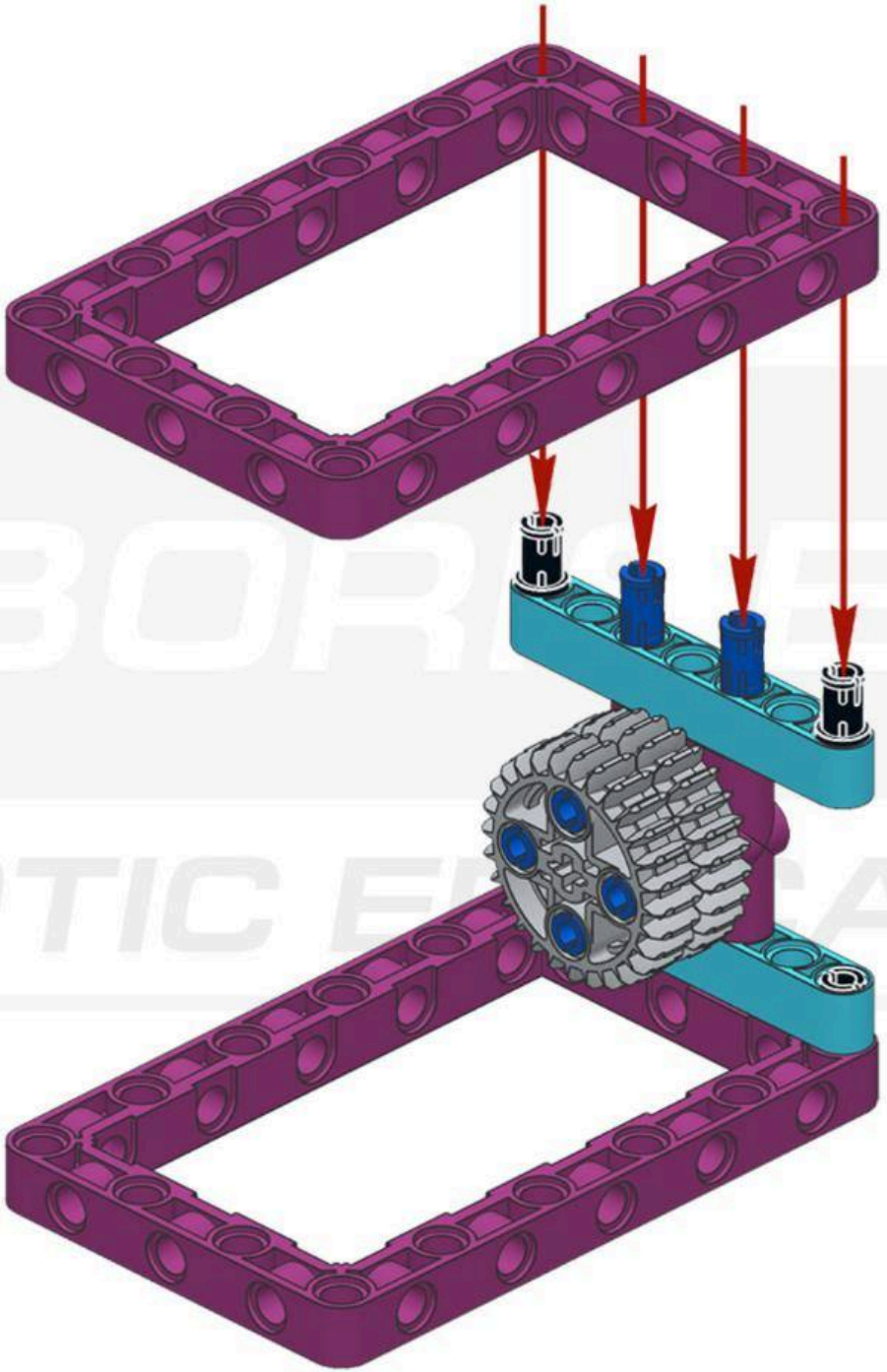
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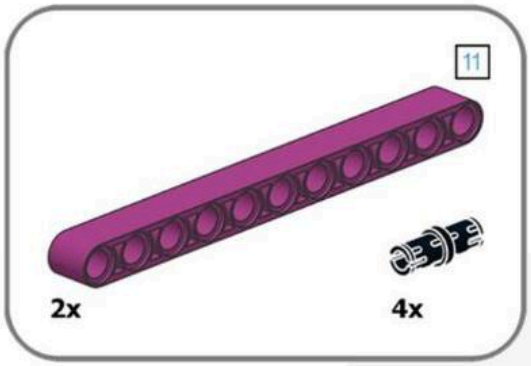
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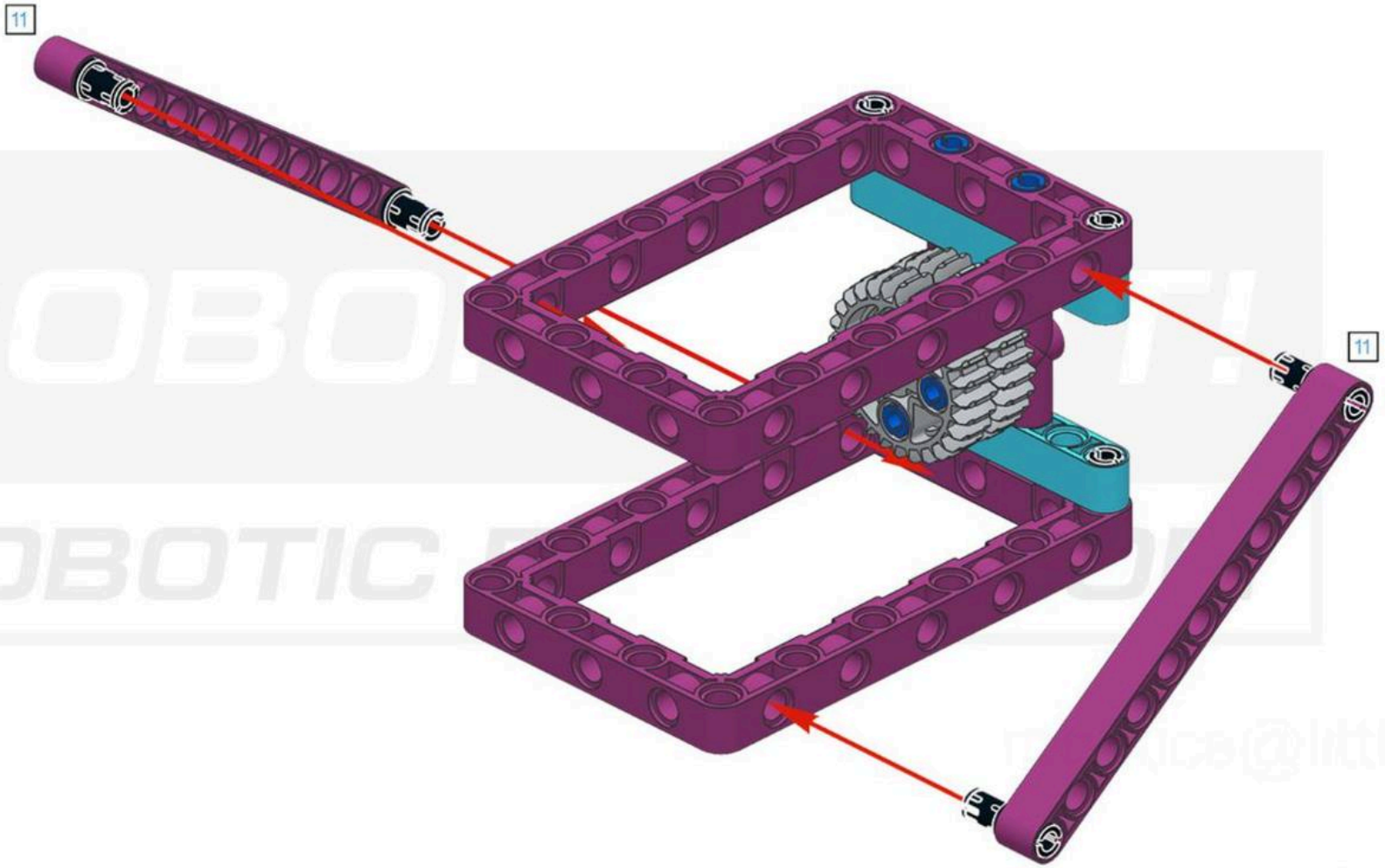
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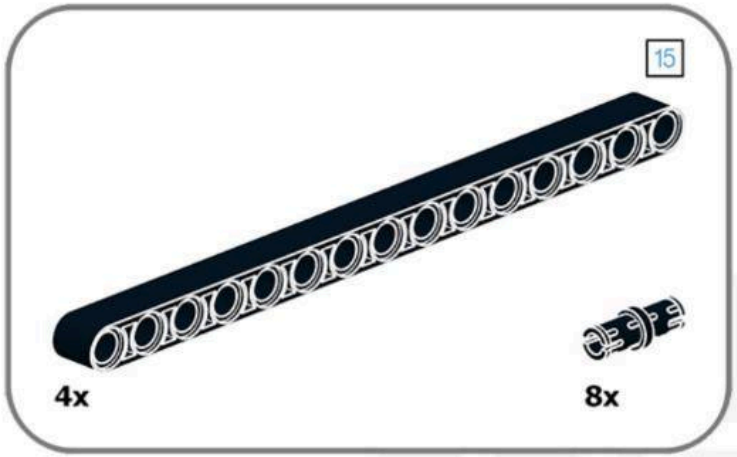


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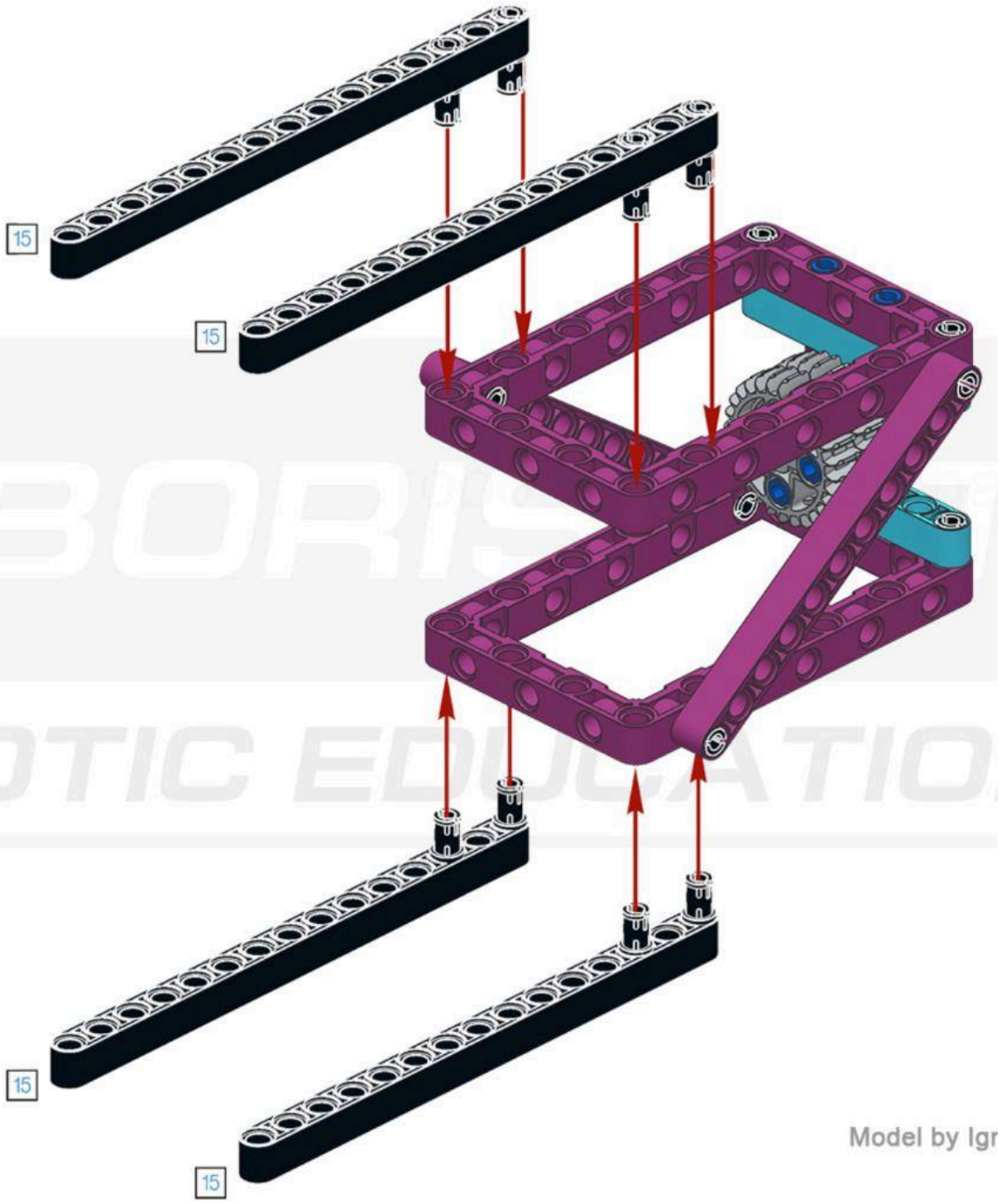


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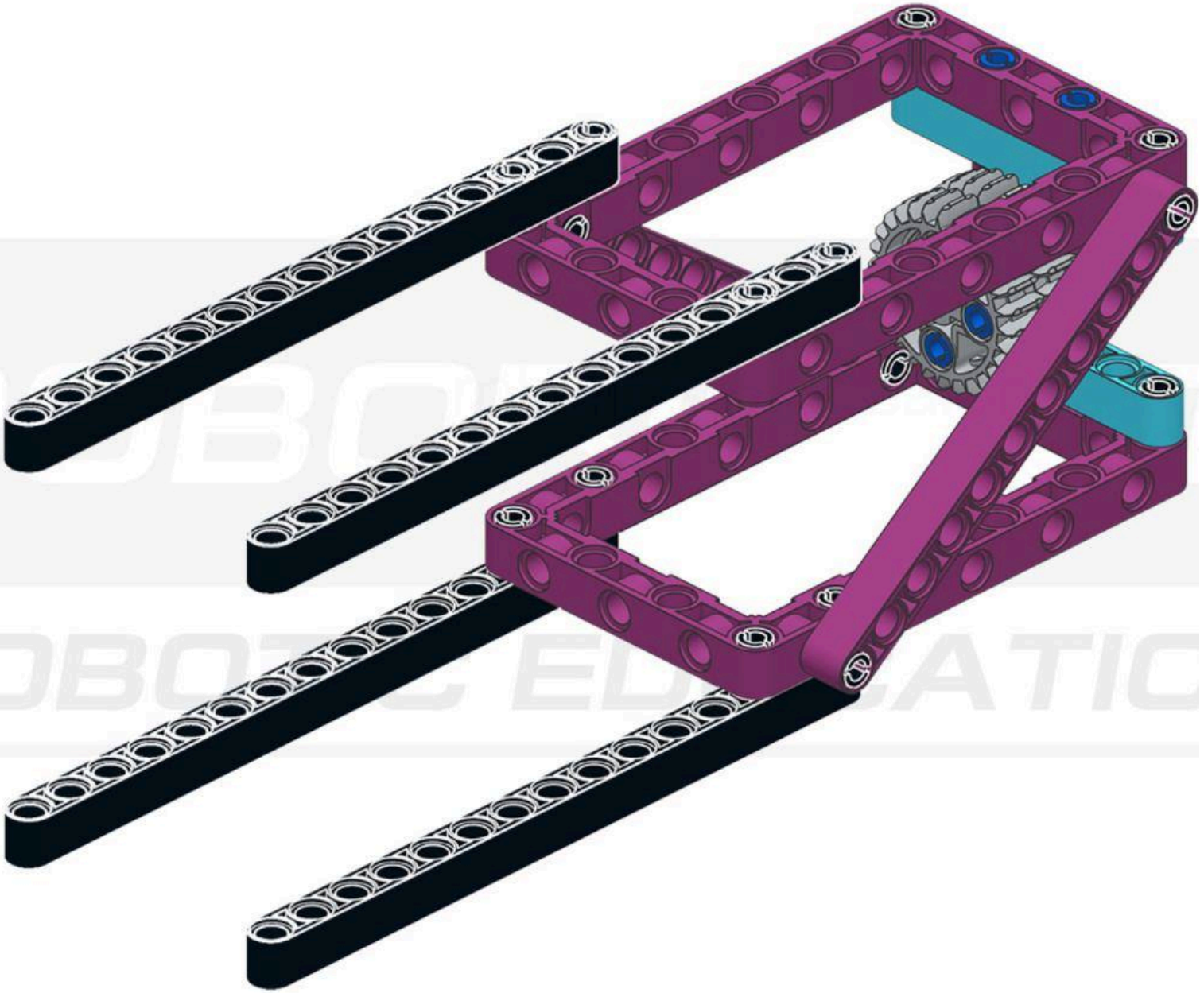


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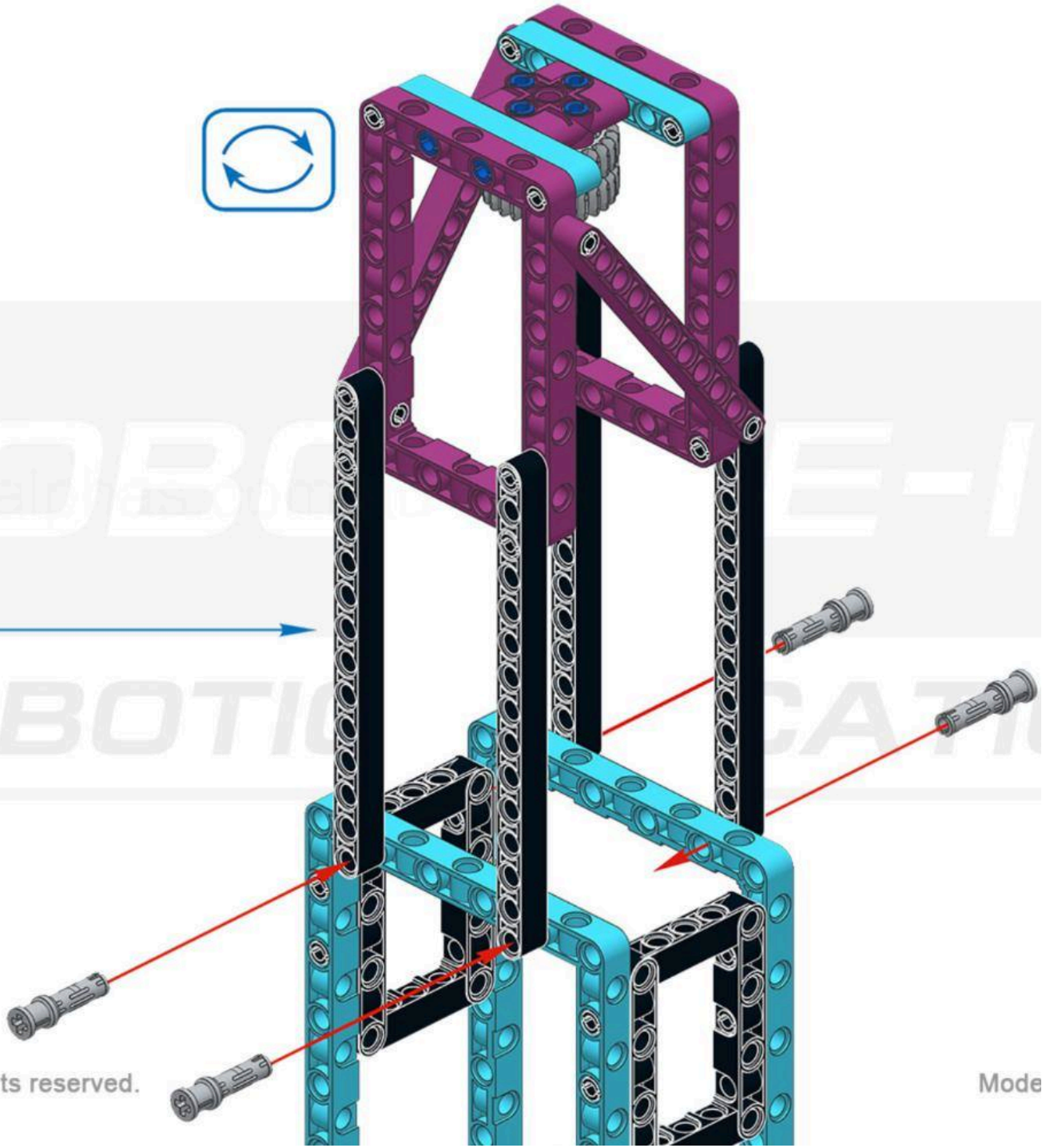
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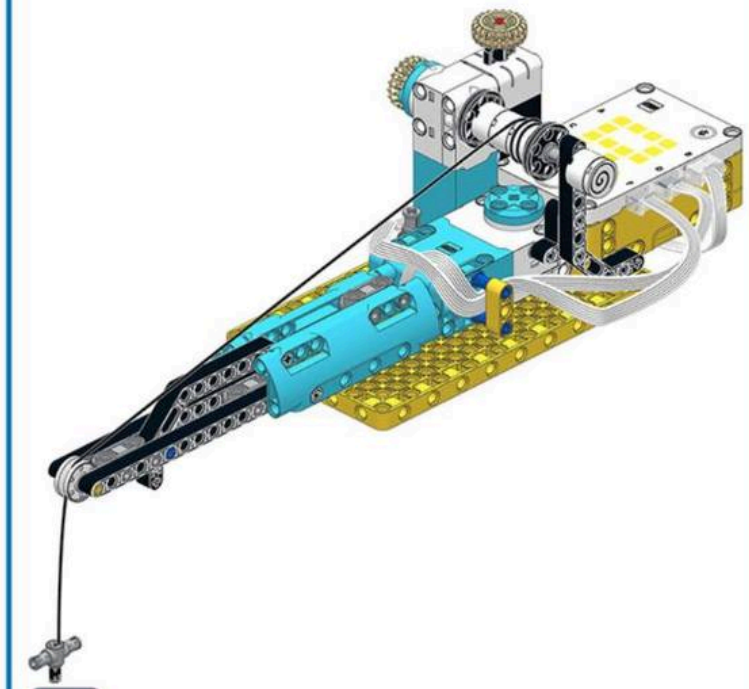
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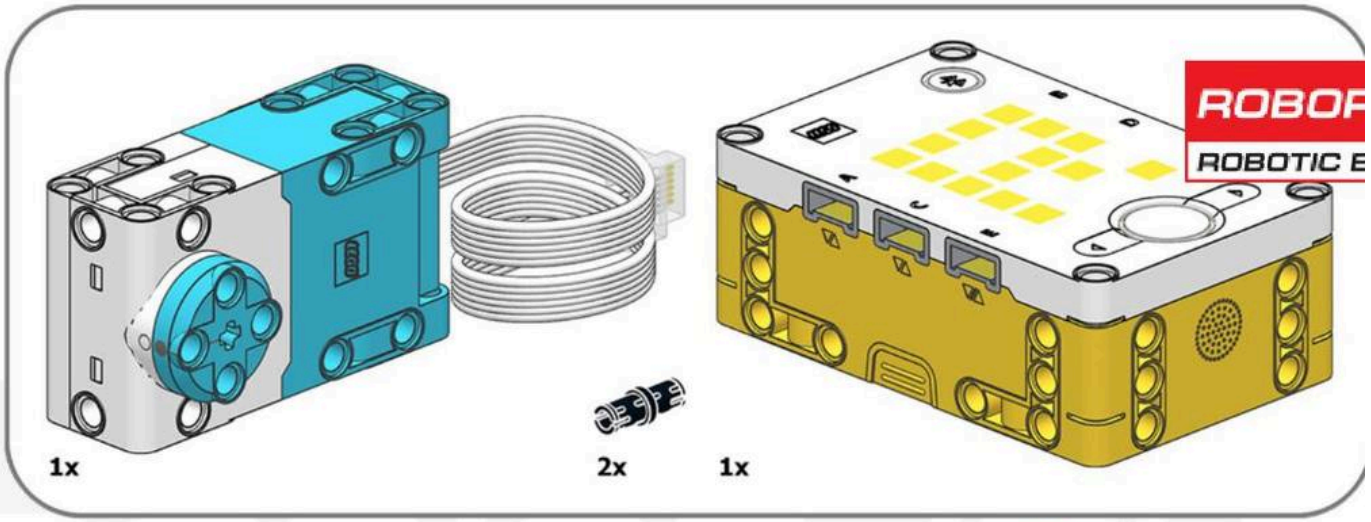
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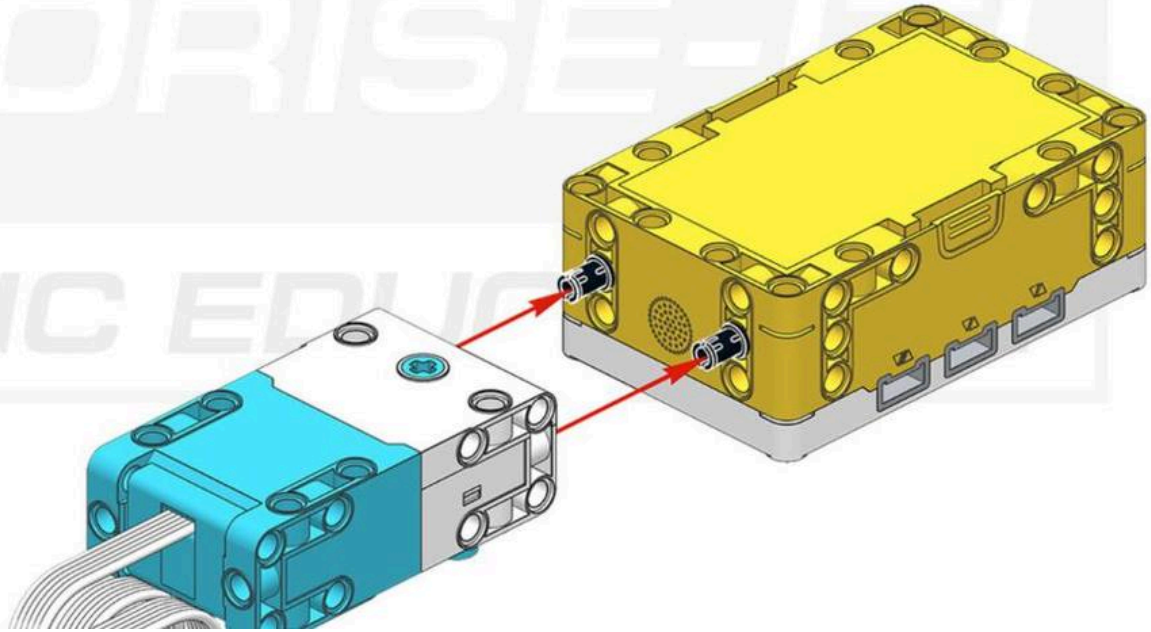


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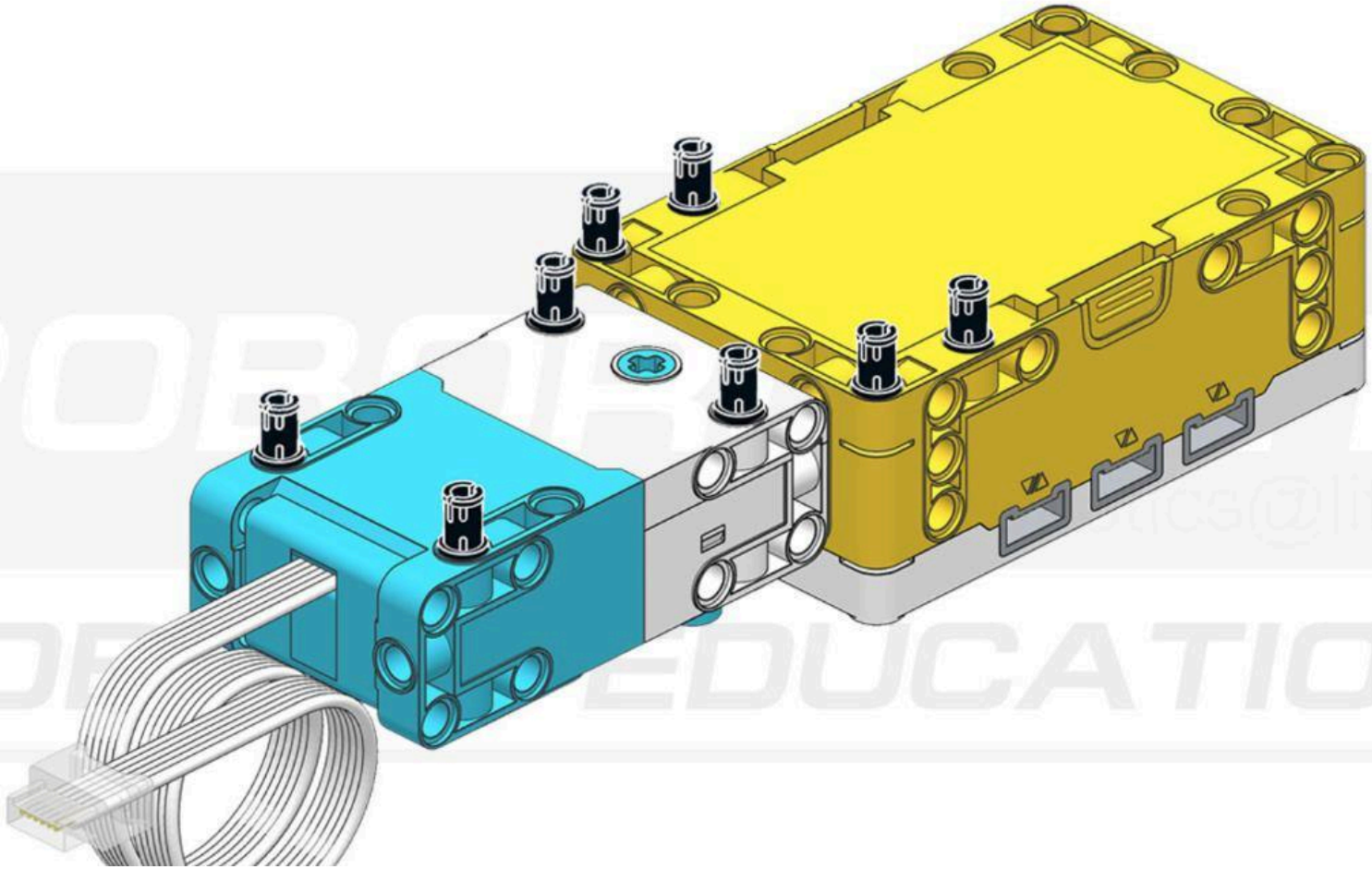


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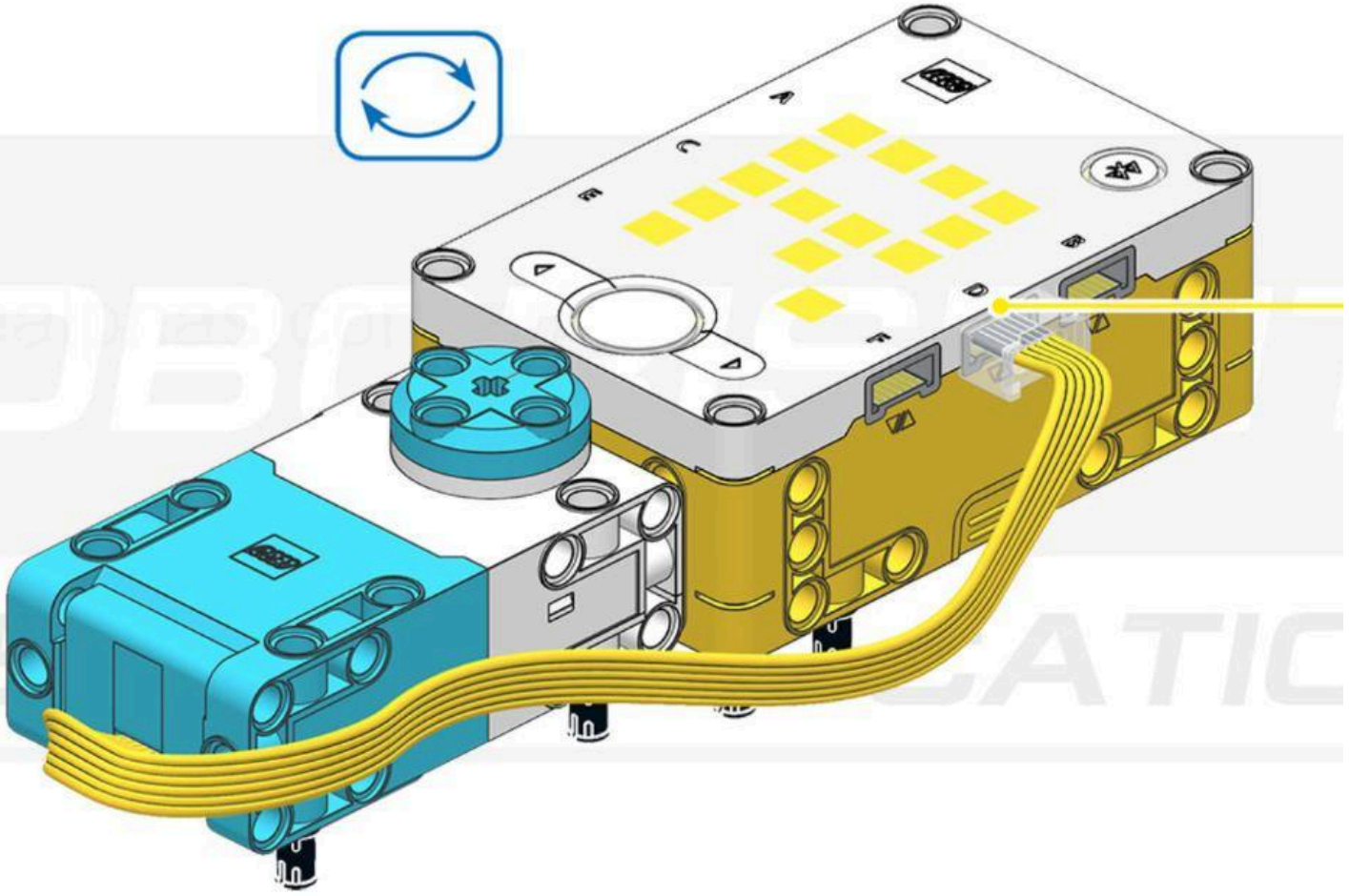




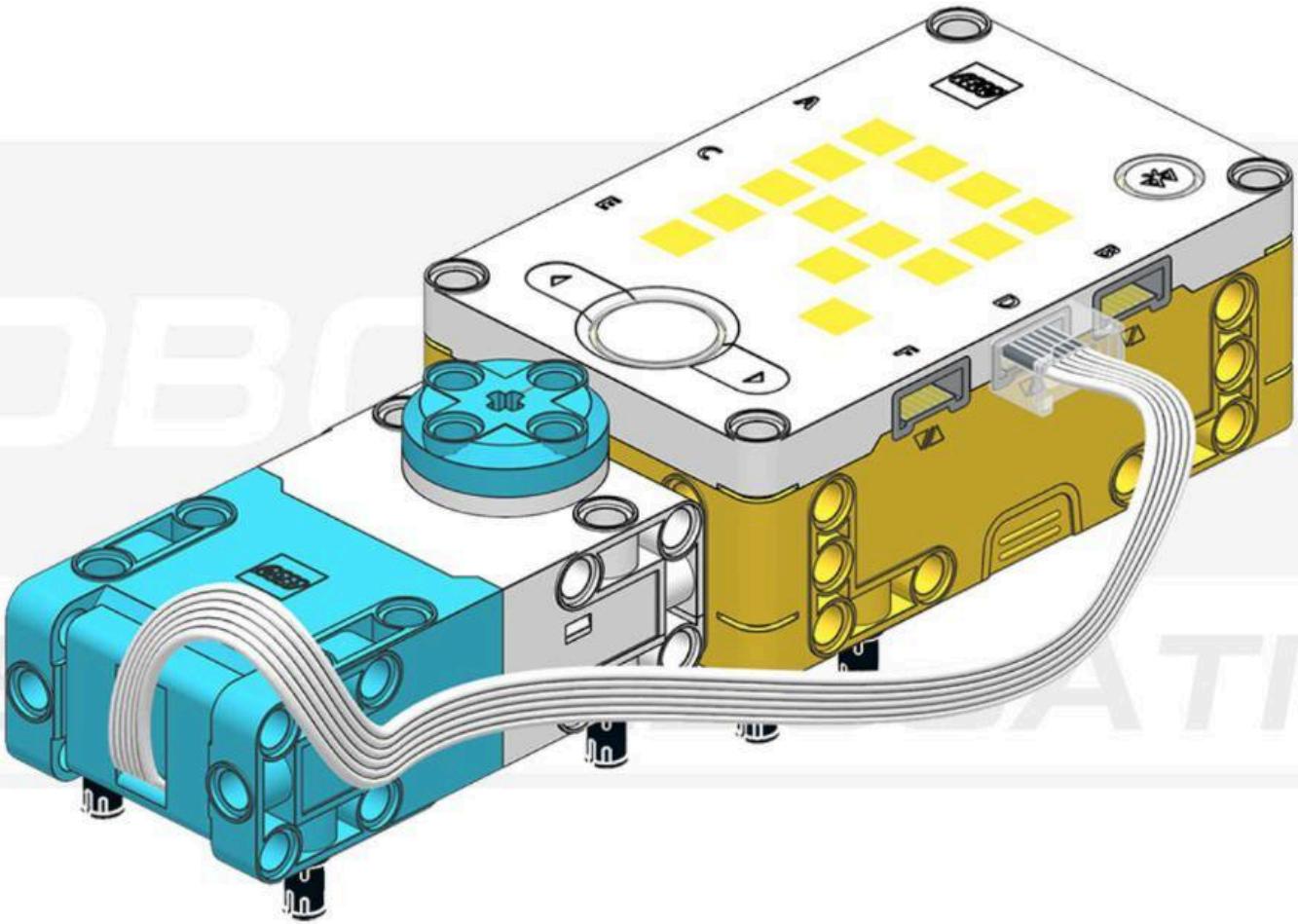
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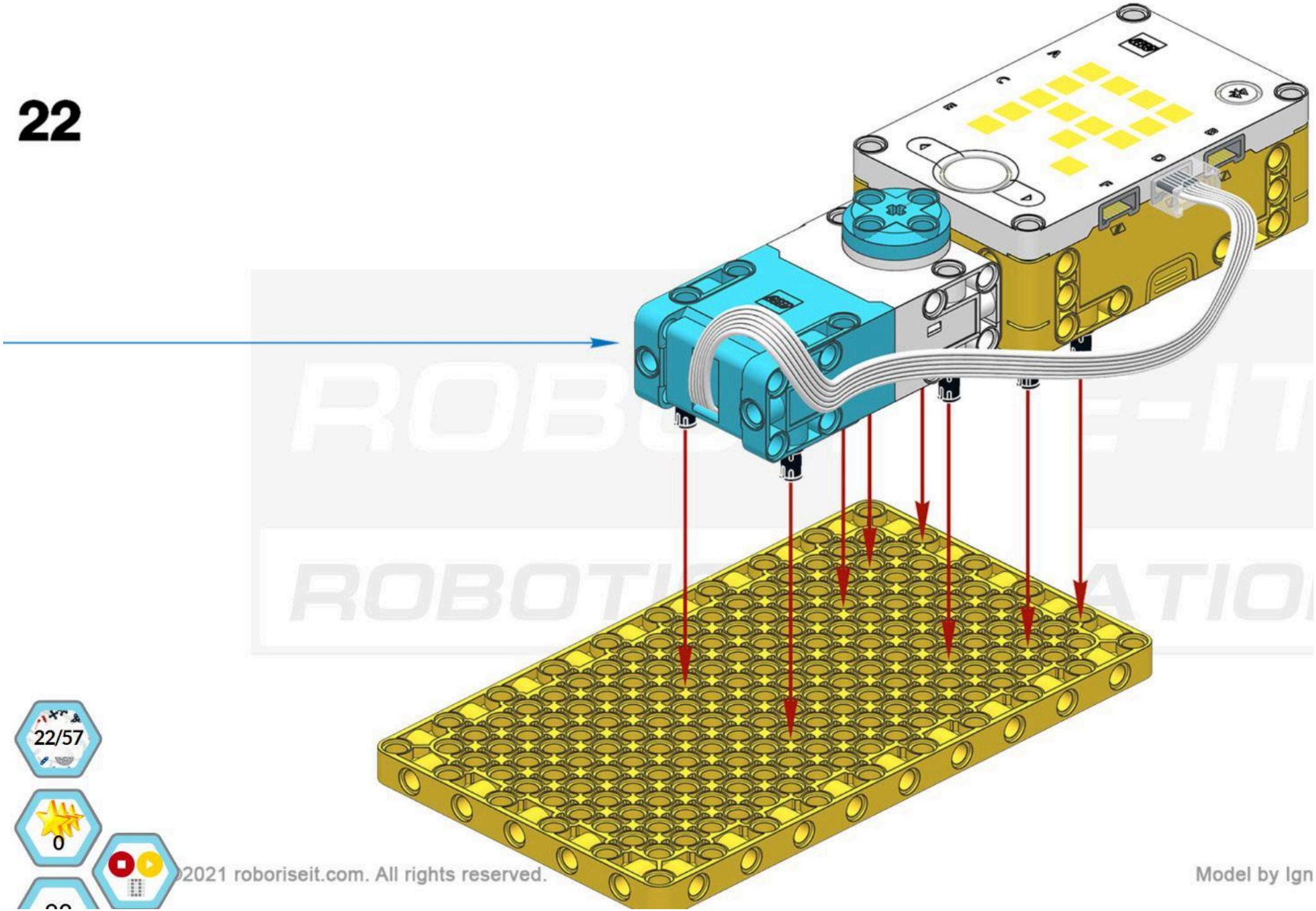
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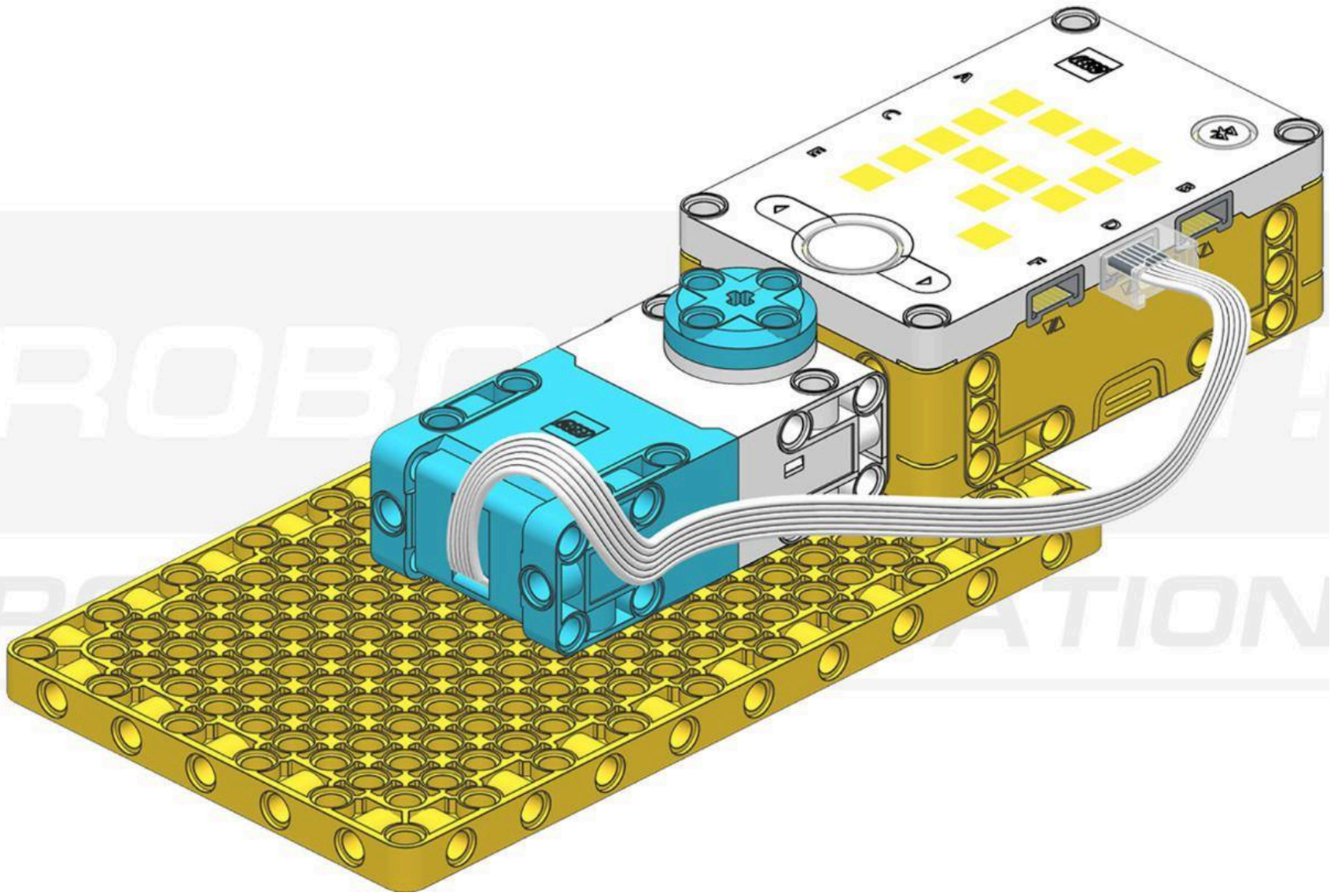
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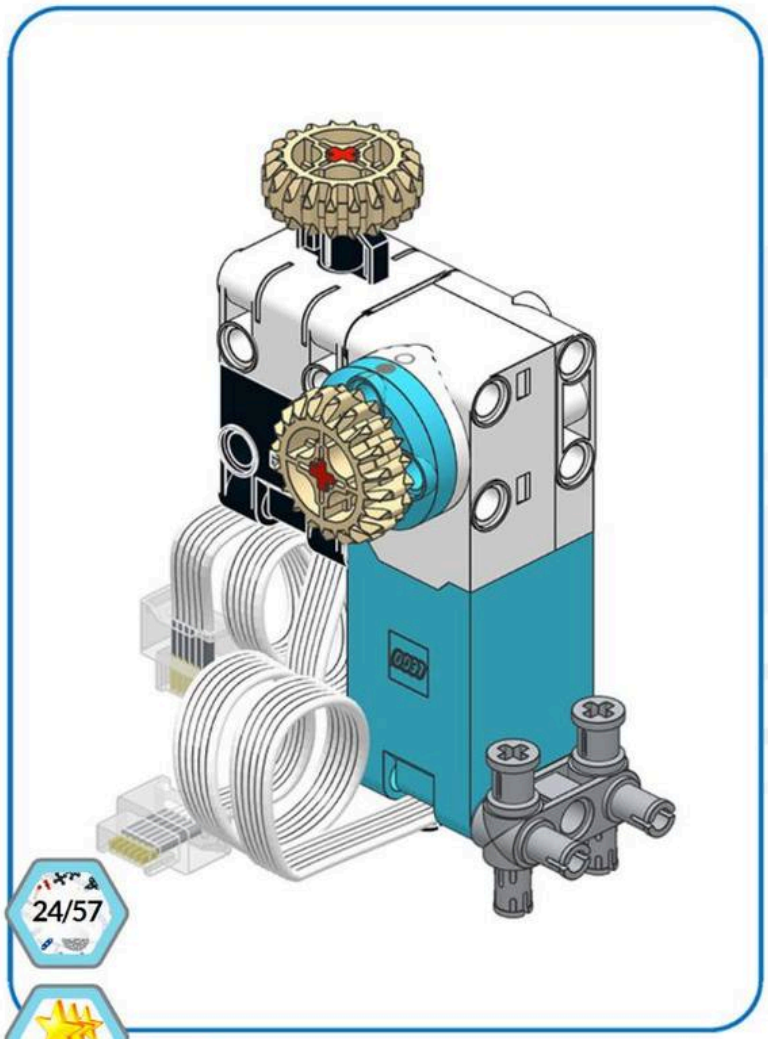


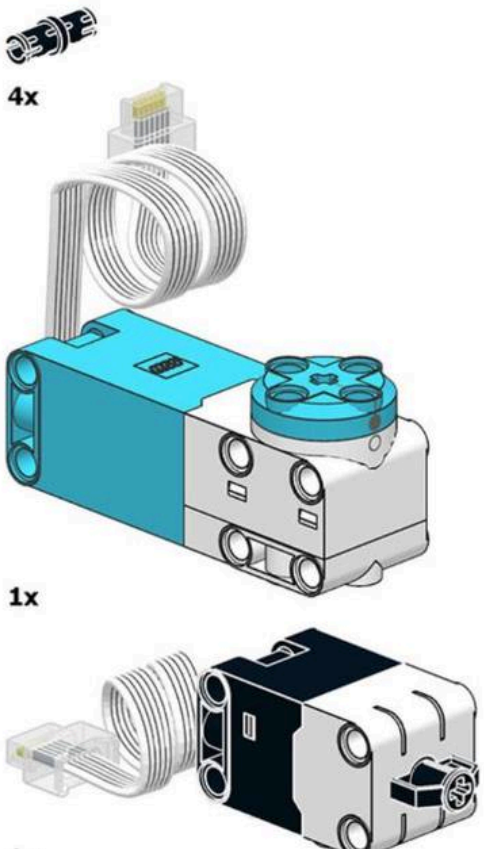
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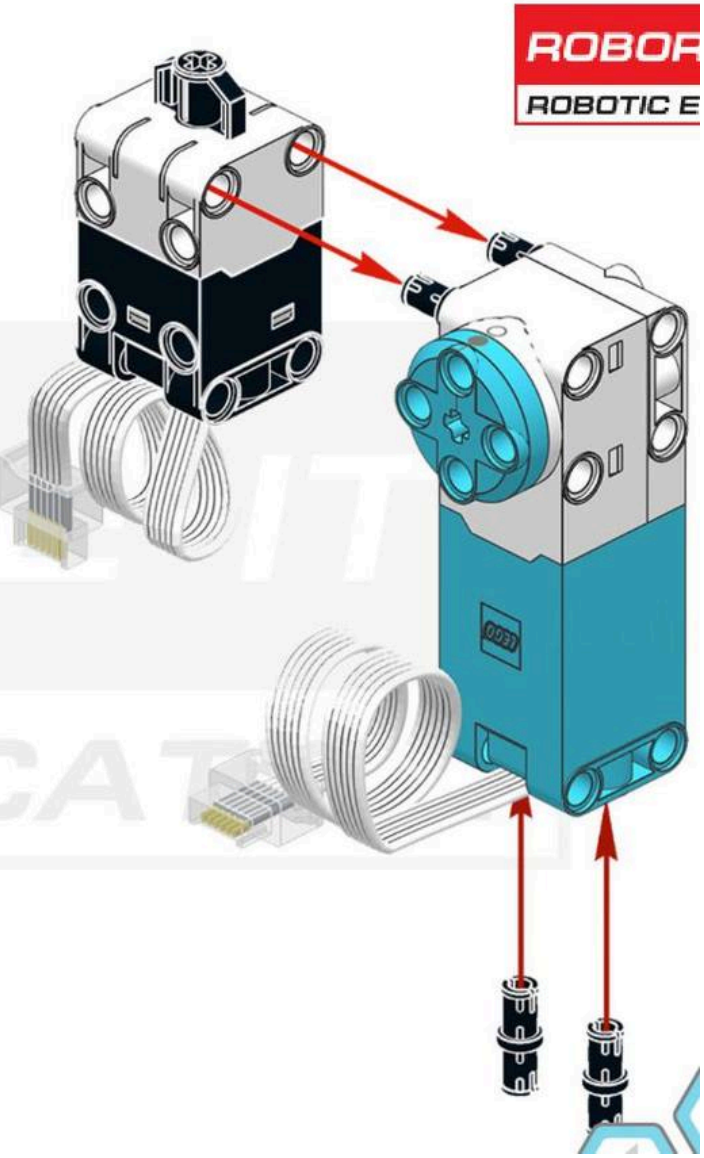
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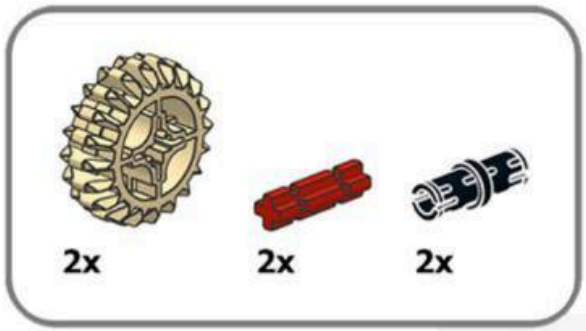
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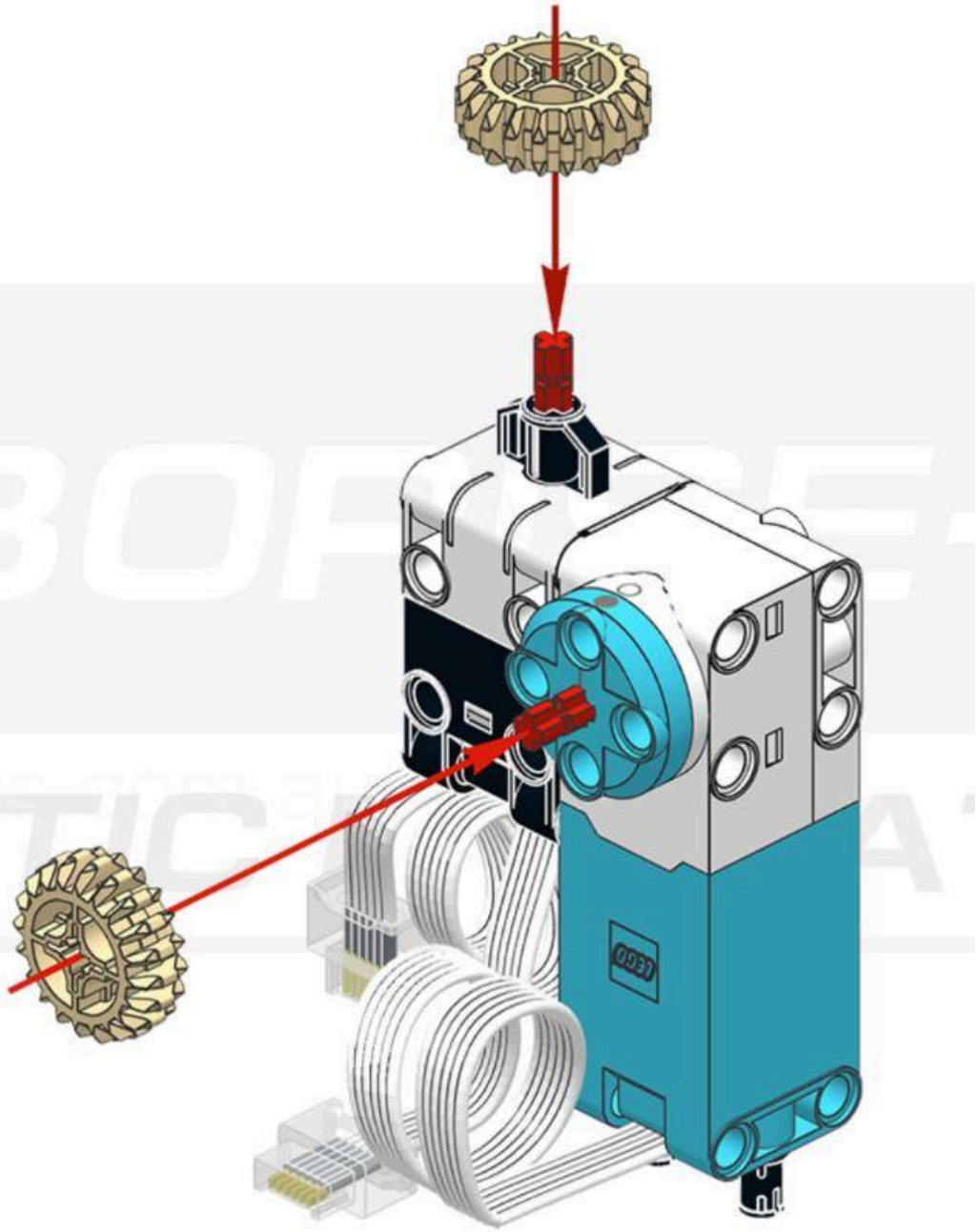


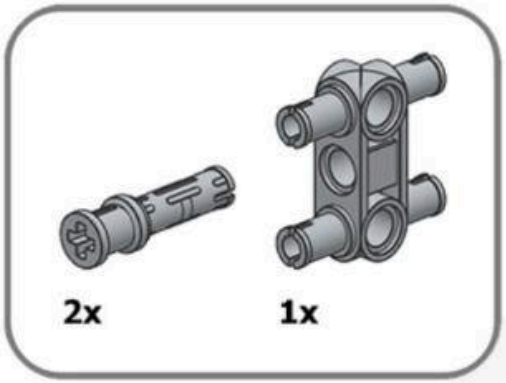

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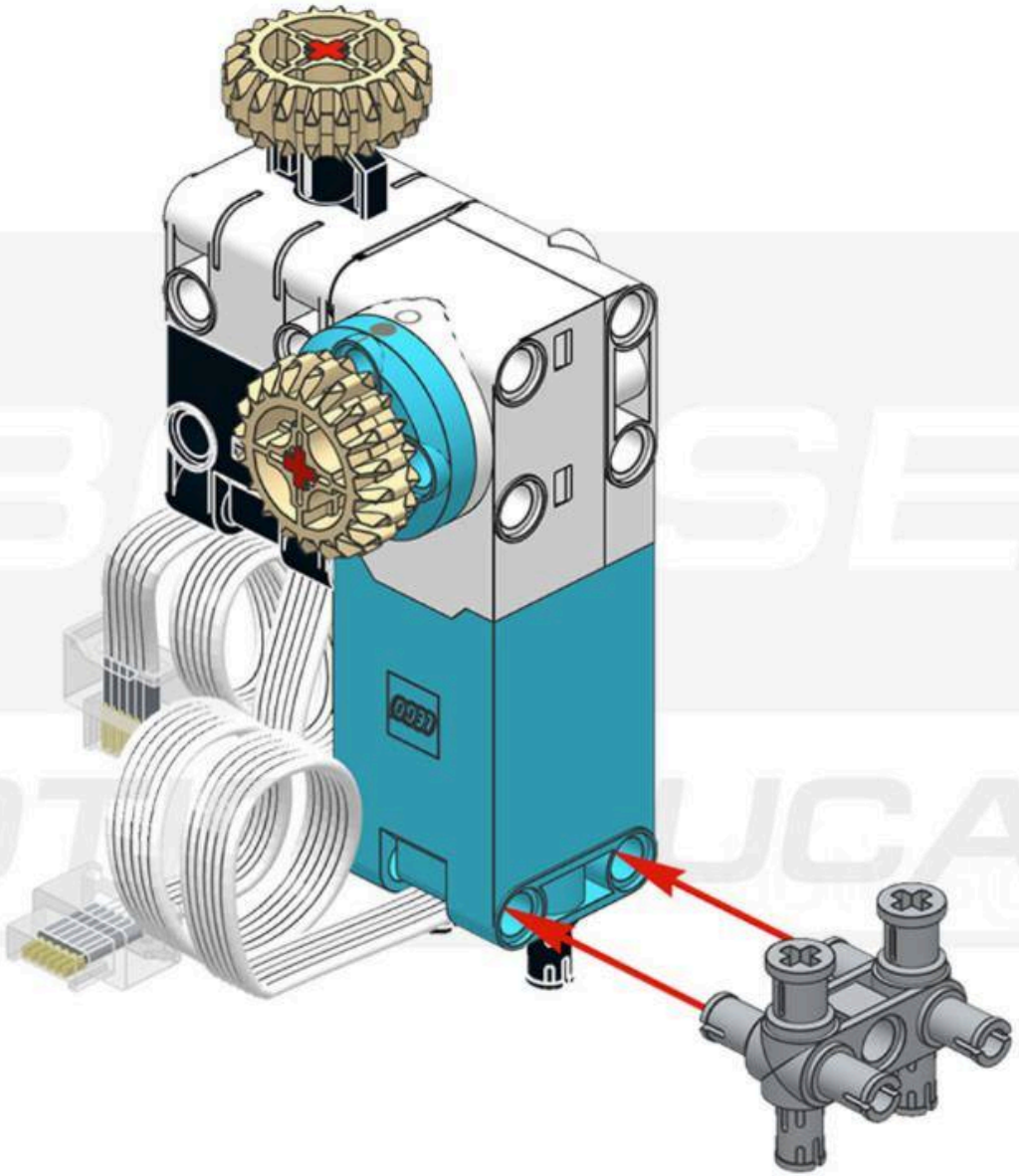


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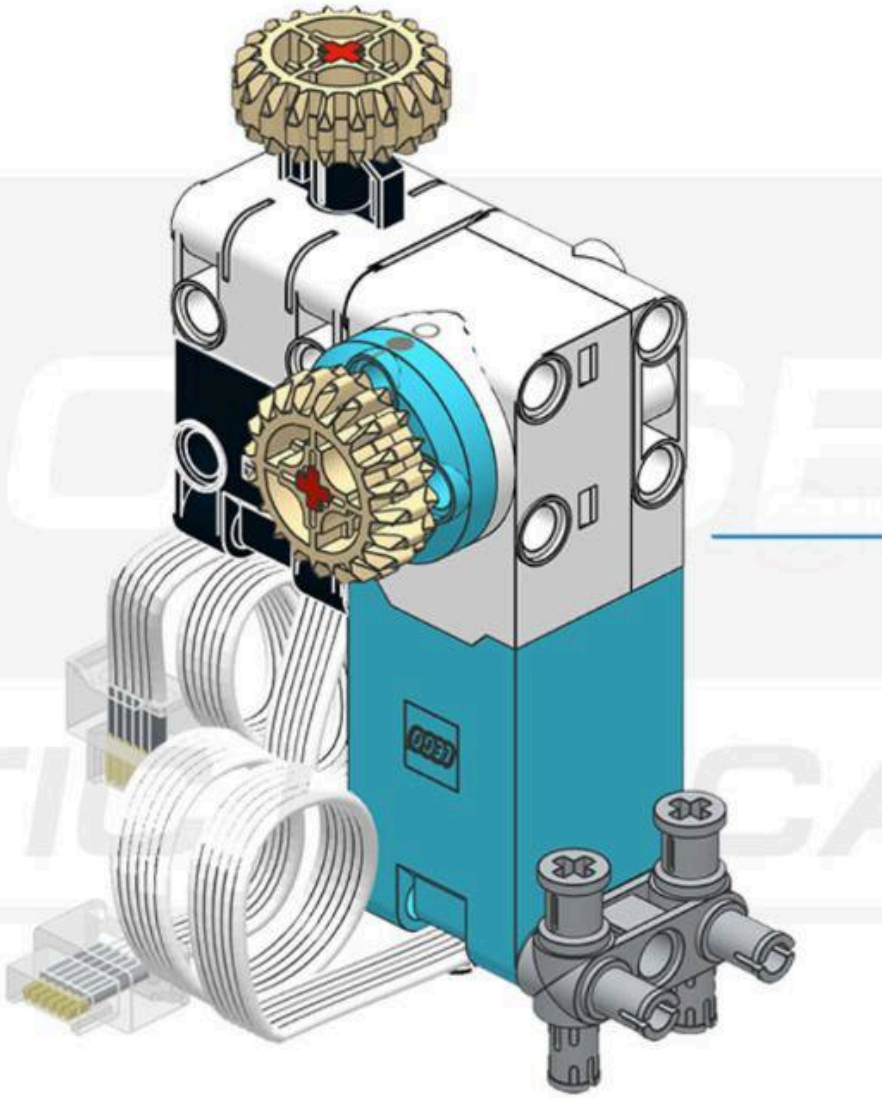




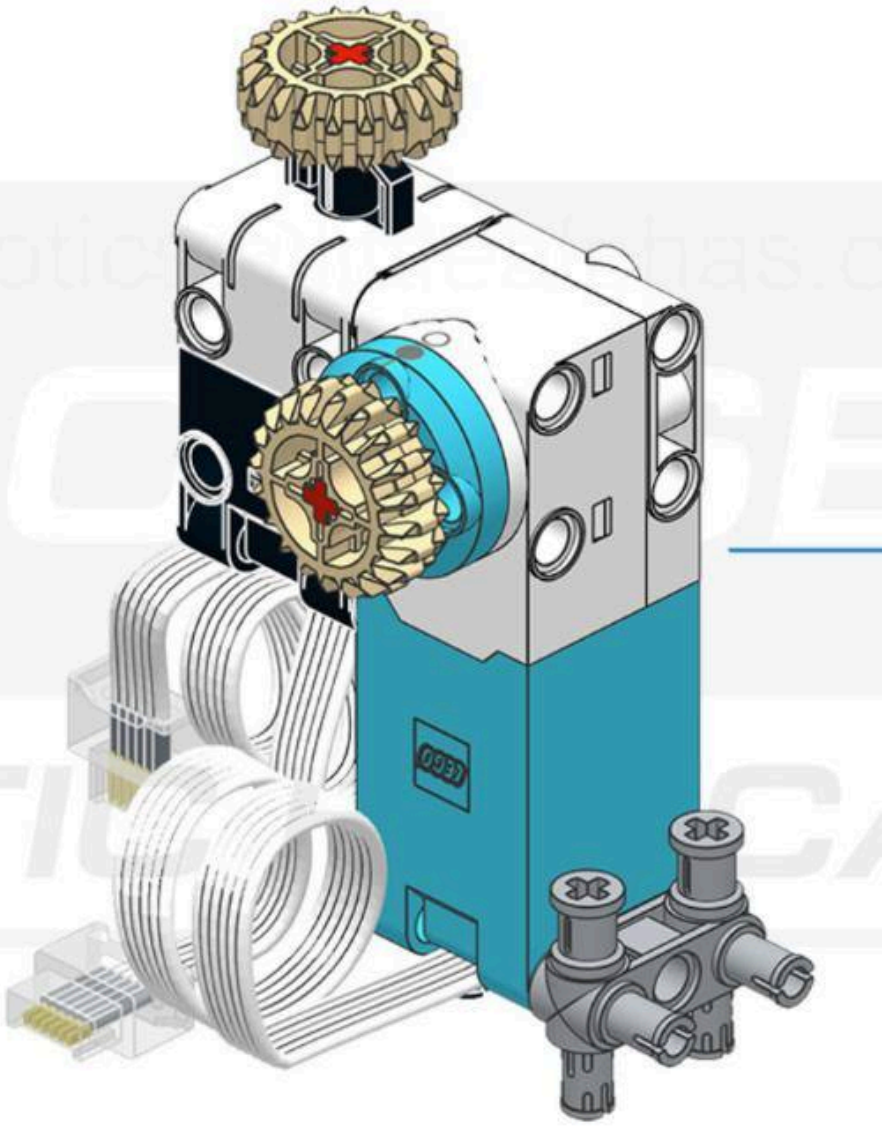
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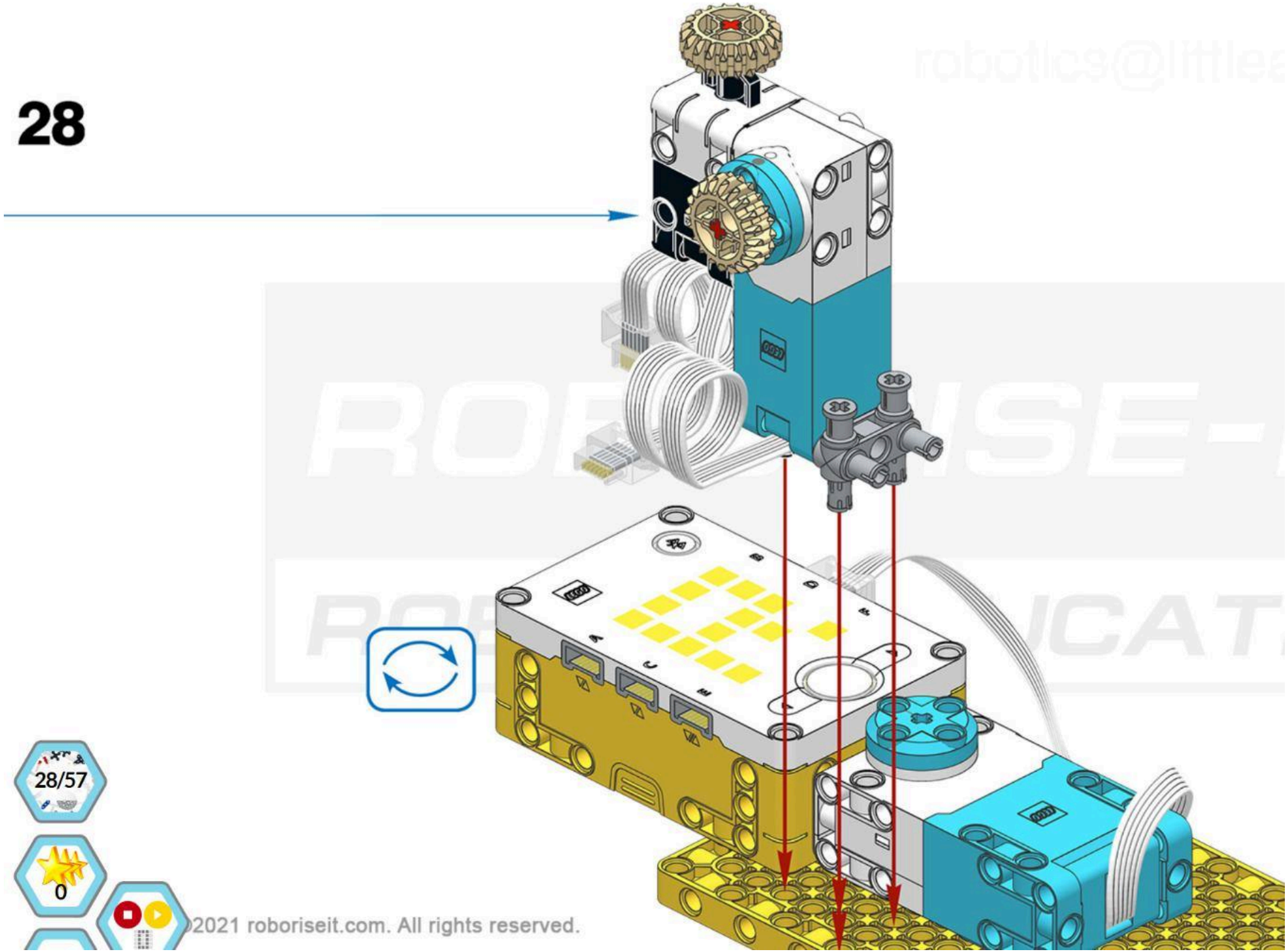
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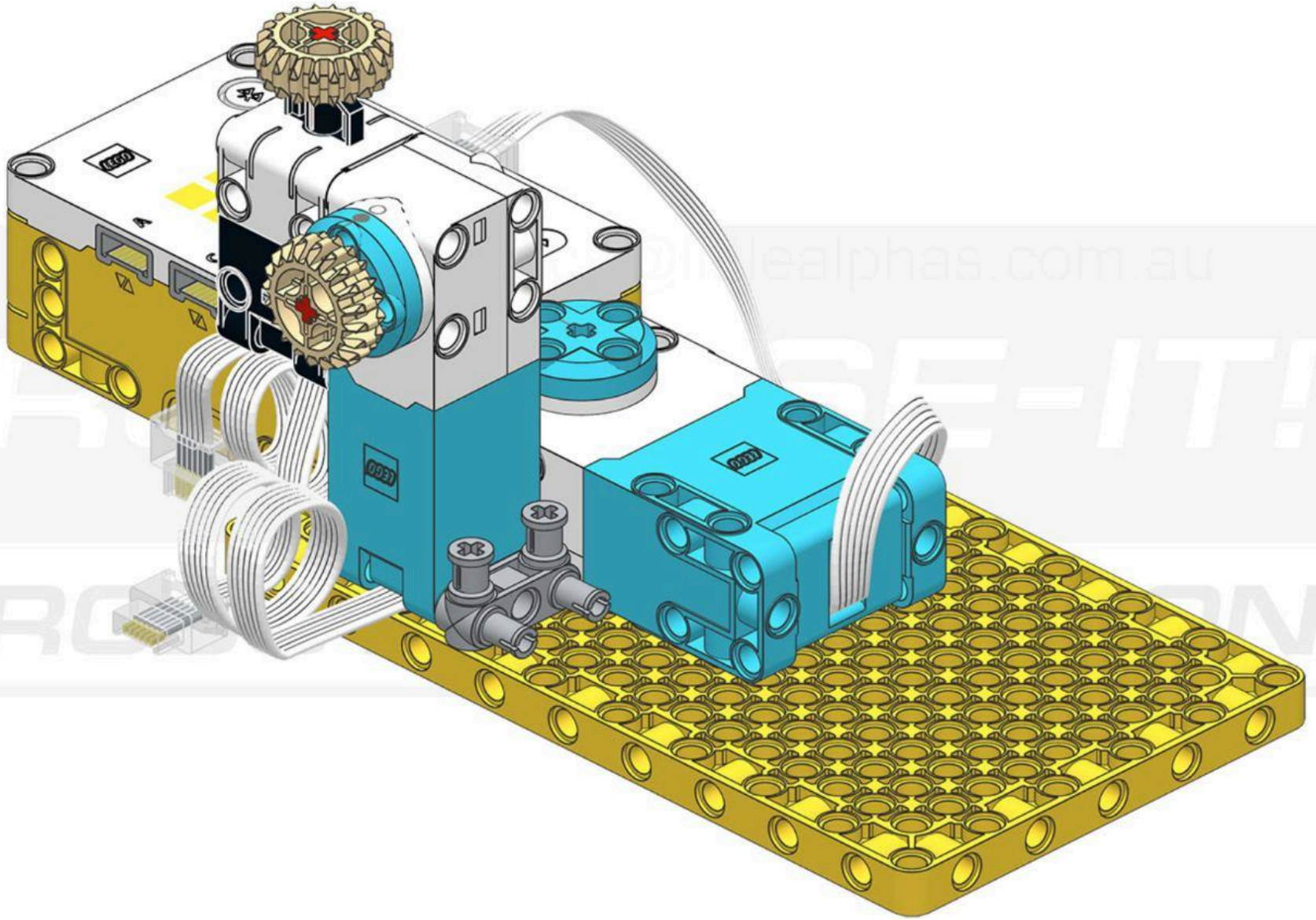


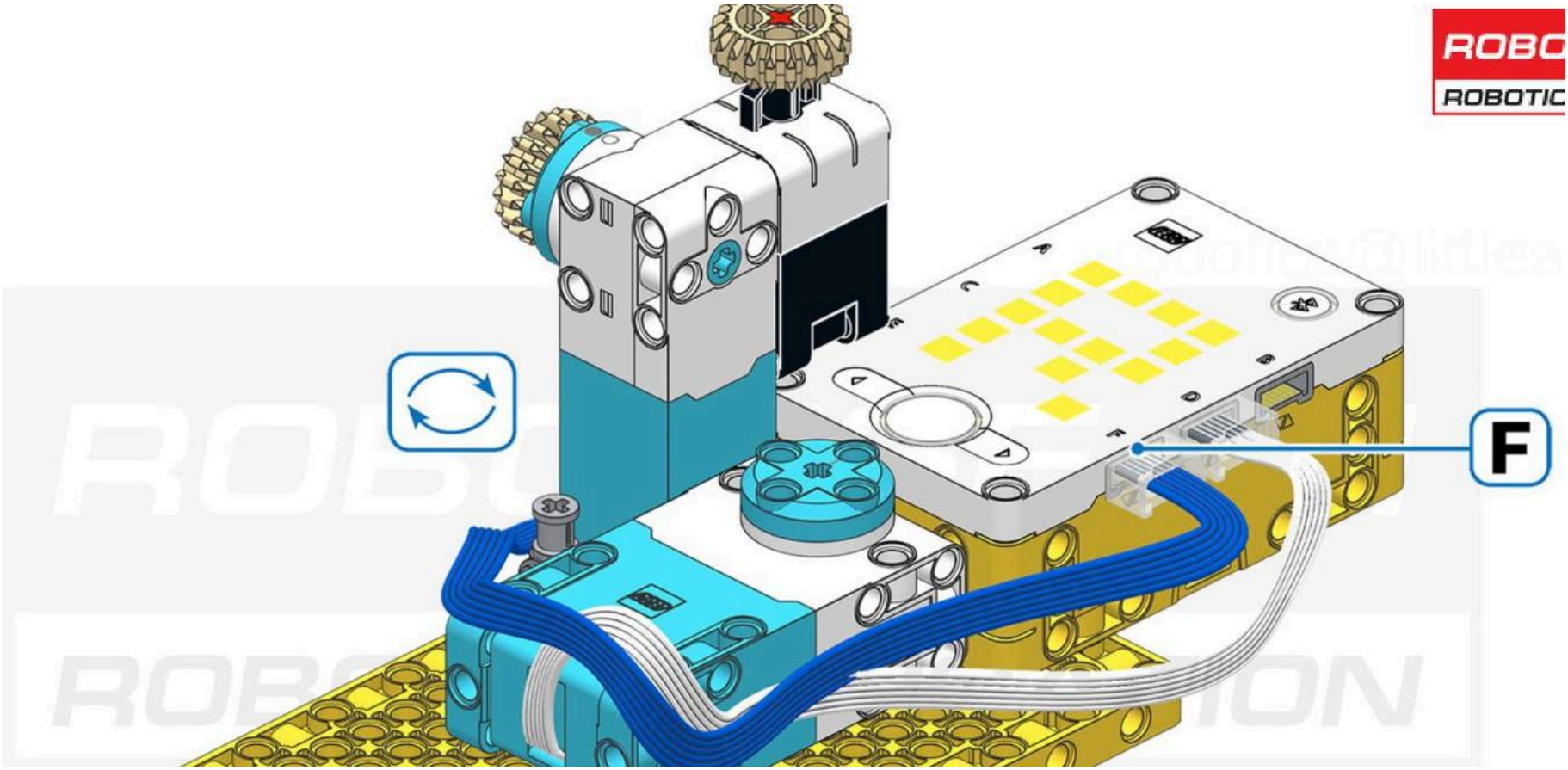
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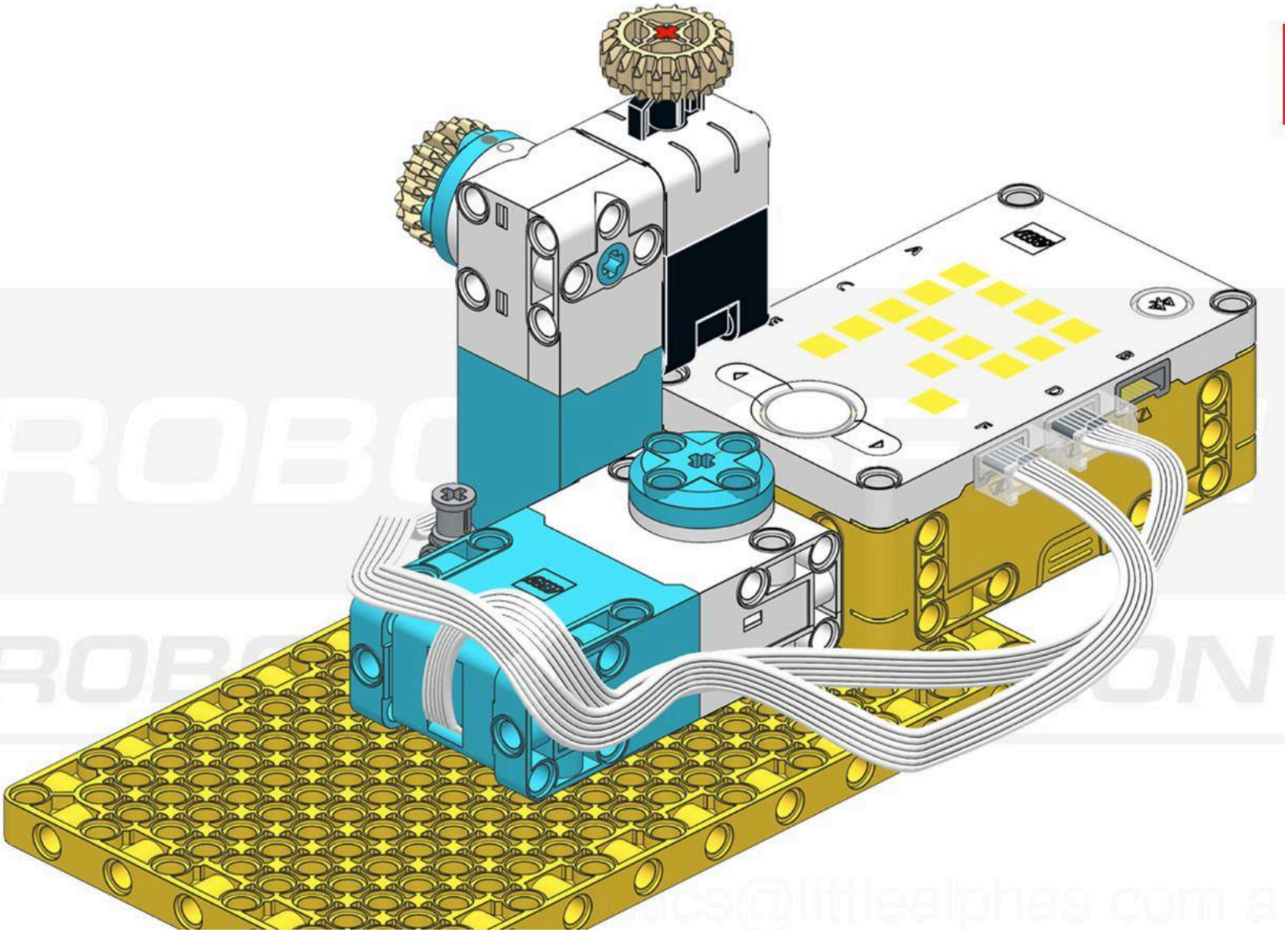
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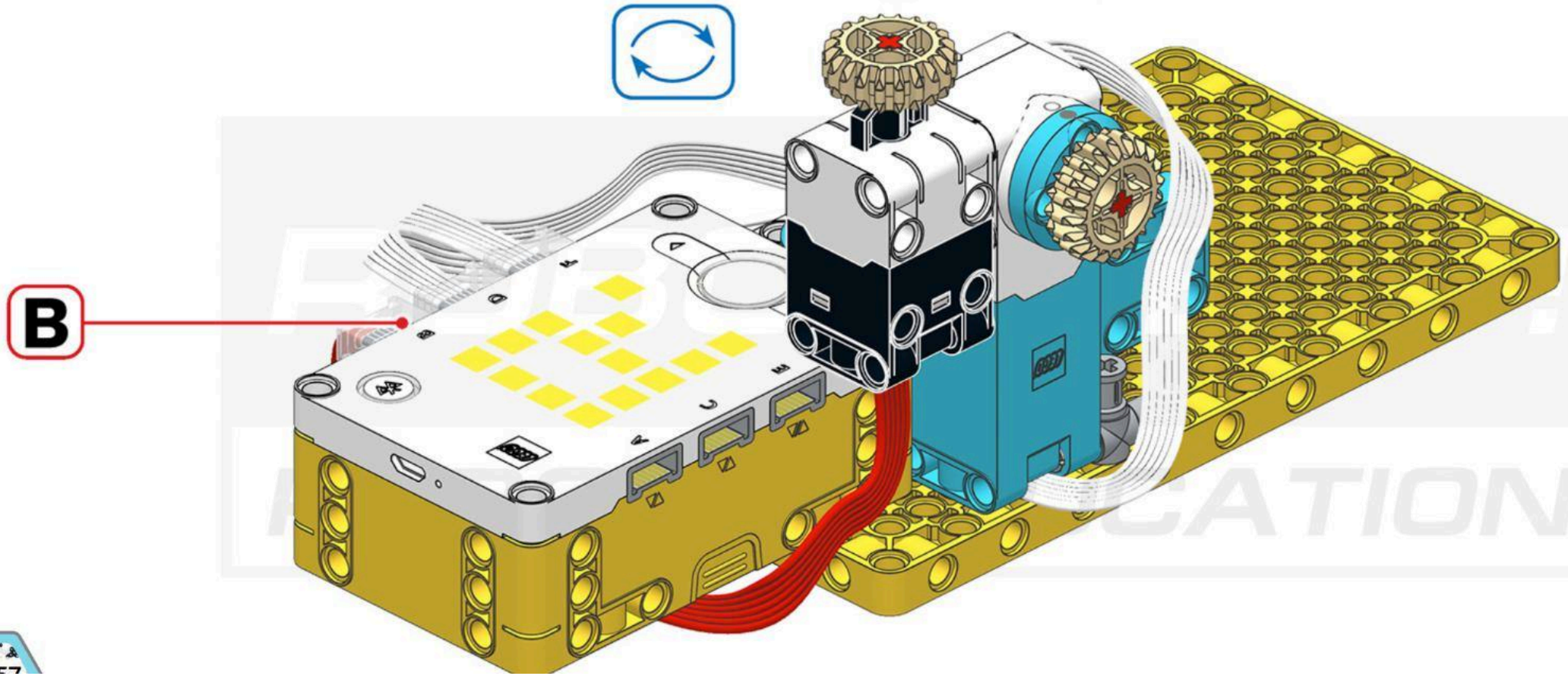


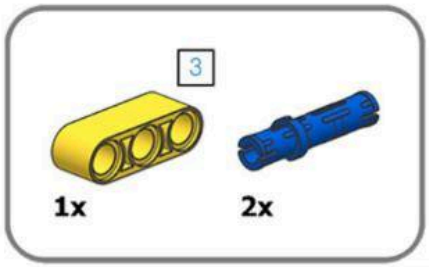

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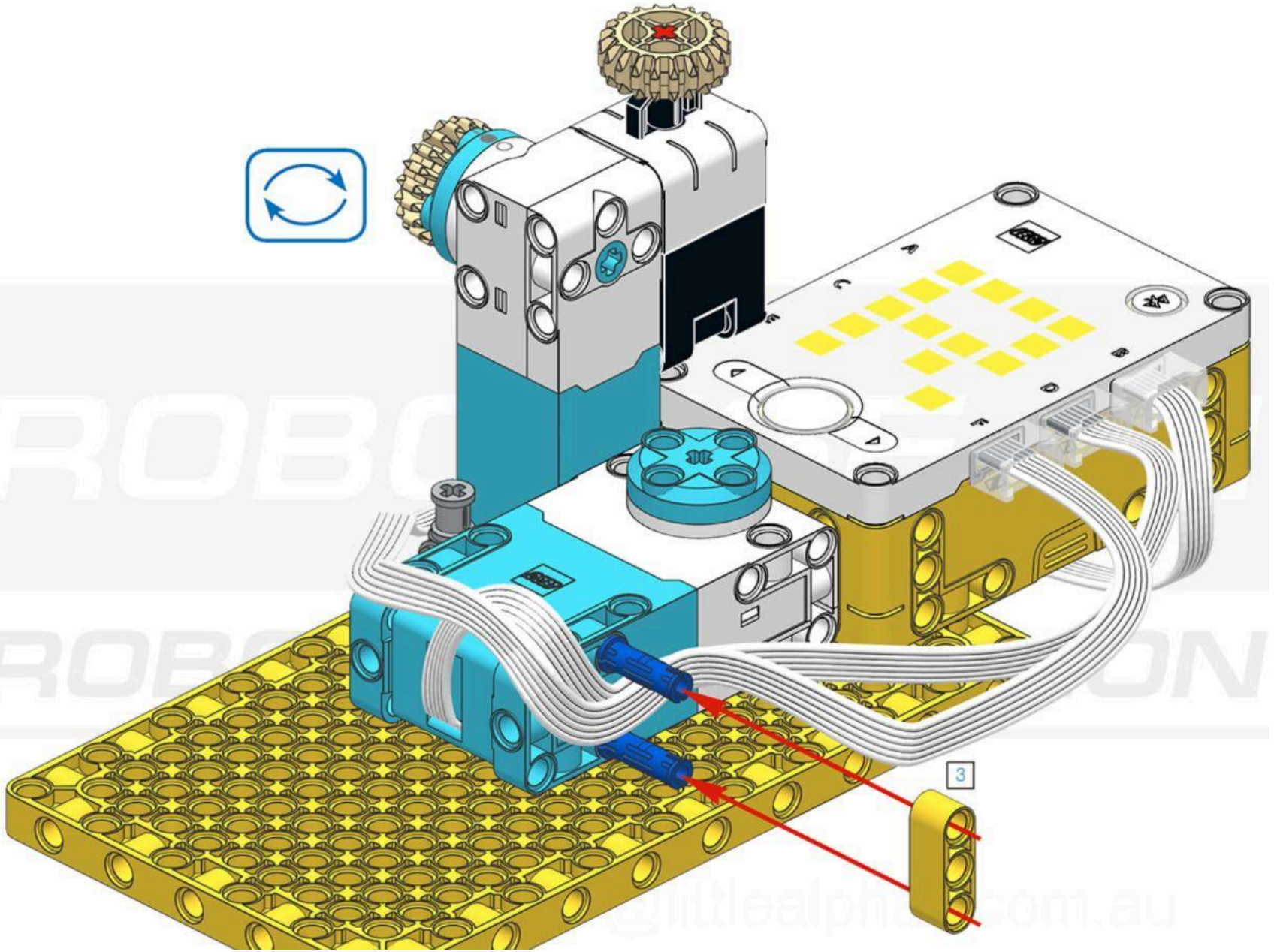


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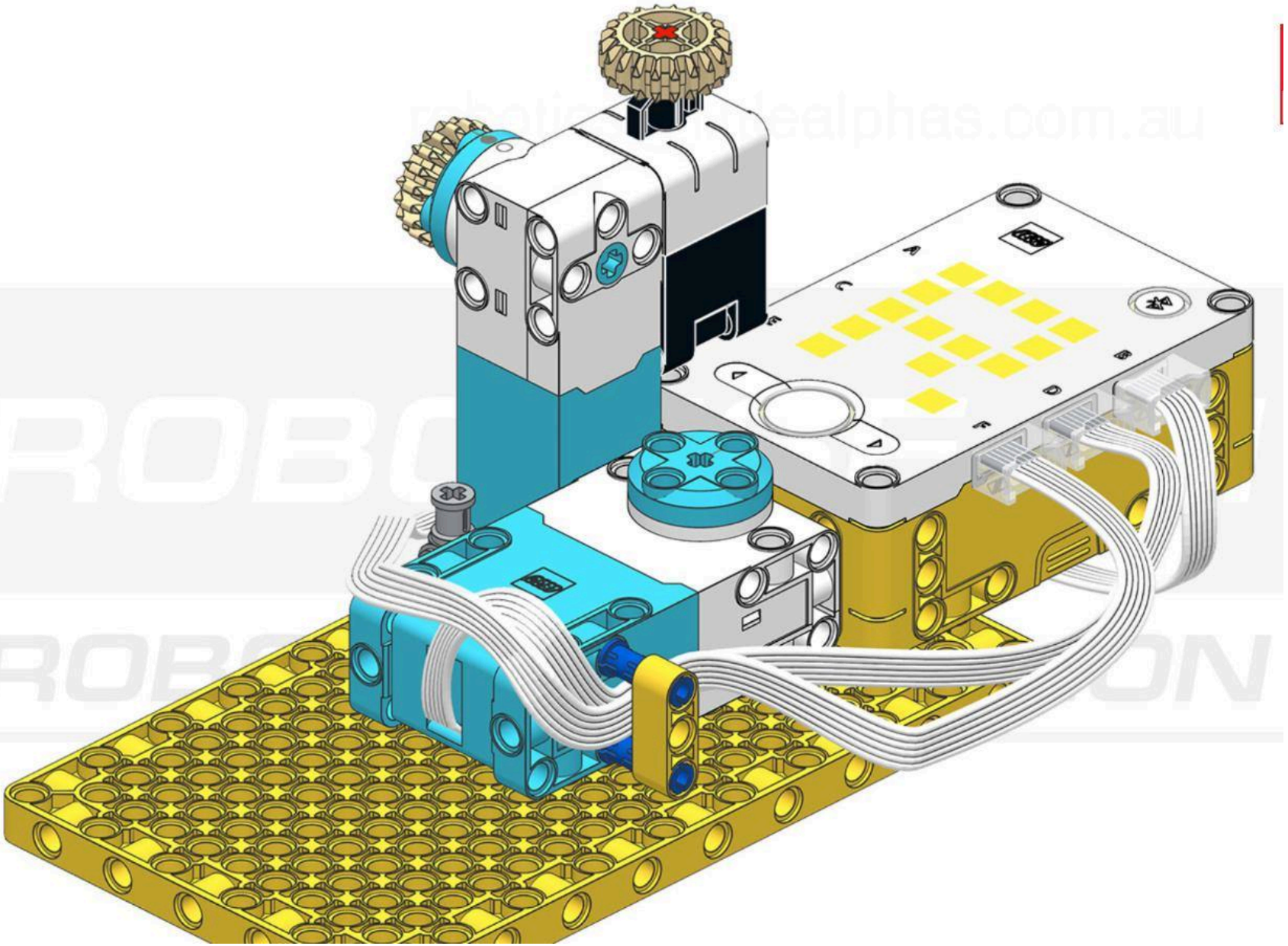




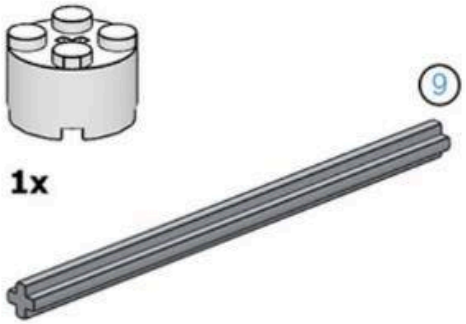
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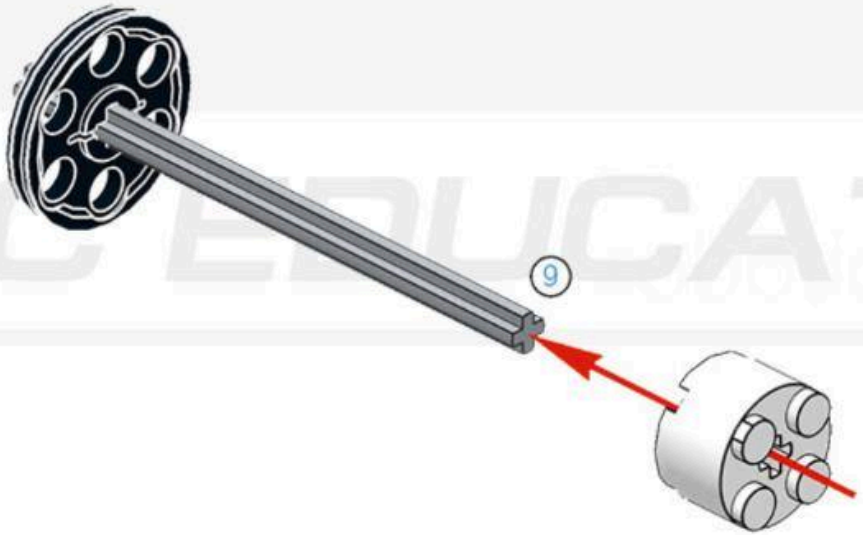
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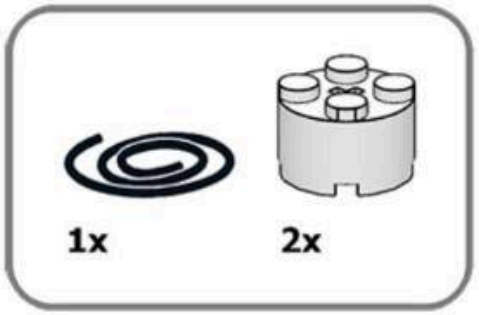




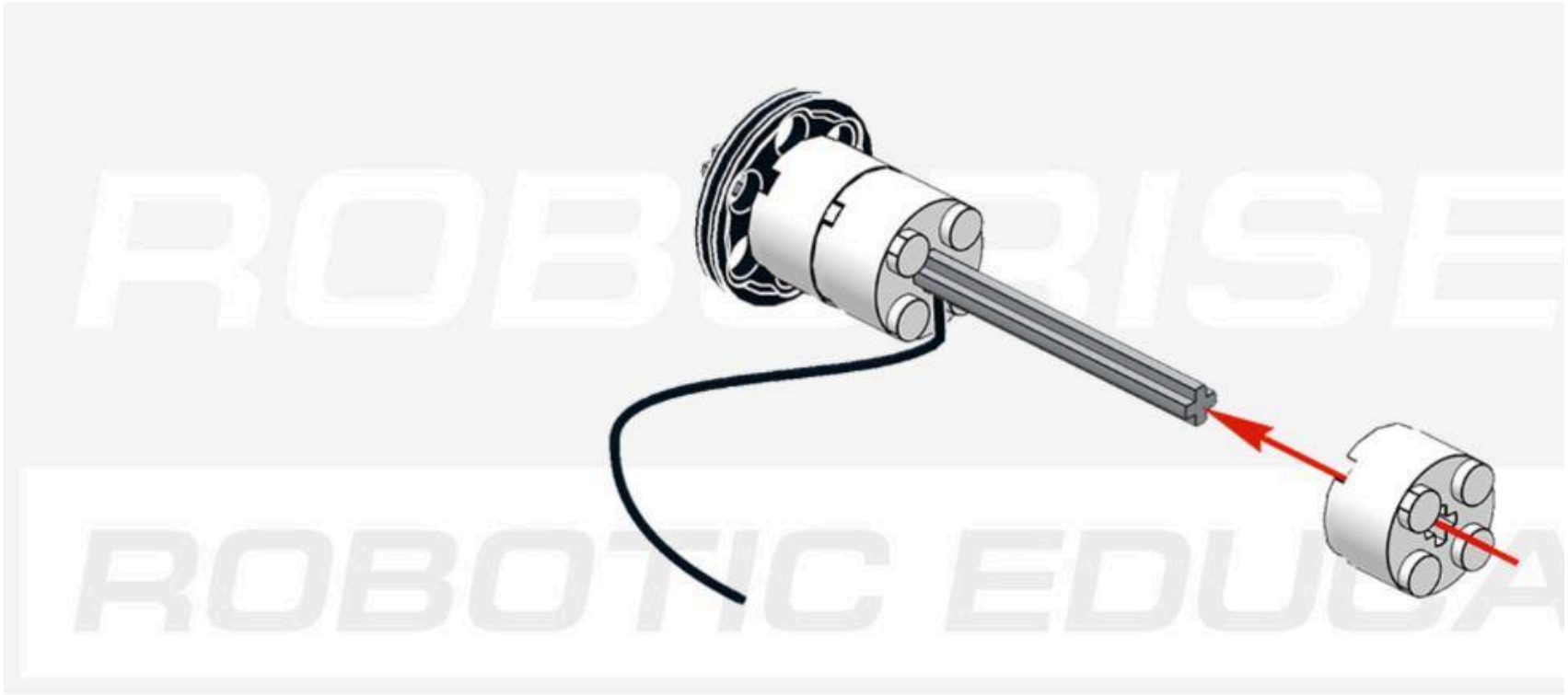
35



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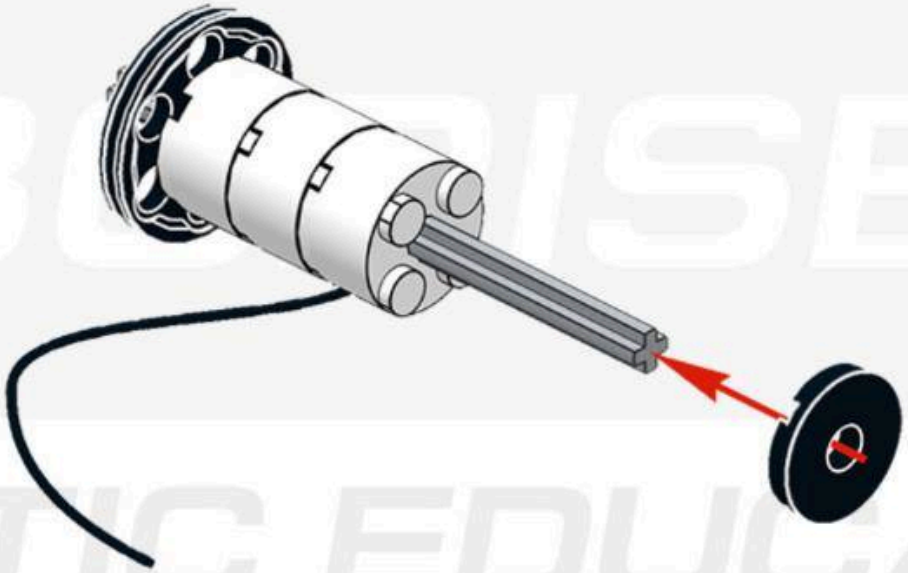


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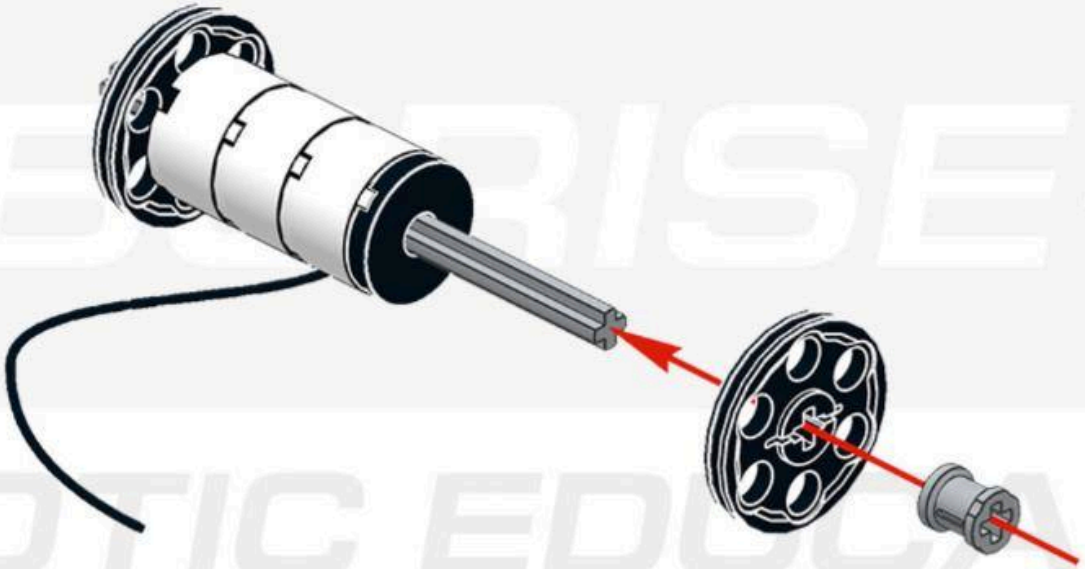


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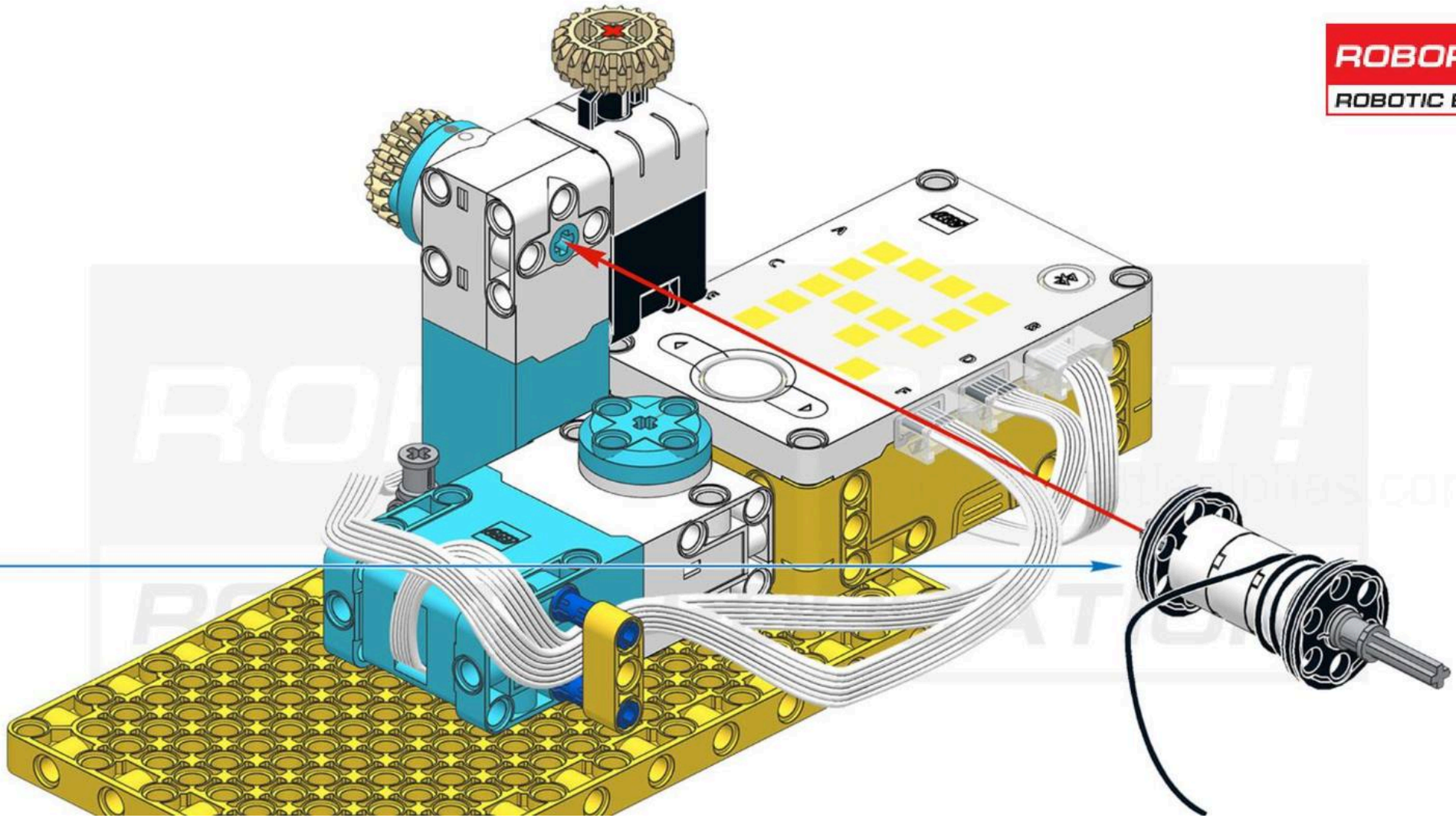
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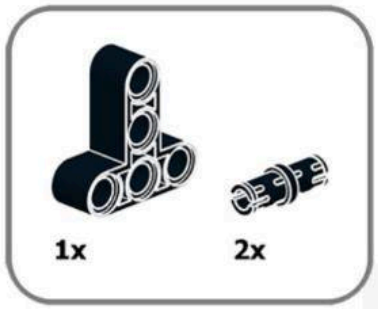


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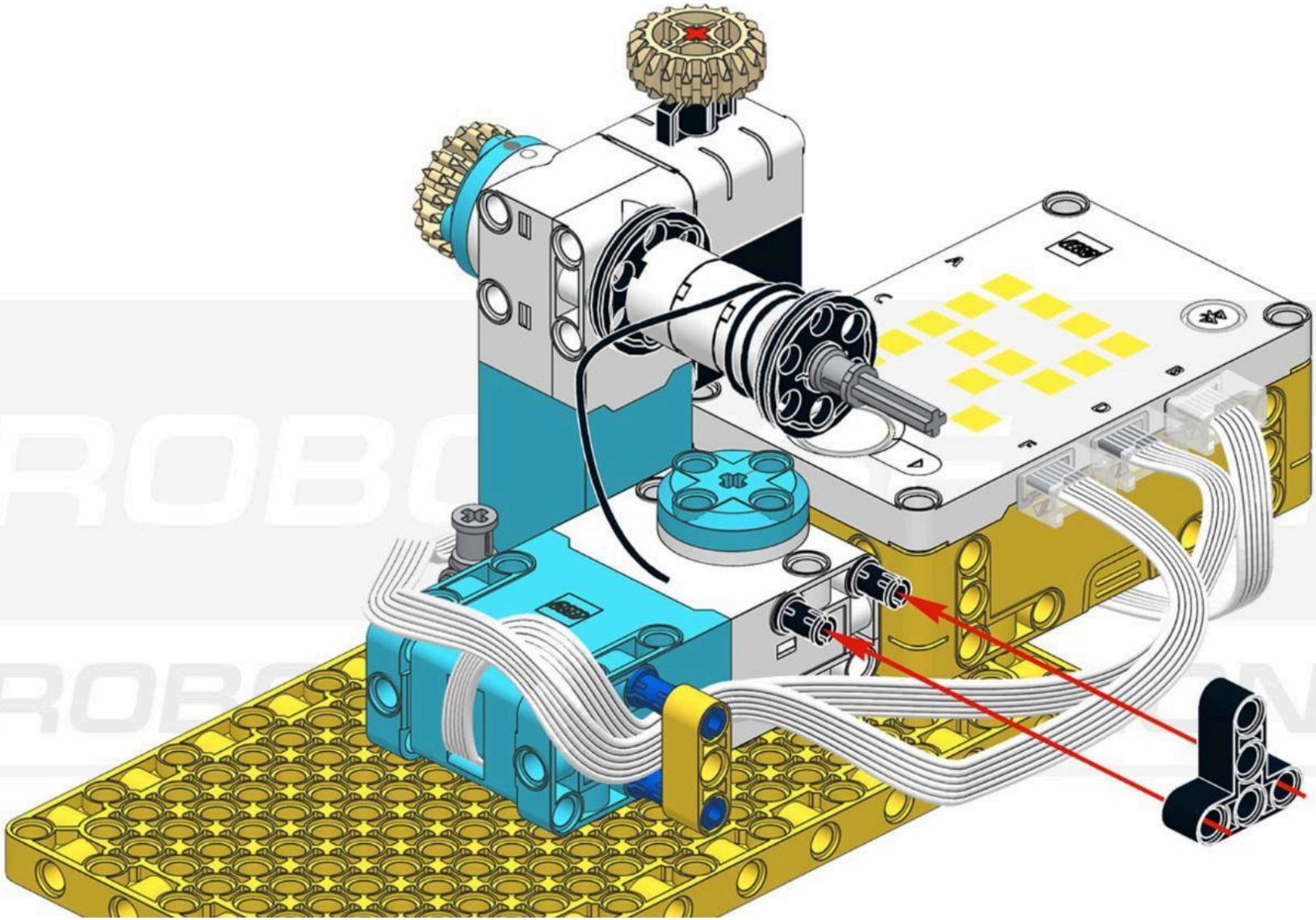


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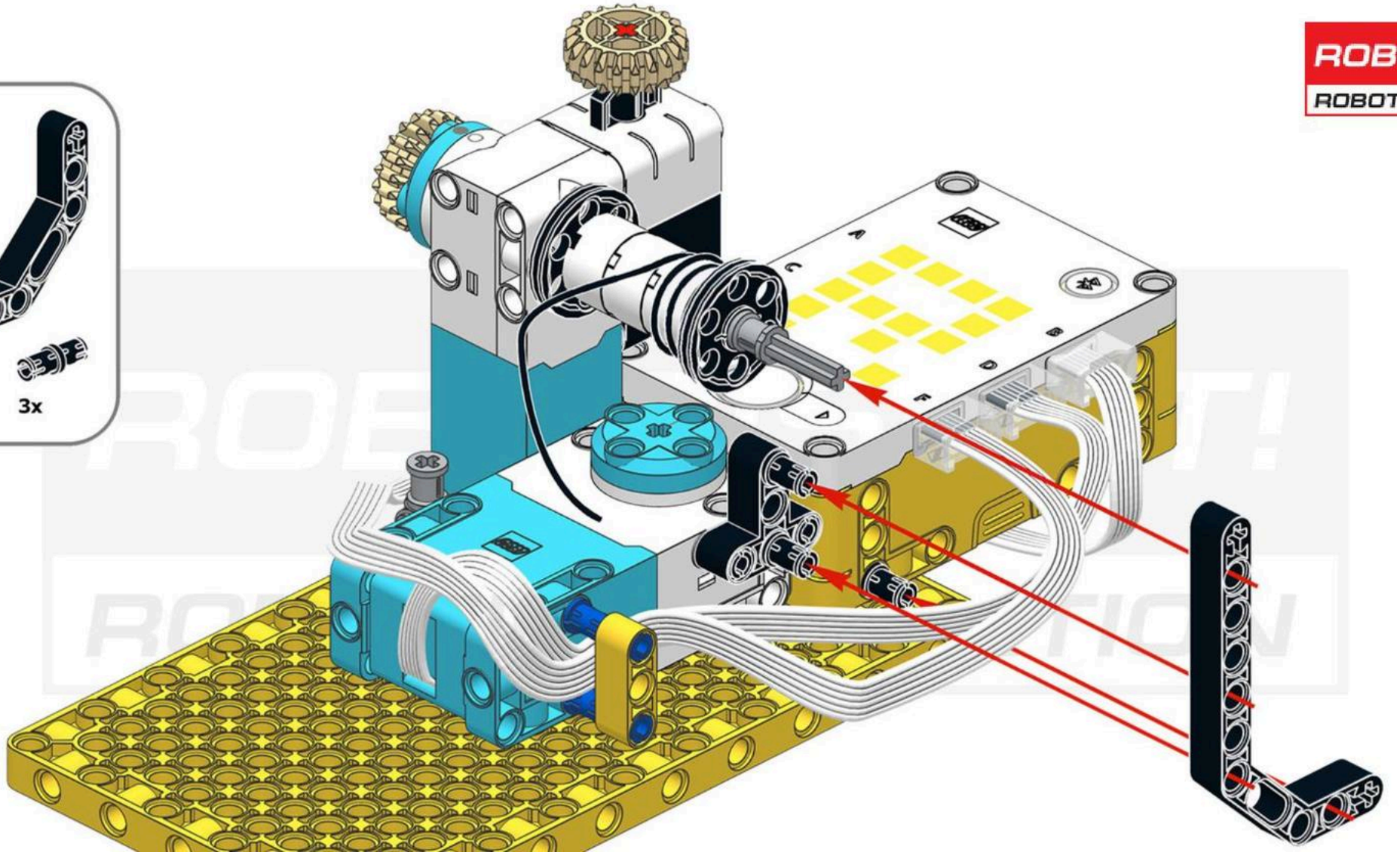


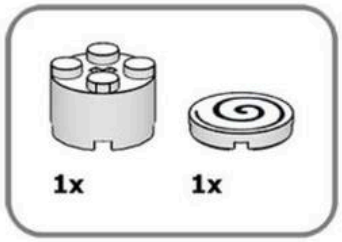
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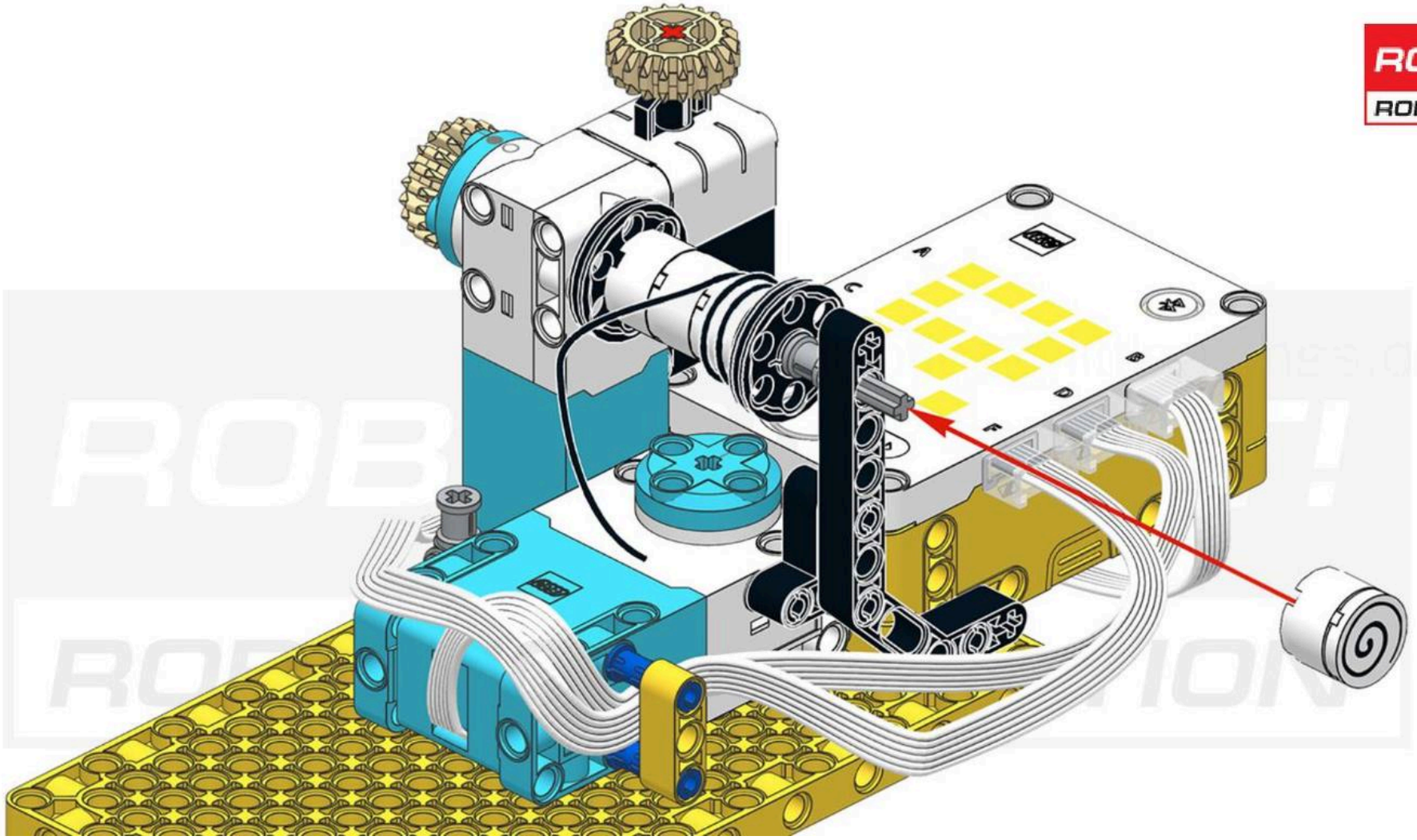


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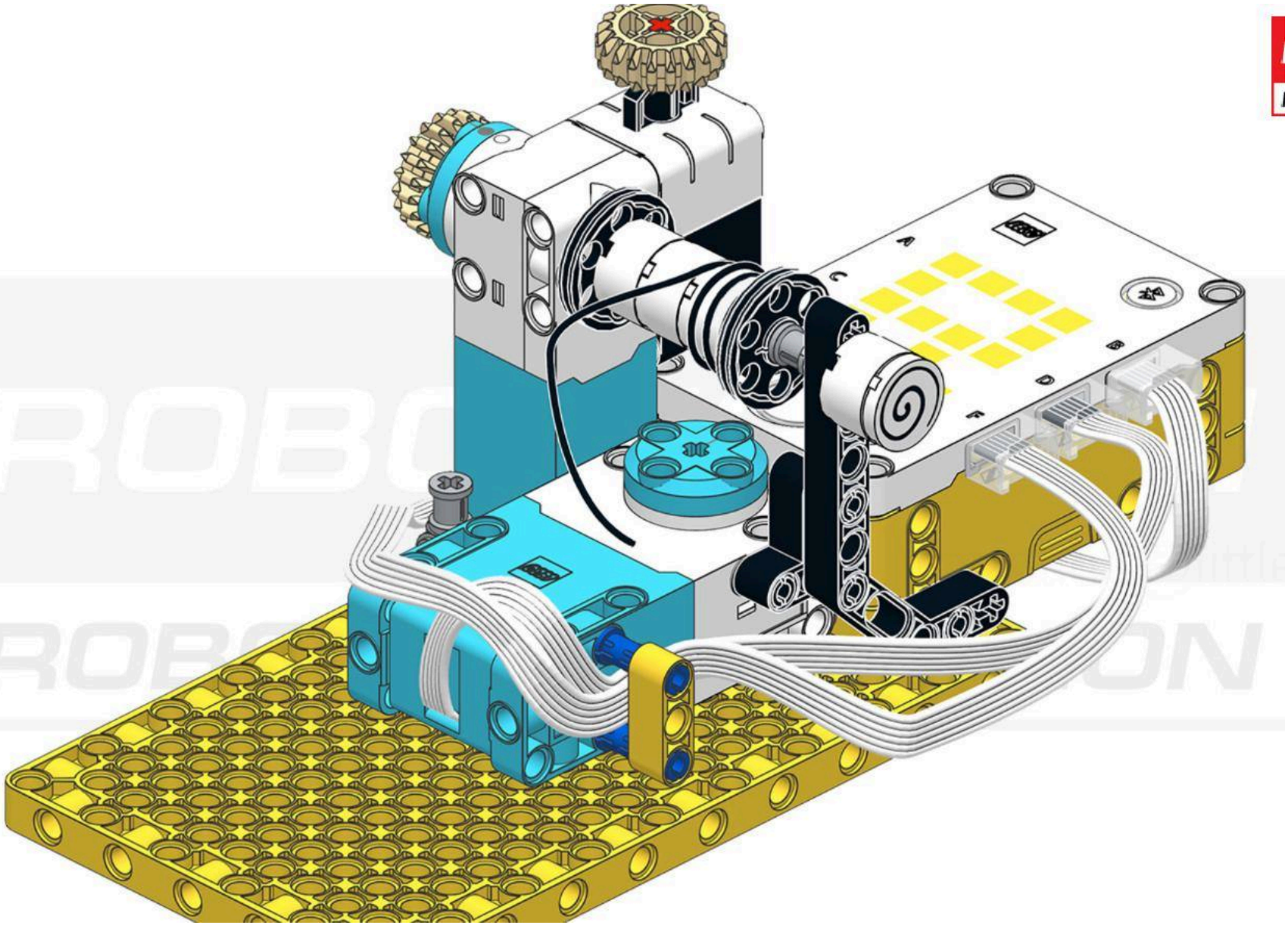


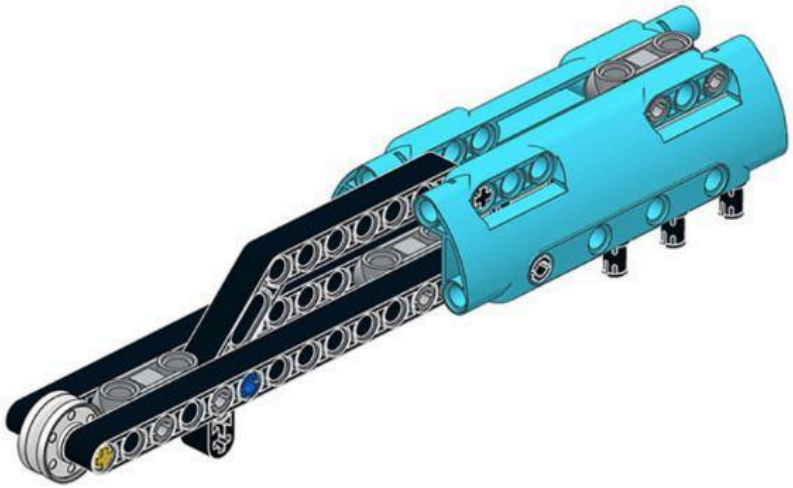


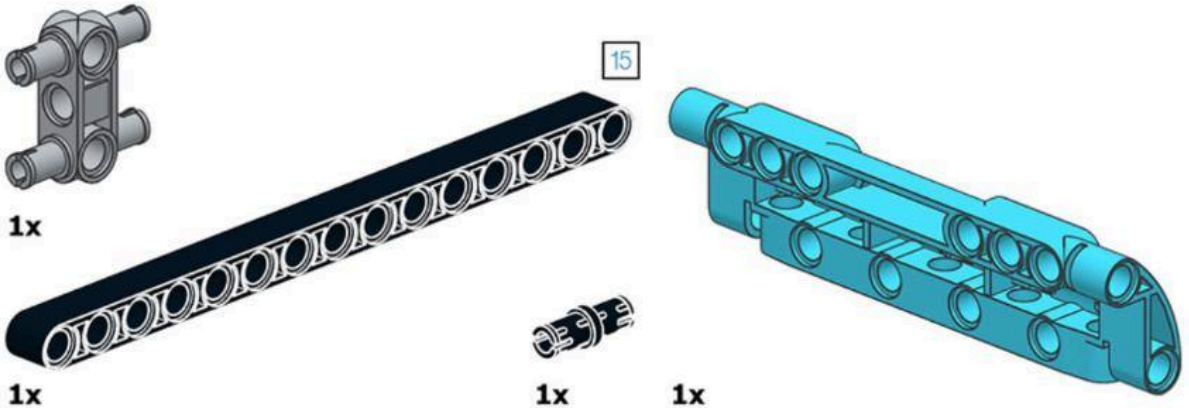
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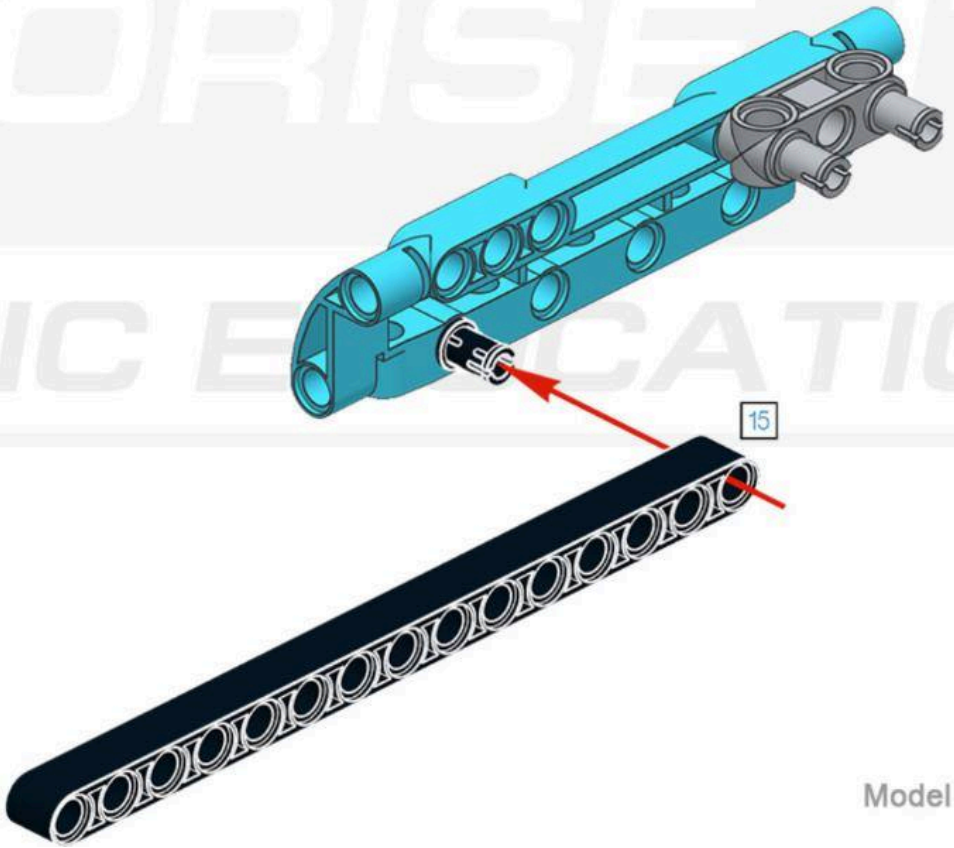
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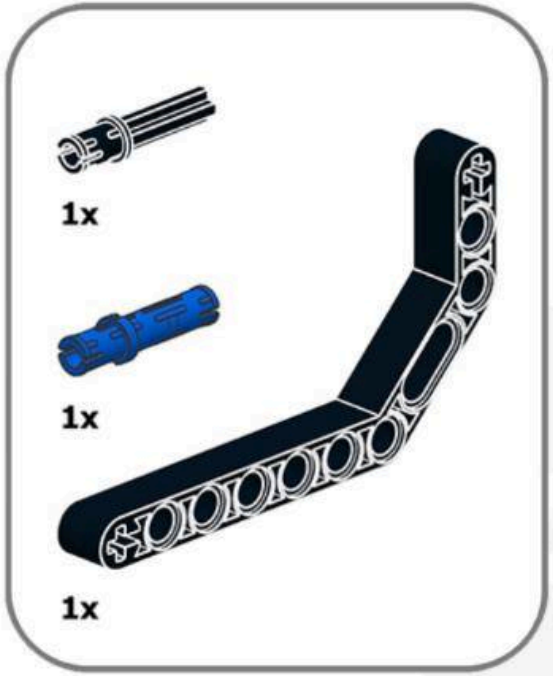




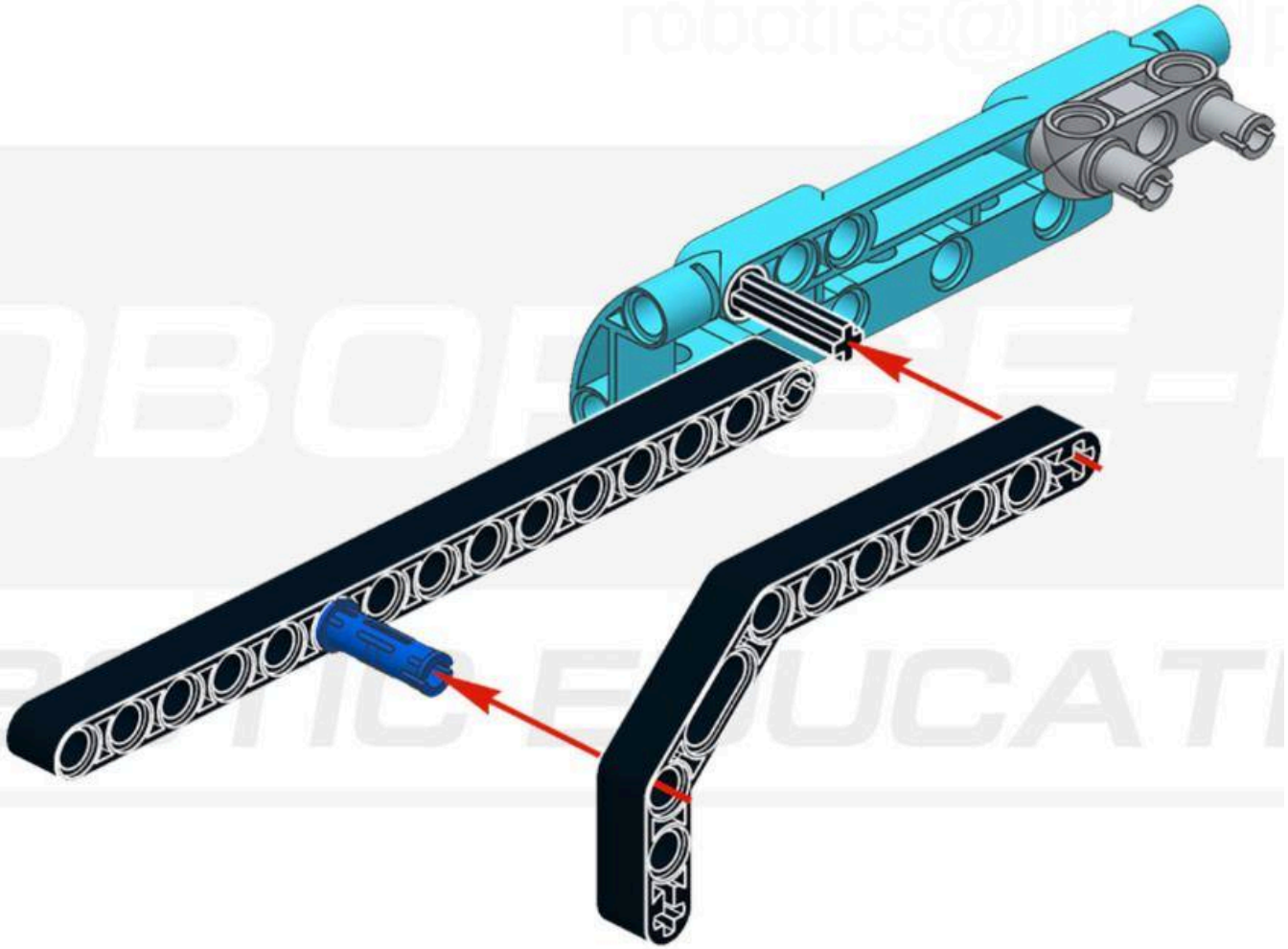
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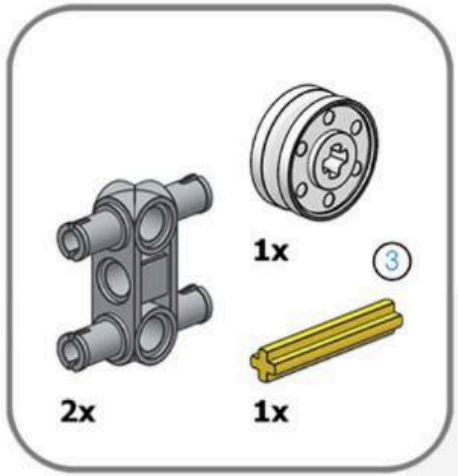


Model by Ignat Khli

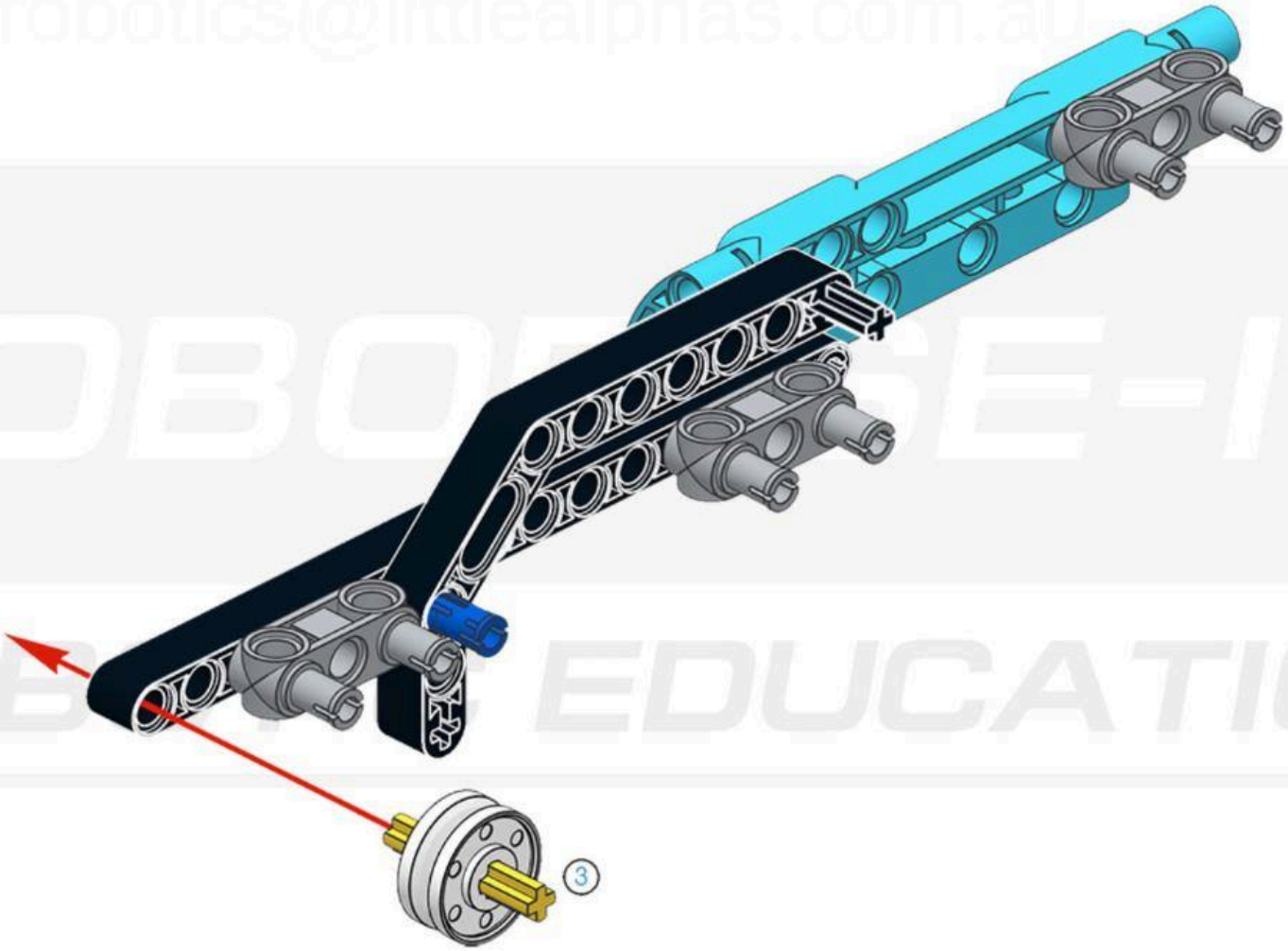


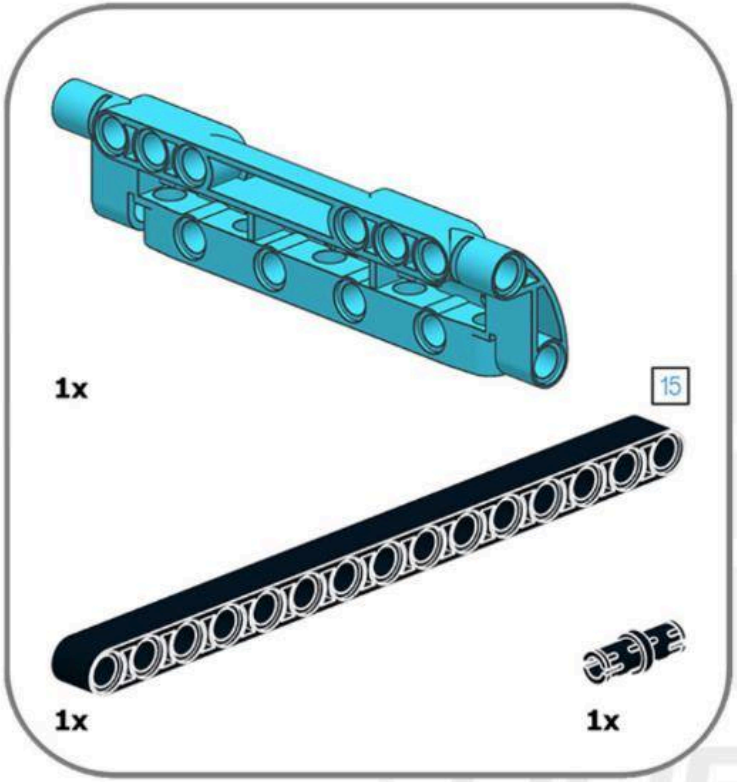
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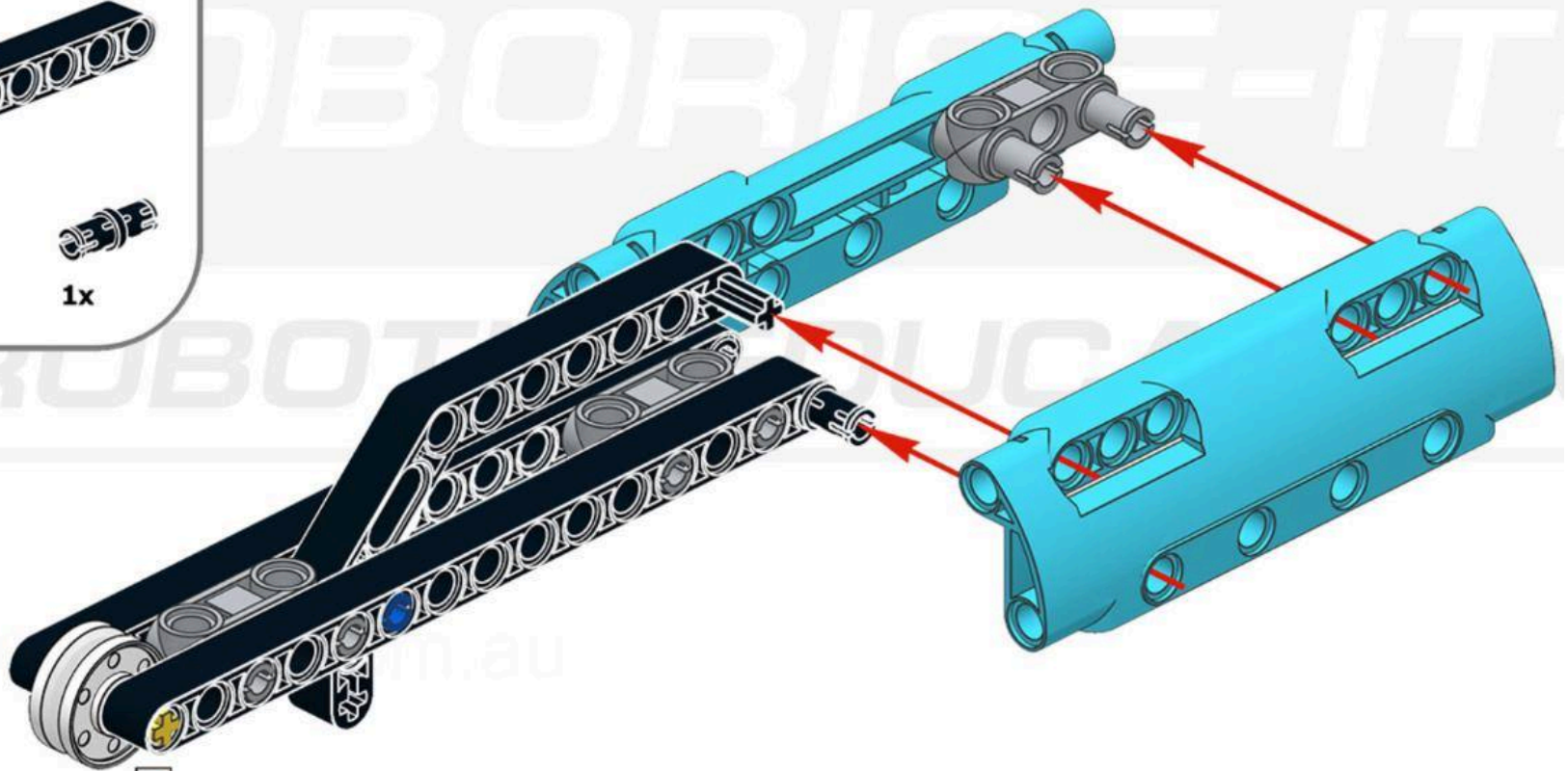


47





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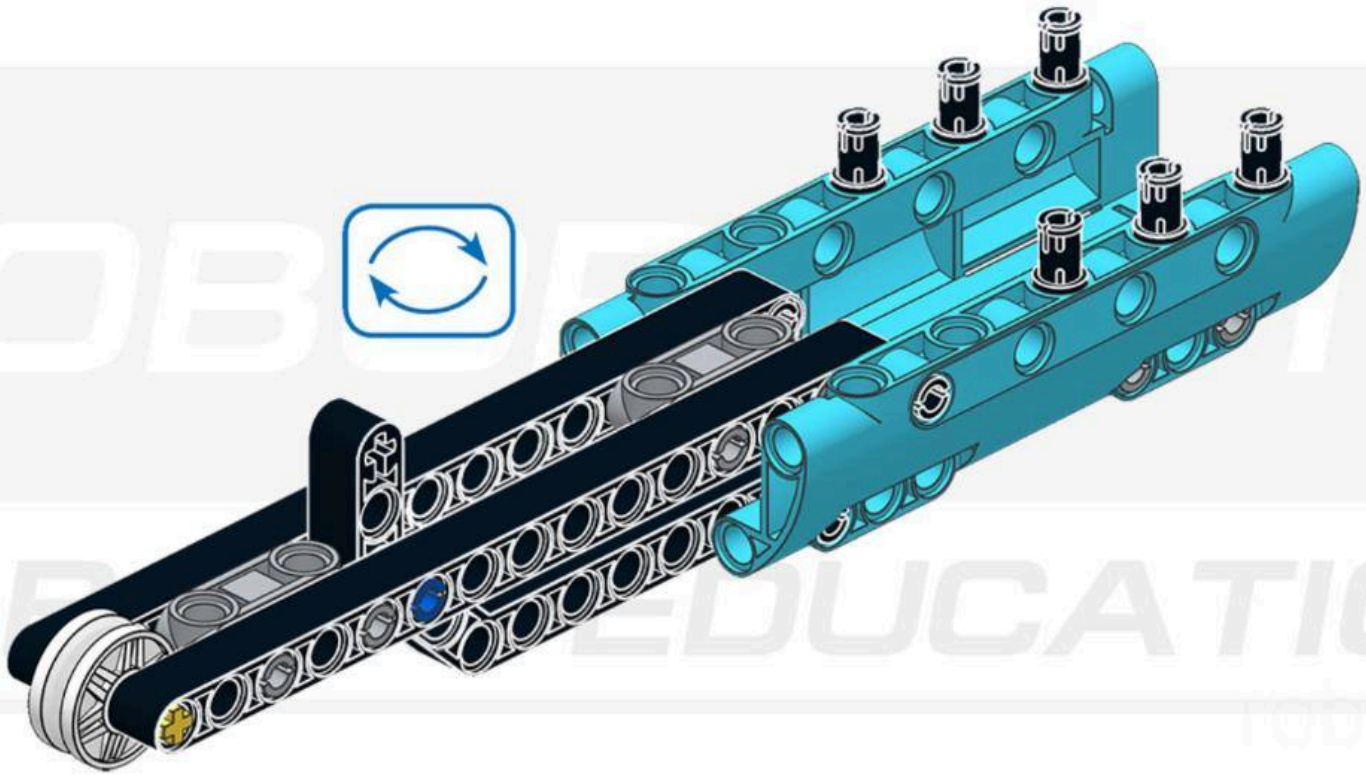


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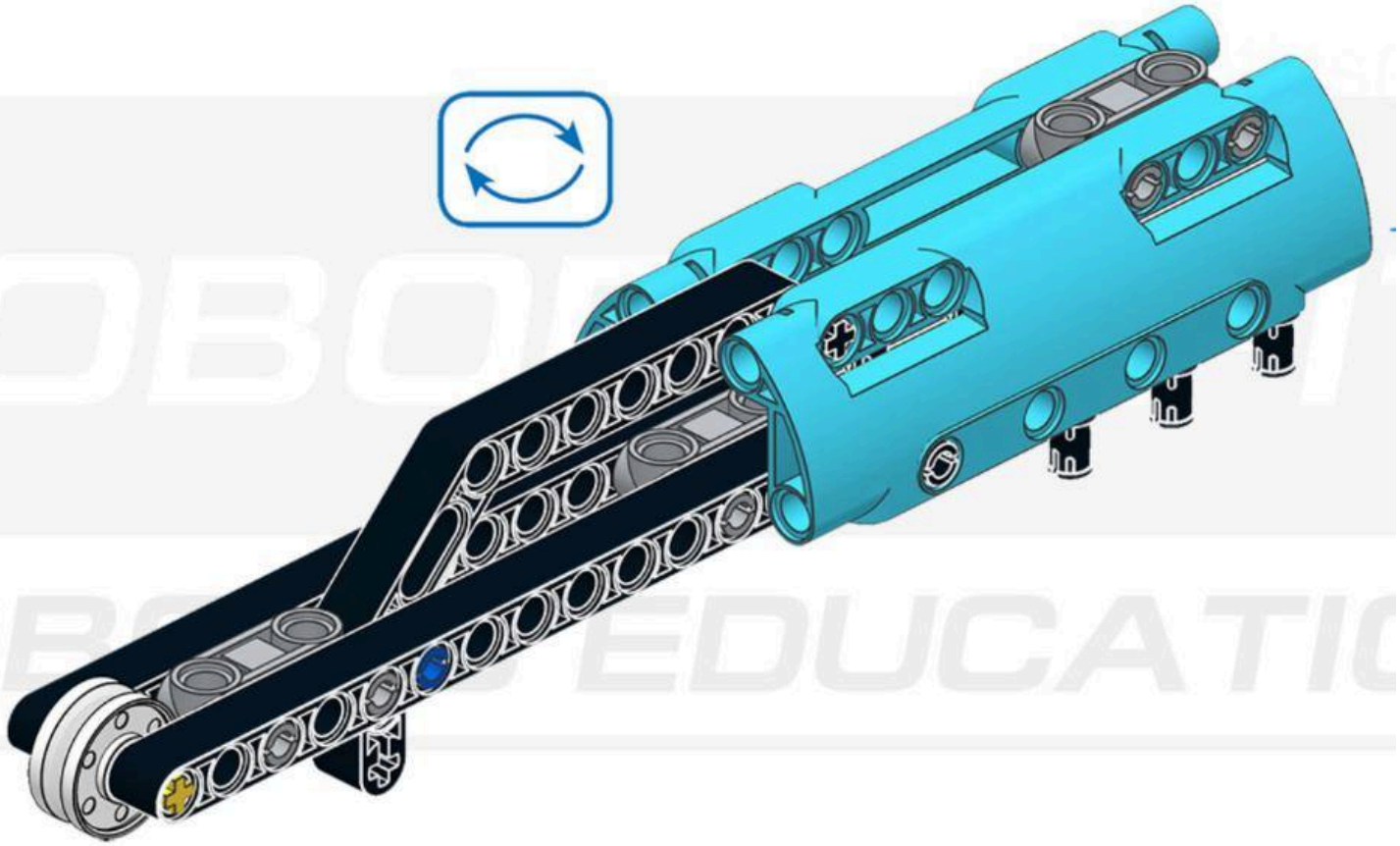
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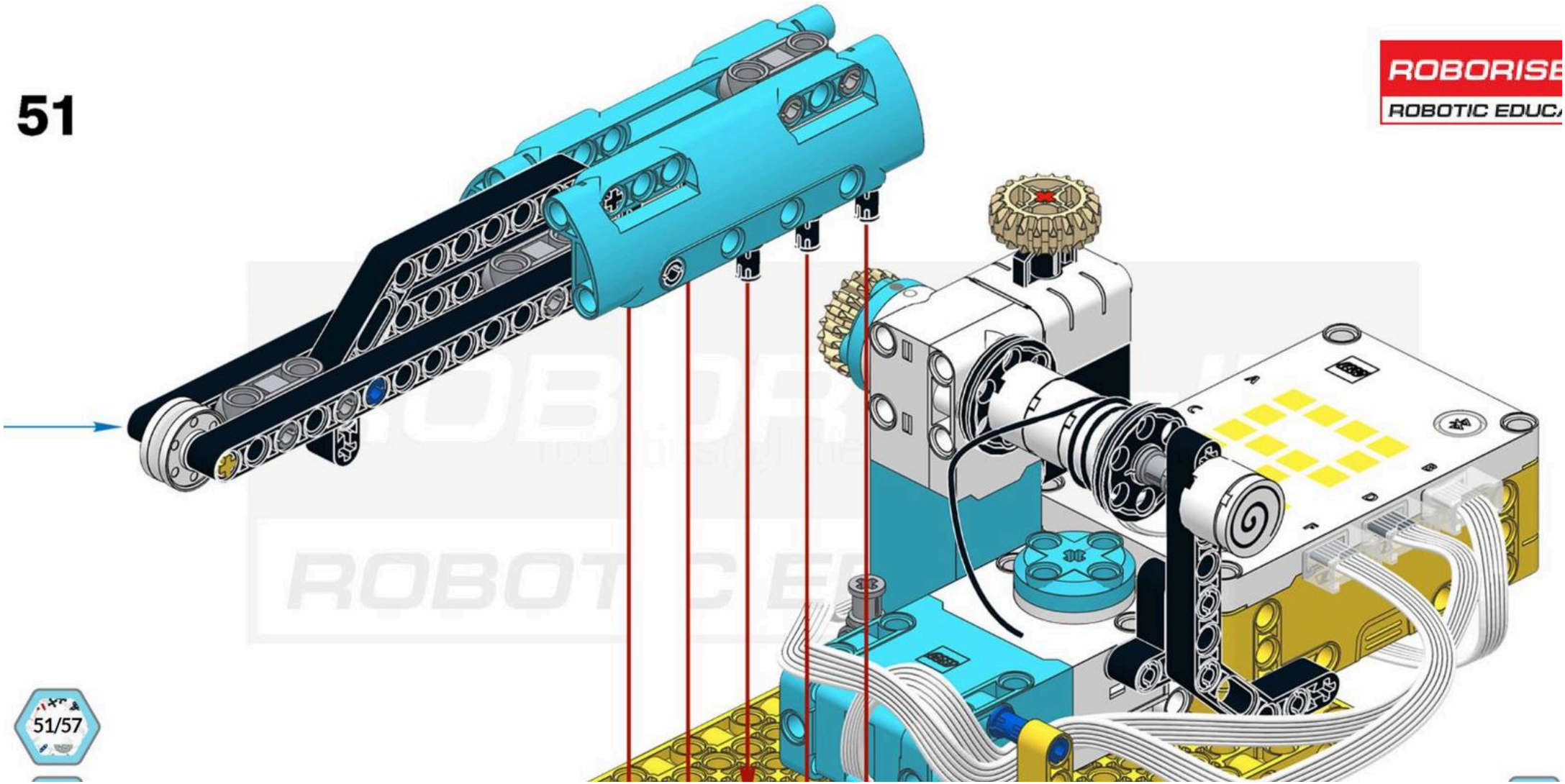
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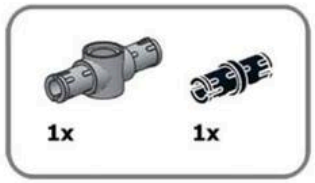


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51





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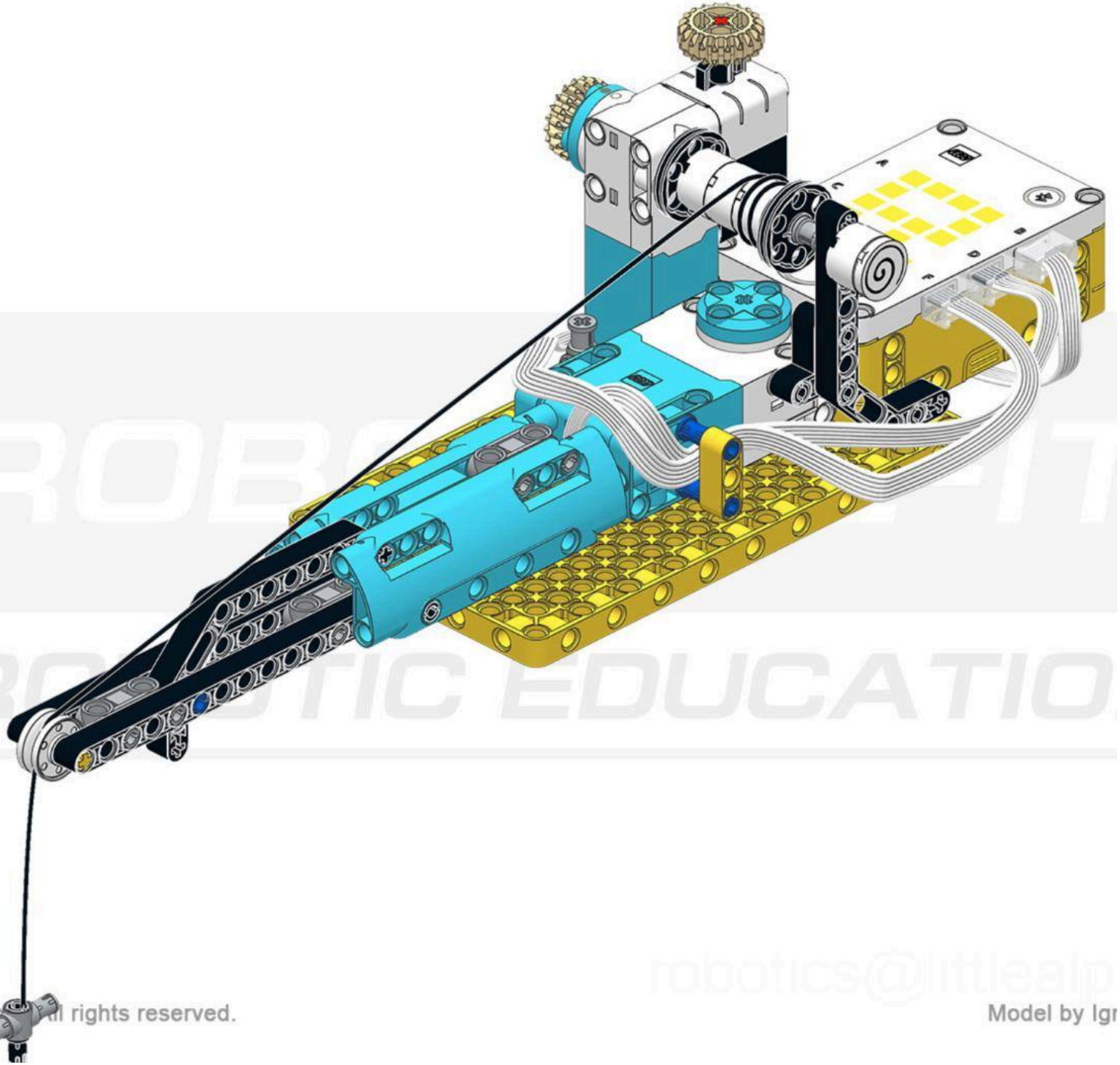


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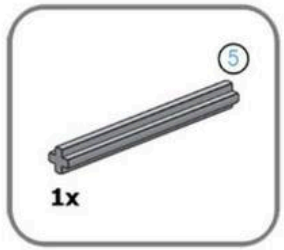
Model by Ignat Khliebniko

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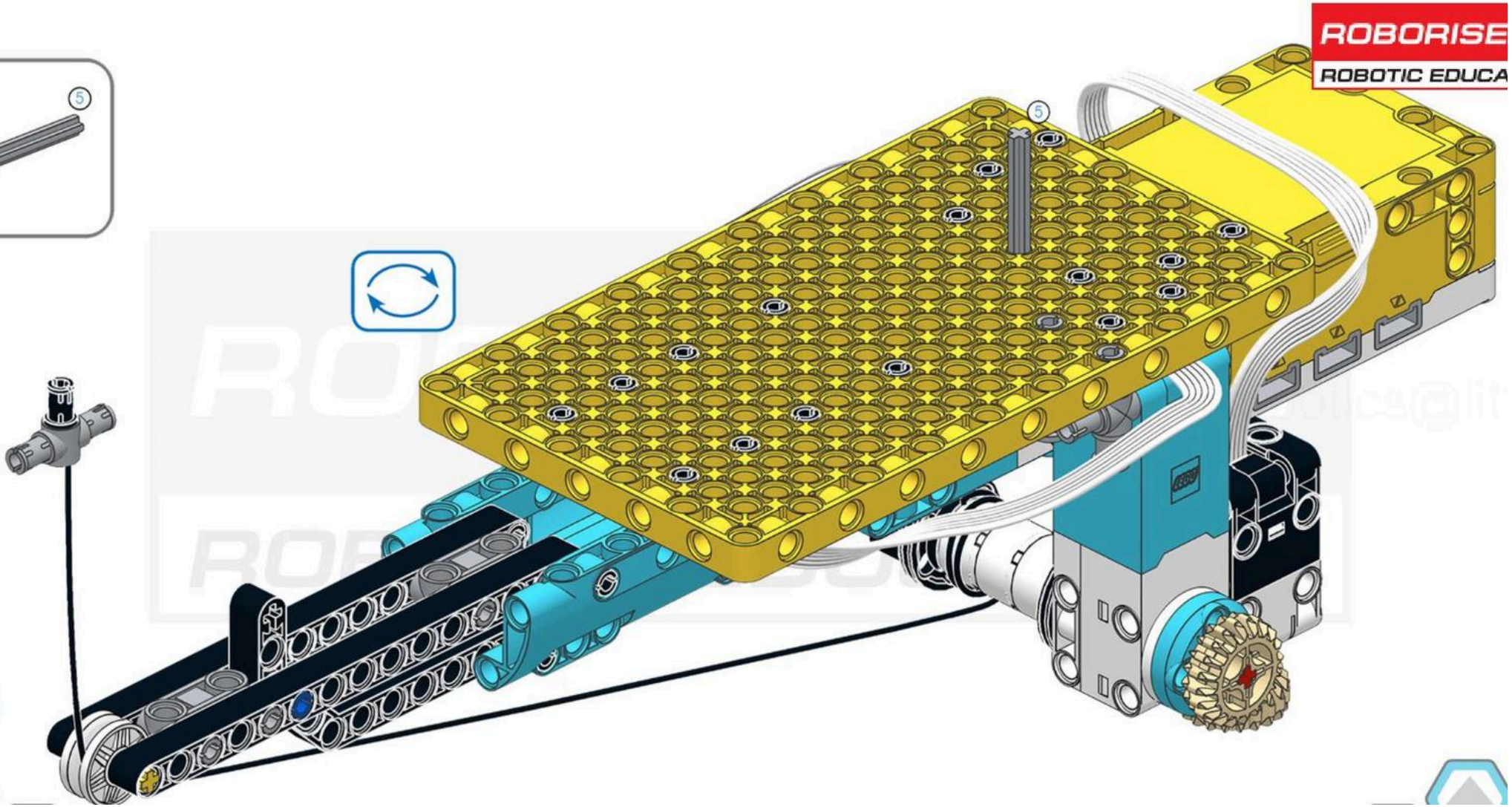


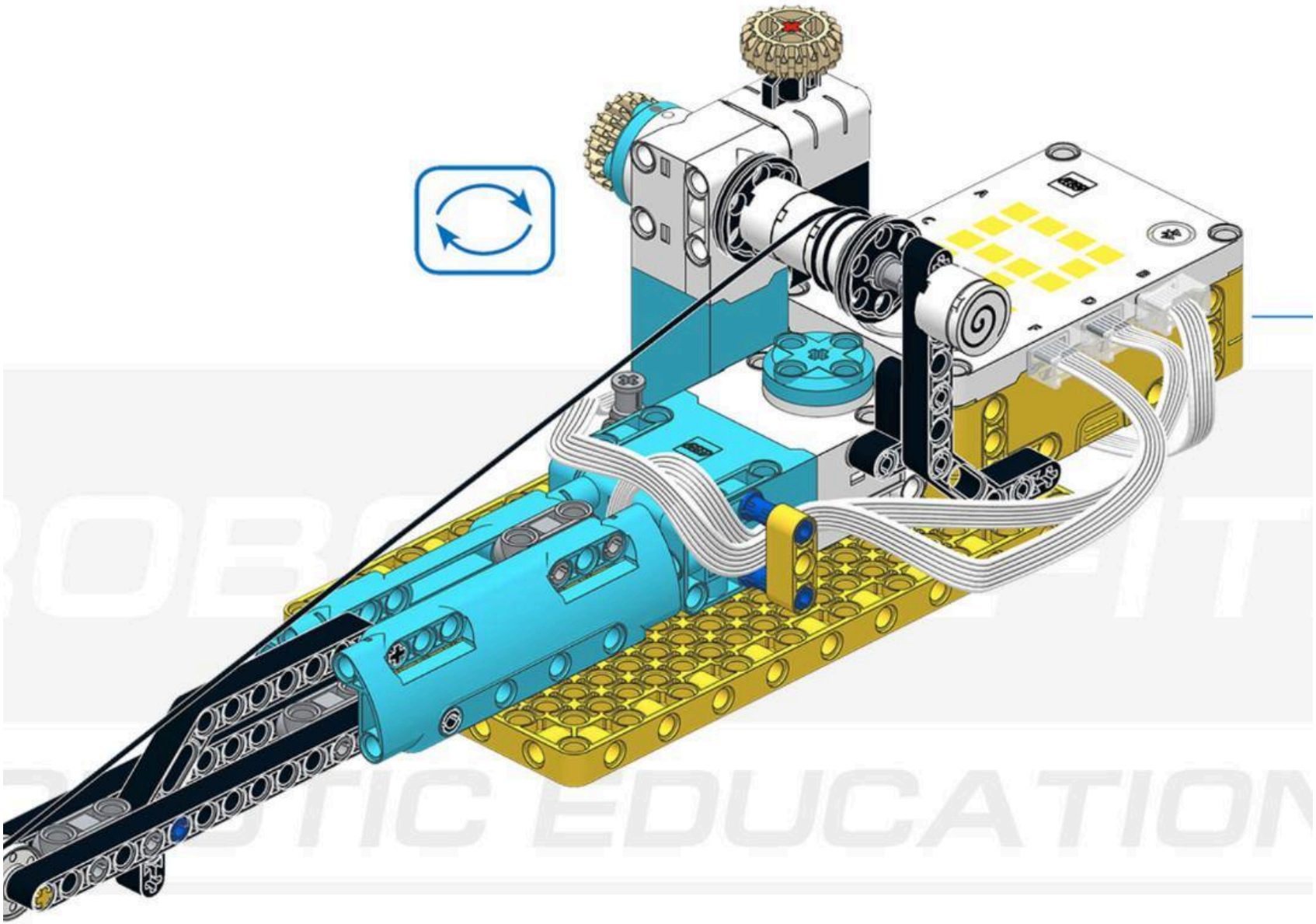
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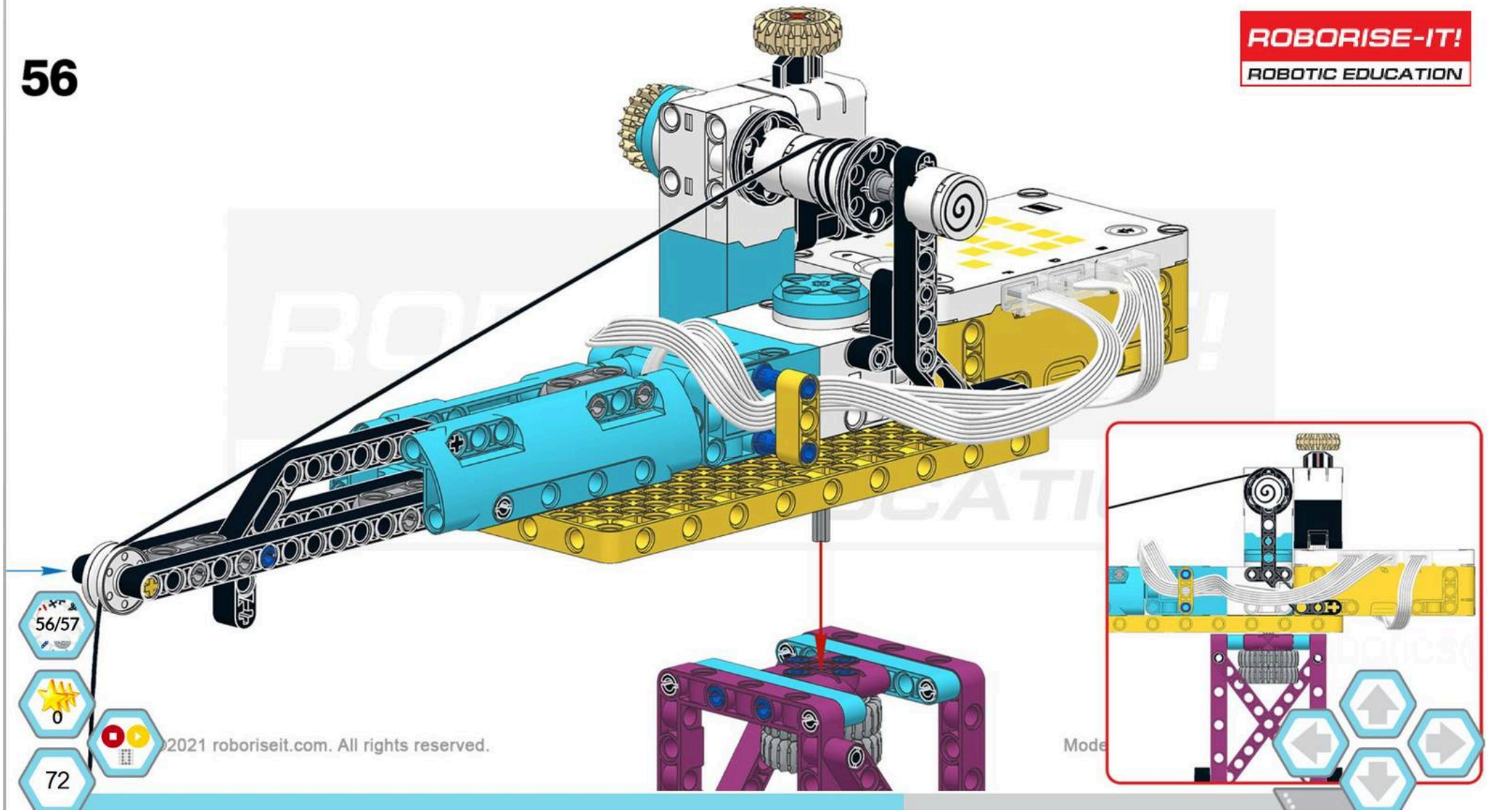


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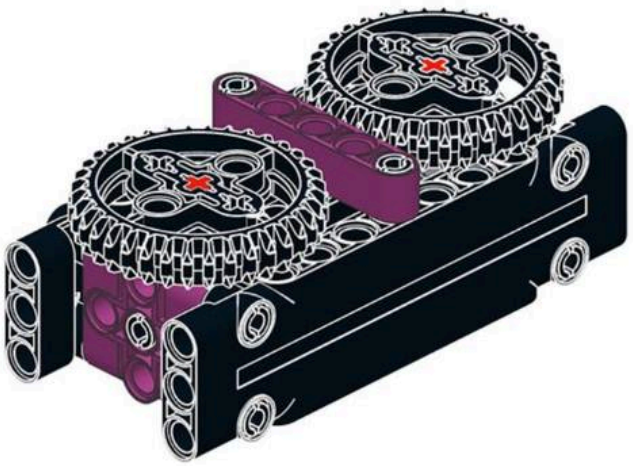
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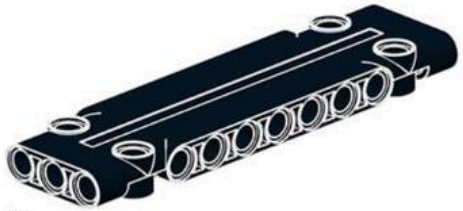
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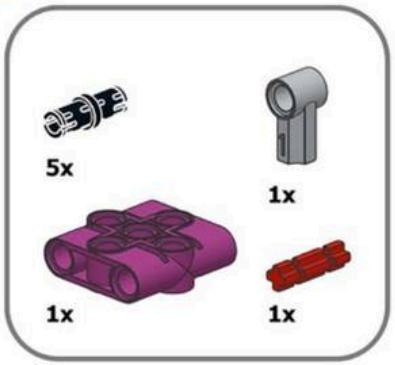


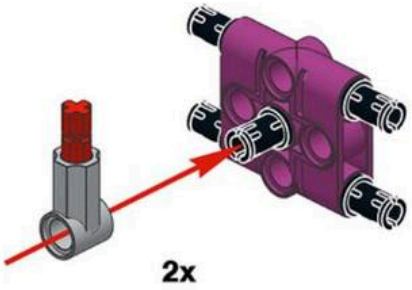
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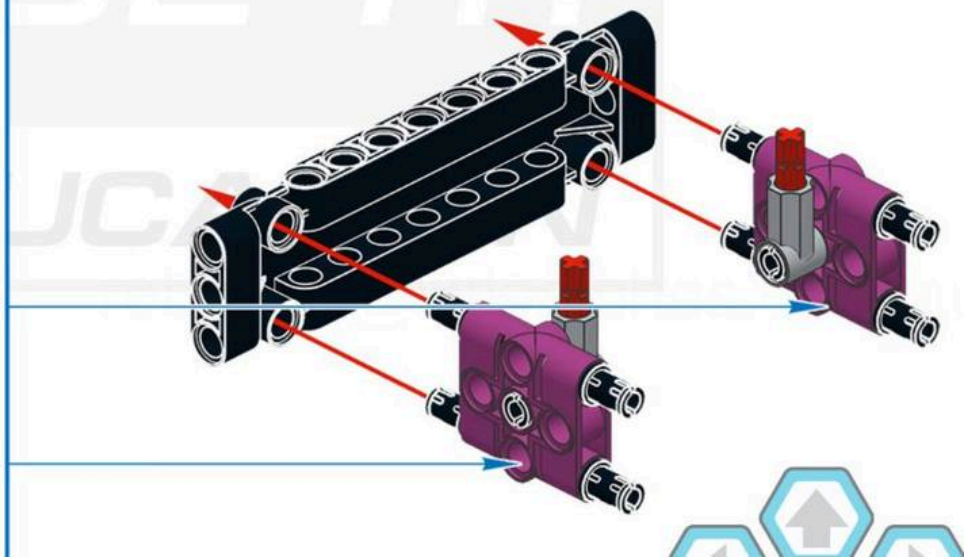
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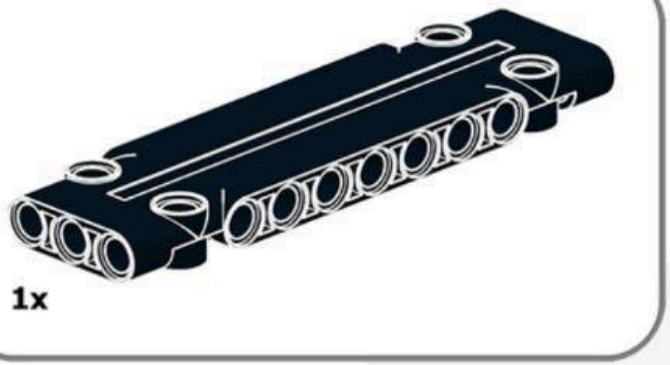




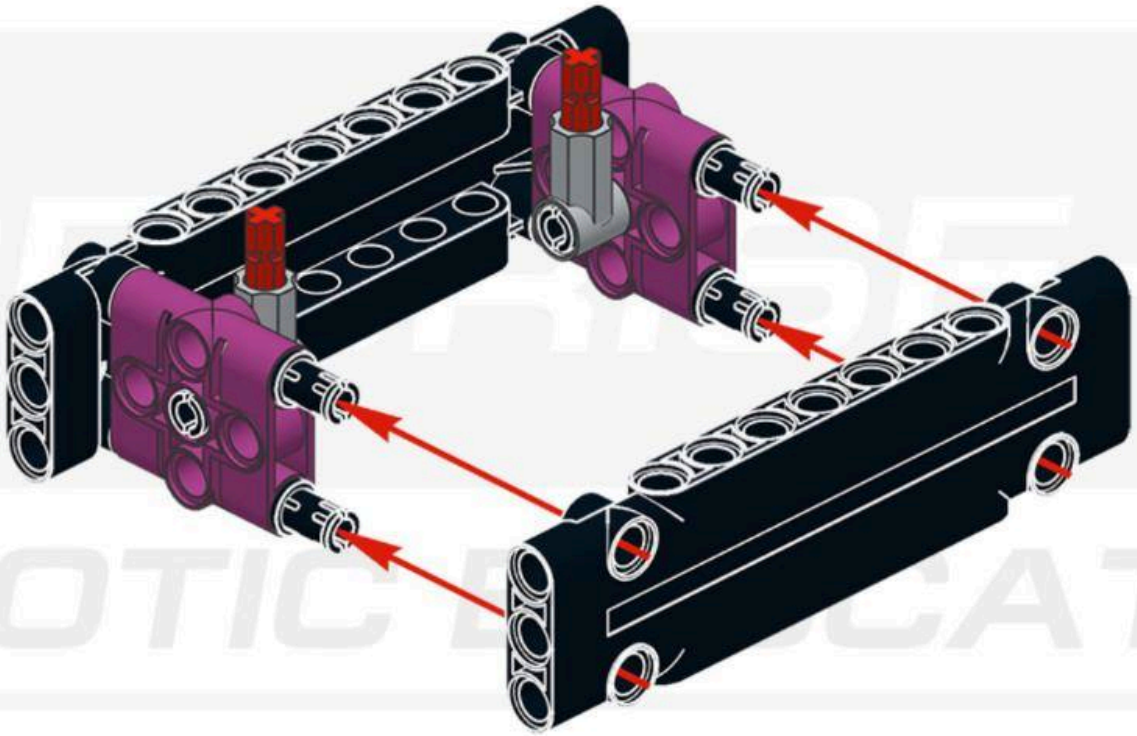


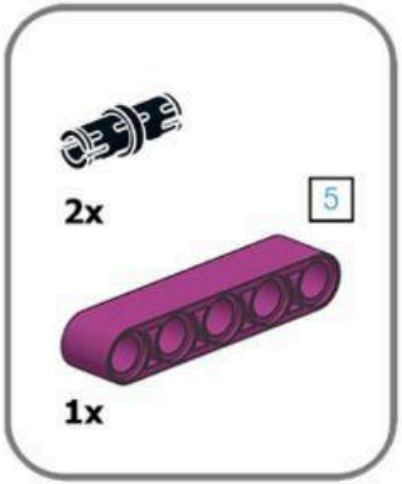
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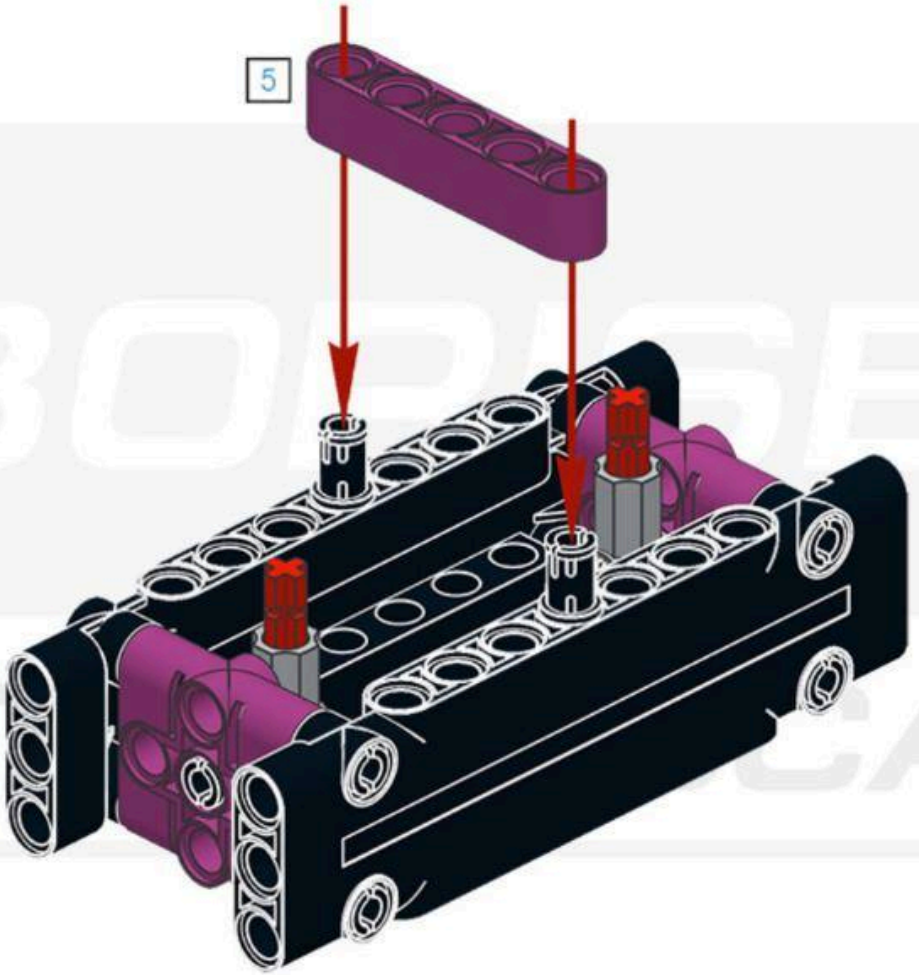


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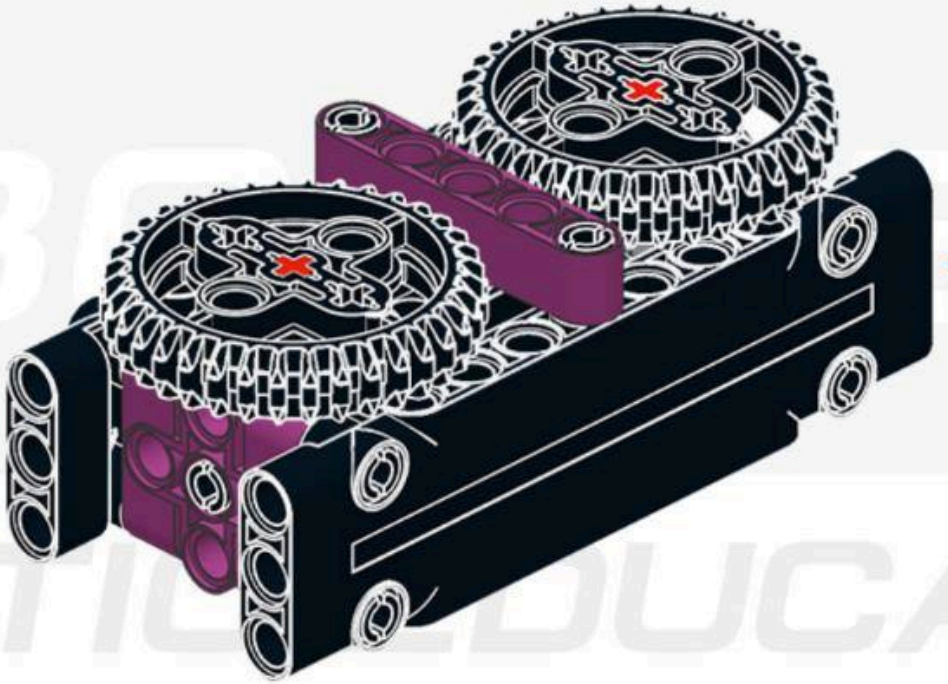


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2x

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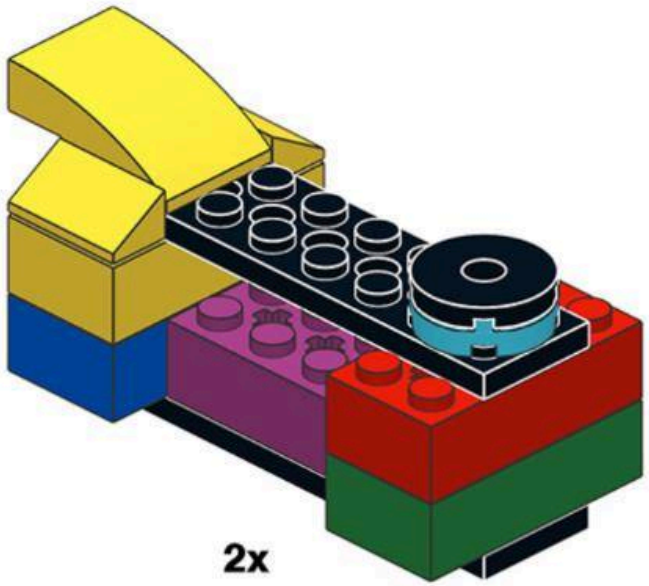
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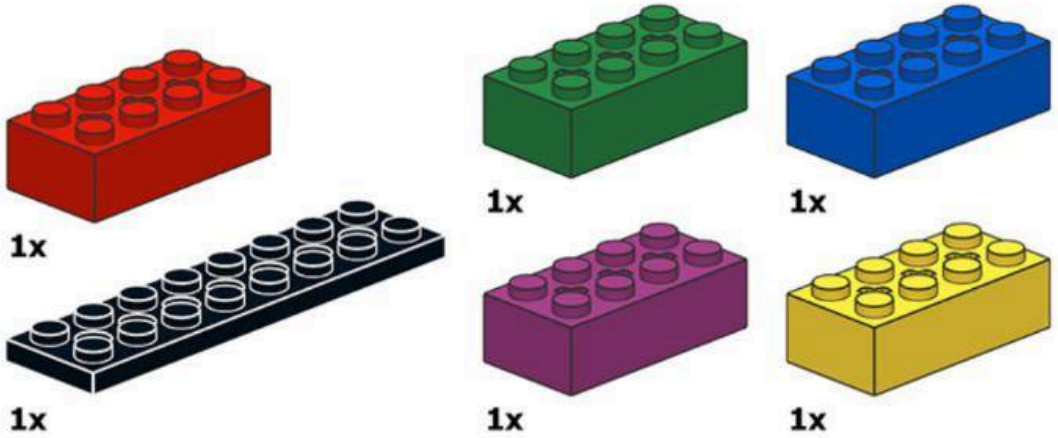


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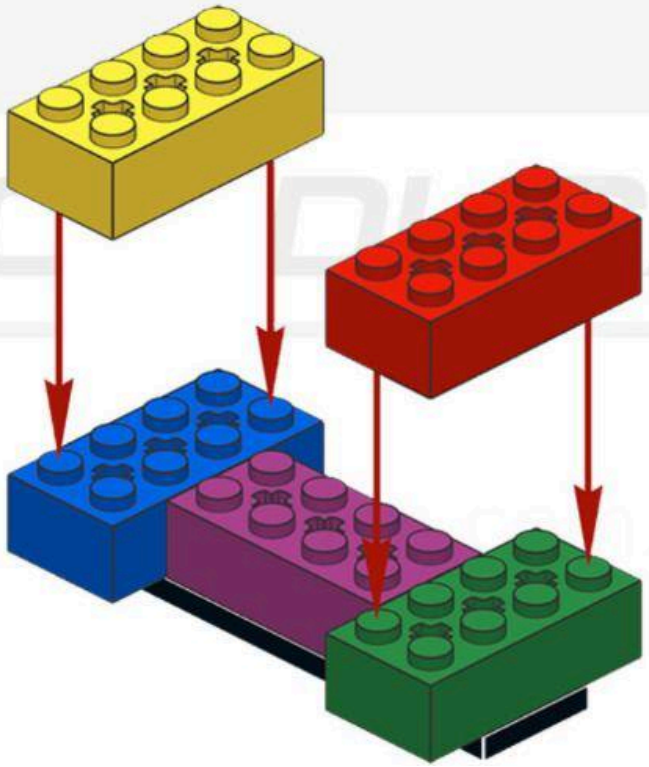
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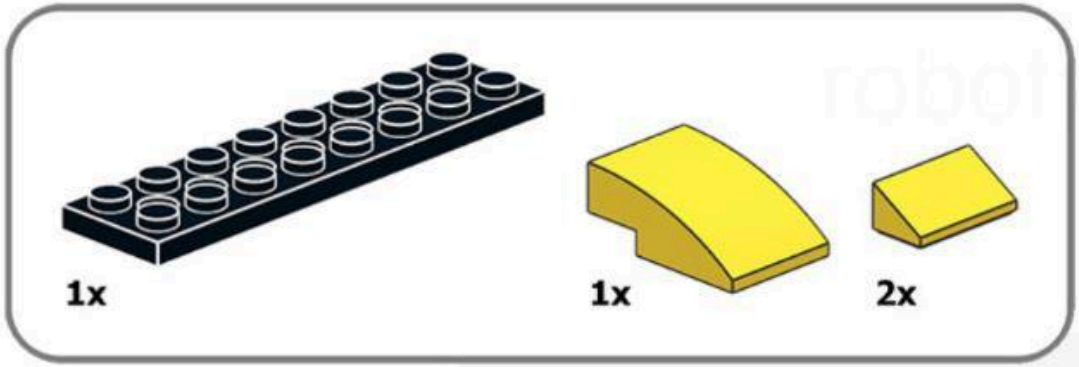




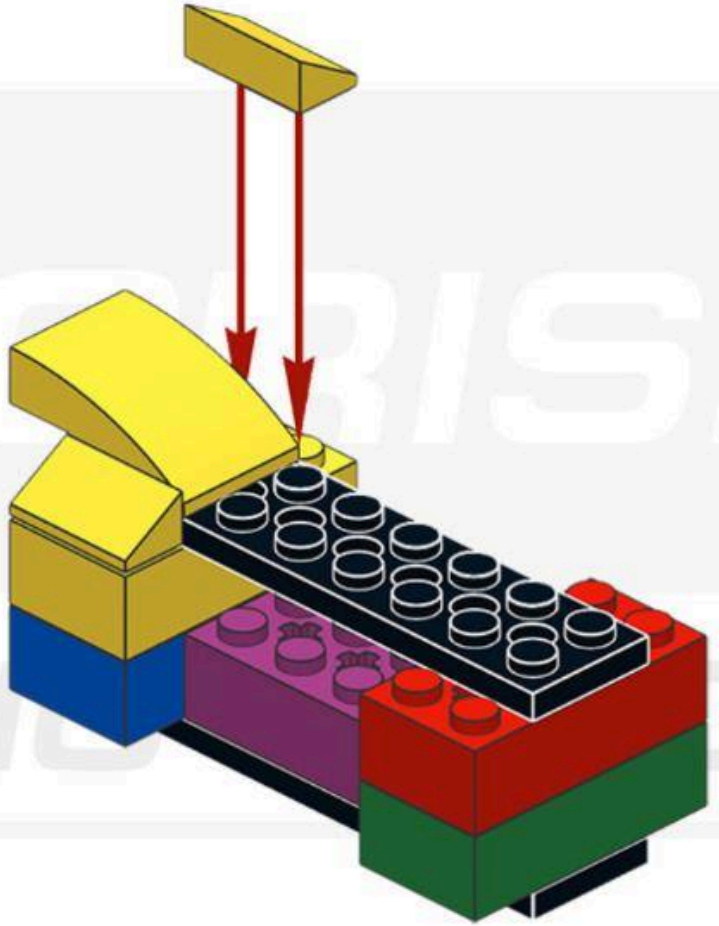
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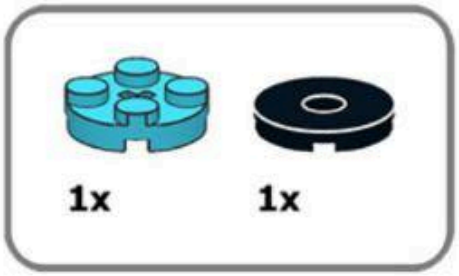


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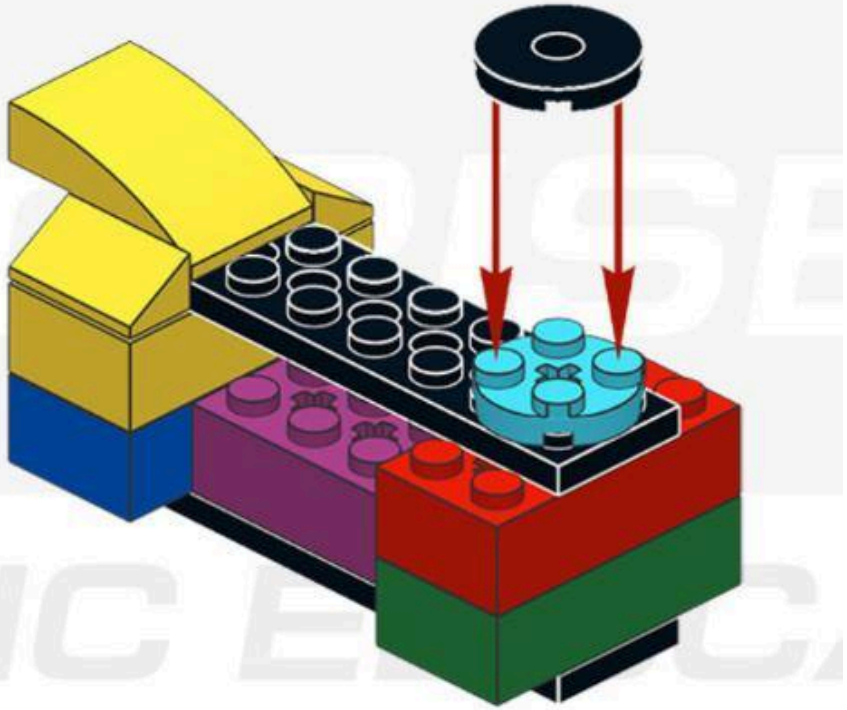


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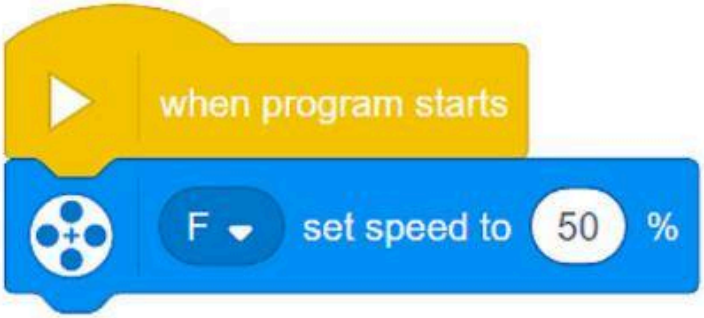




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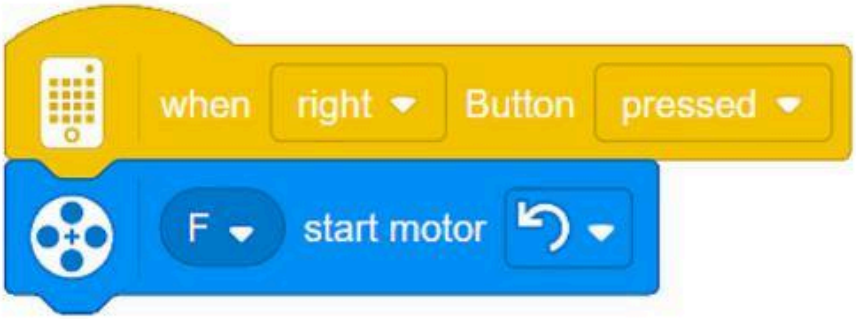


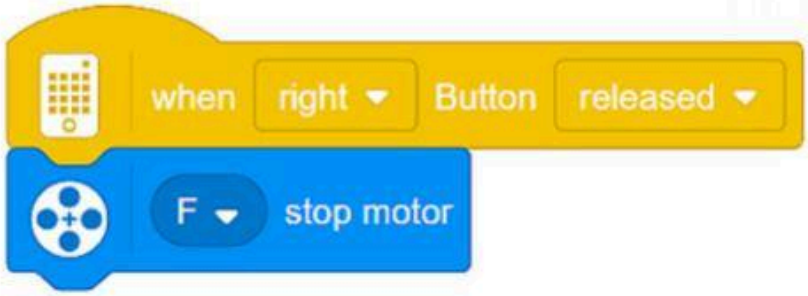
The program might look like this:







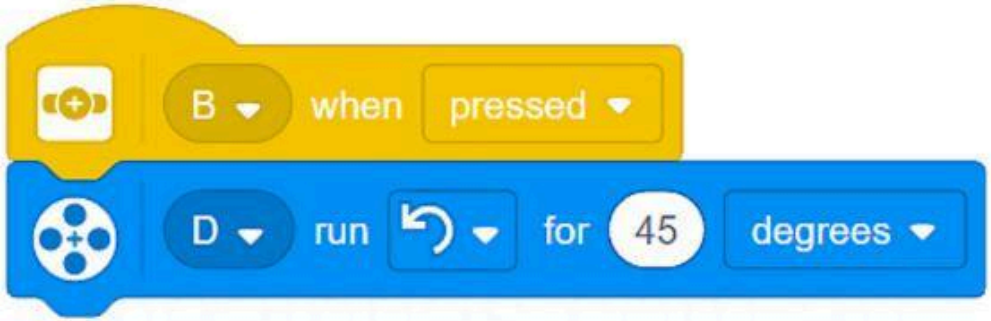




Program the robot arm to swivel manually. After pressing the touch sensor, the boom should return 45 degrees

Add the program as follows:





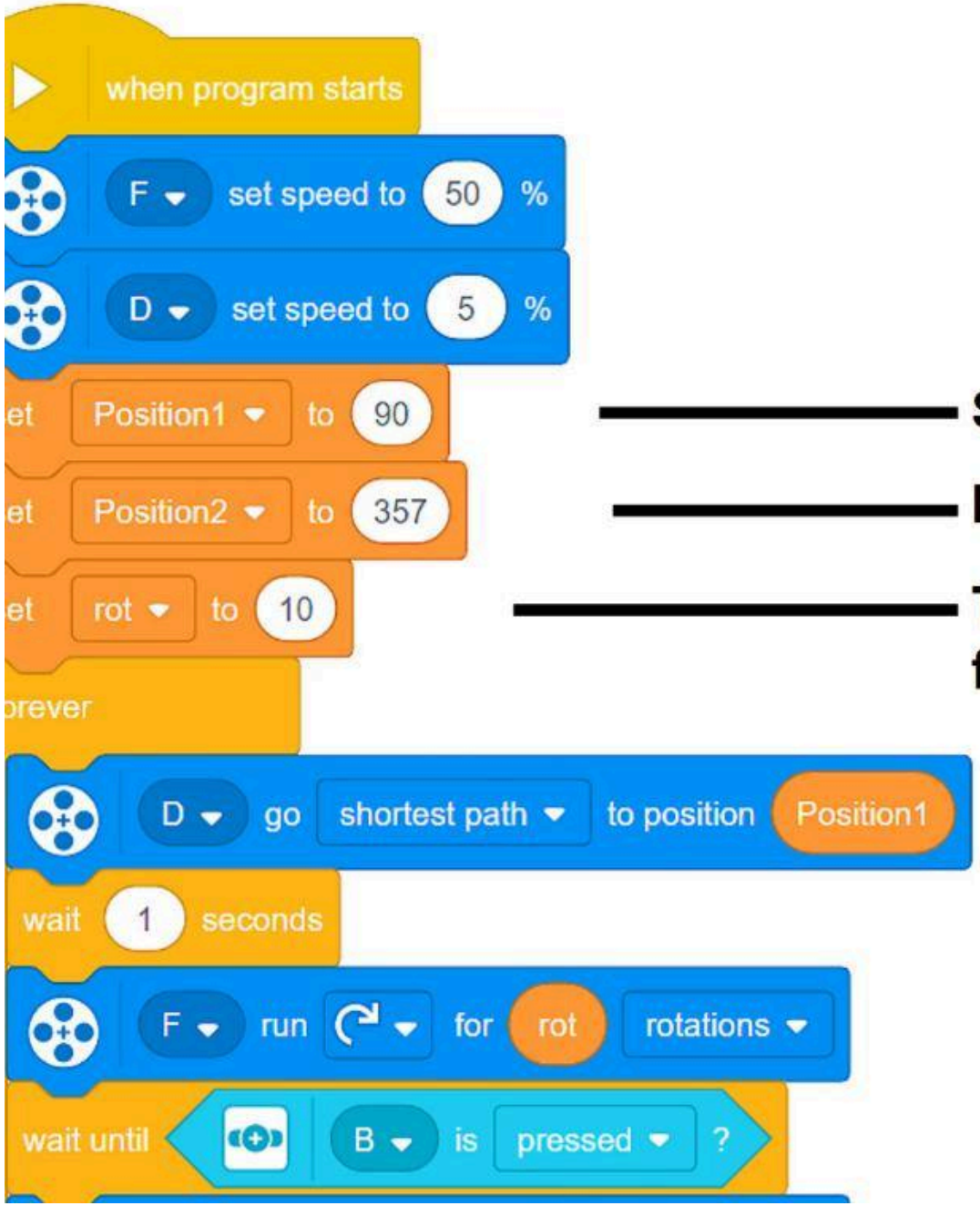
Move the weights between two points in manual robot control mode. What is the minimum time you need to move all three loads?

In order to program the robot for autonomous operation, you need to know how many turns of the reel are needed to raise or lower the load. For a convenient calculation of the number of turns, replace the part as shown in the video:

The next step is to determine the starting and ending positions of the robot. Rotate the robot arm to its original position and in the feedback window find the current number of degrees of the servo motor D. Repeat to determine the final position of the robot arm.

Program the autonomous operation of the crane. To do this, use the data obtained in the past tests. The touch sensor is convenient to use to confirm the attachment of the load.

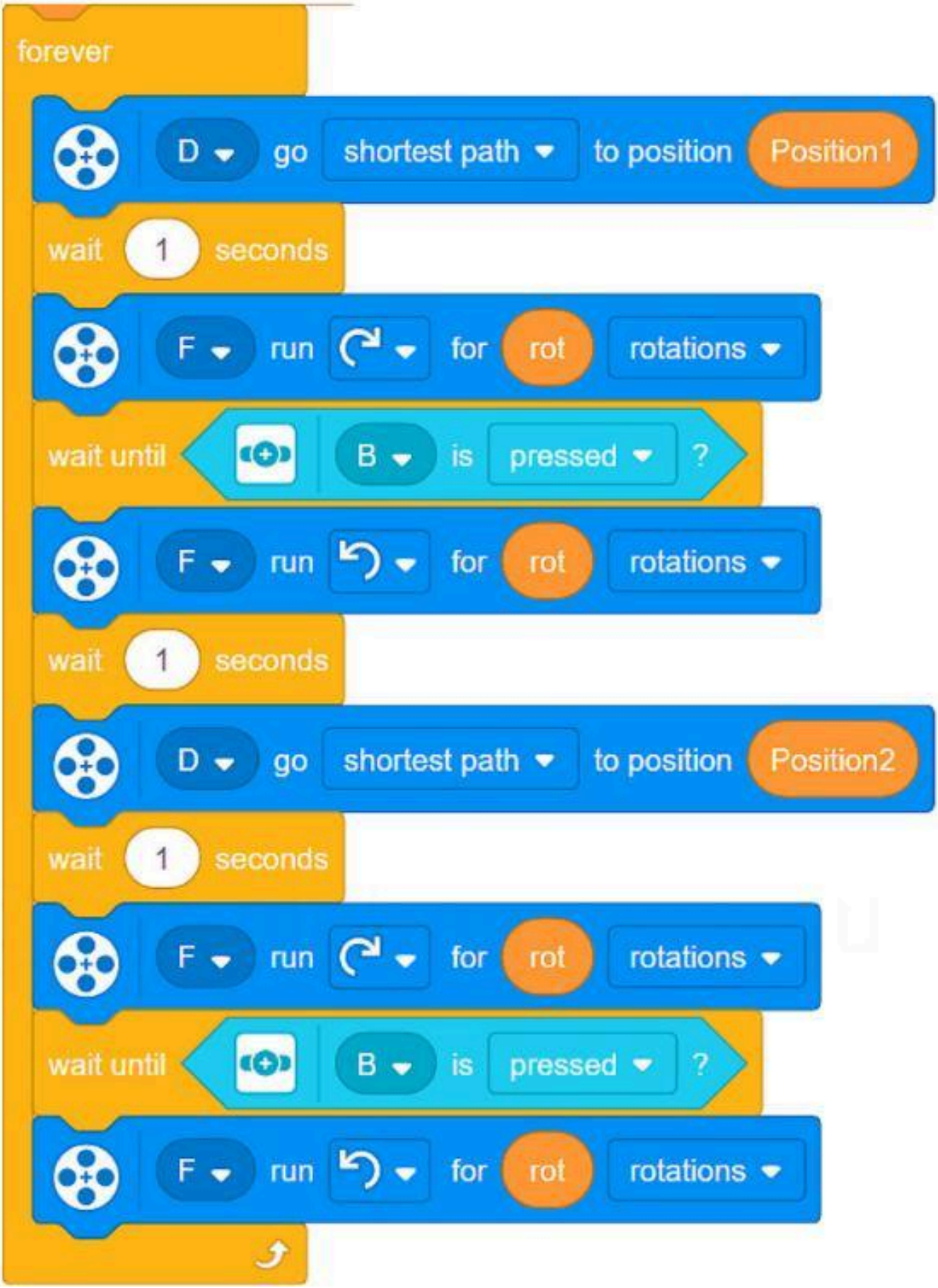
beginning of the program for the task is as follows:



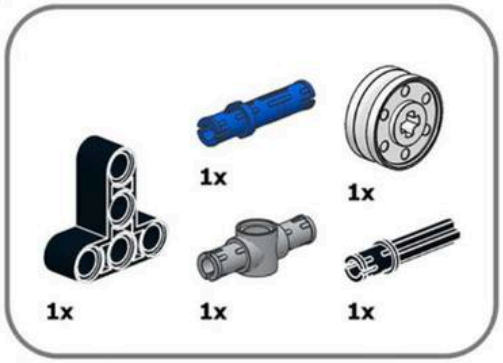
Starting position

End position

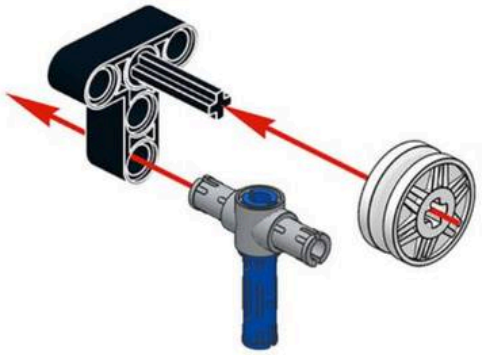
The number of turns for lifting the load from the lowest point to the top


forever
D go shortest path to position Position1
wait 1 seconds
F run for rot rotations
wait until B is pressed ?
F run for rot rotations
wait 1 seconds
D go shortest path to position Position2
wait 1 seconds
F run for rot rotations
wait until B is pressed ?
F run for rot rotations

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1



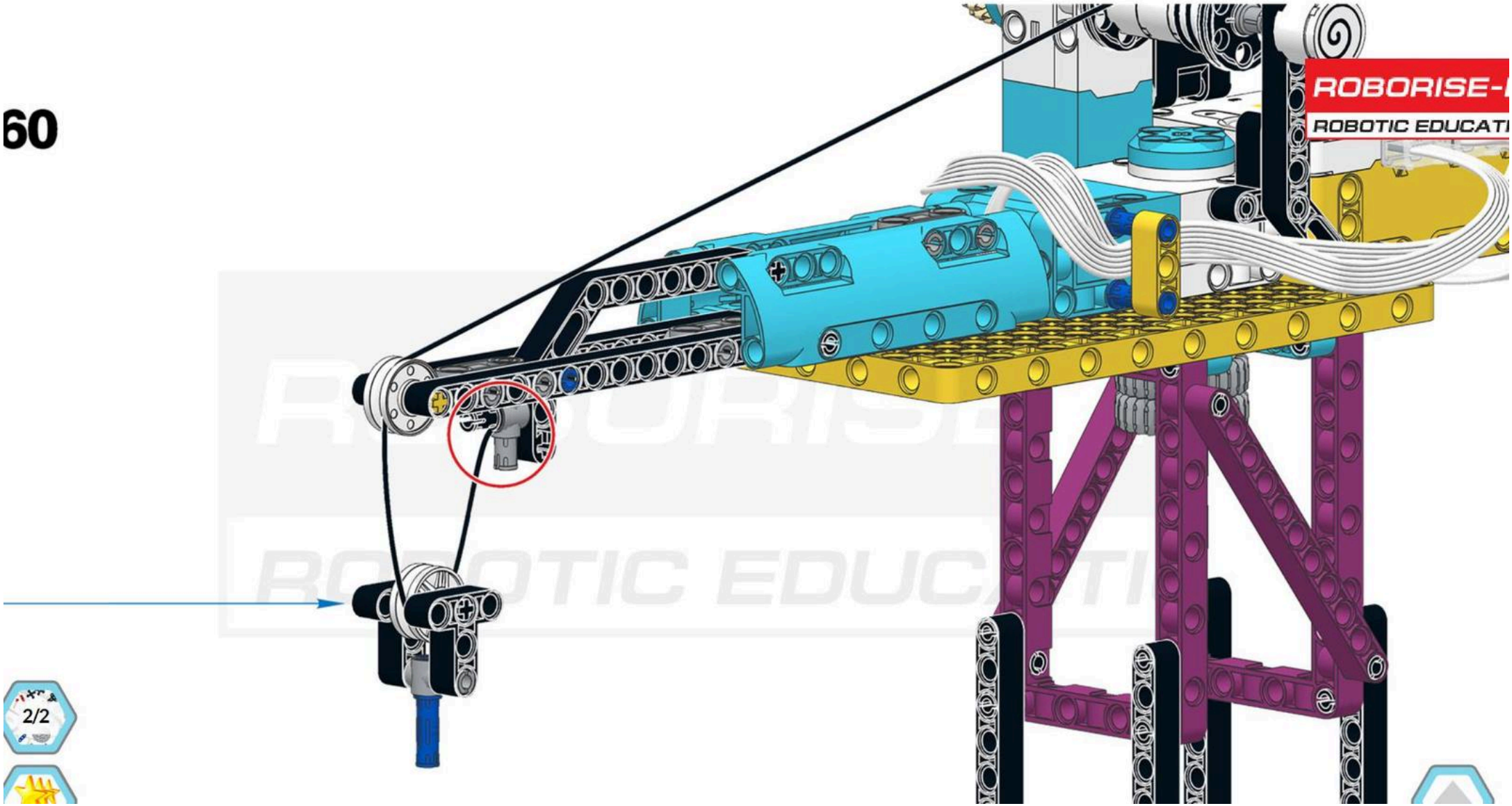


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